INTERNATIONAL STANDARD

IEC 60774-5

First edition
2004-04

Helical-scan video tape cassette system using 12,65 mm (0,5 in) magnetic tape on type VHS –

Part 5: D-VHS

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HELICAL-SCAN VIDEO TAPE CASSETTE SYSTEM USING 12,65 mm (0,5 in) MAGNETIC TAPE ON TYPE VHS –

Part 5: D-VHS

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International Standard IEC 60774-5 has been prepared by IEC technical committee 100: Audio, video and multimedia systems and equipment.

The text of this standard is based on the following documents:

CDV	Report on voting
100/664/CDV	100/748/RVC

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

IEC 60774 consists of several parts under the general title *Helical-scan video tape cassette system using 12,65 mm (0,5 in) magnetic tape on type VHS:*

Part 1: VHS and compact VHS video cassette system

Part 2: FM audio recording

Part 3: S-VHS

Part 4: S-VHS video cassette system - ET mode

Part 5: D-VHS

The committee has decided that the contents of this publication will remain unchanged until 2007. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version of this standard may be issued at a later date.

HELICAL-SCAN VIDEO TAPE CASSETTE SYSTEM USING 12,65 mm (0,5 in) MAGNETIC TAPE ON TYPE VHS –

Part 5: D-VHS

1 Scope

This part of IEC 60774 applies to the MPEG TS (Transport Stream) packet recording for helical-scan video tape cassette system using 12,65 mm magnetic tape on type VHS.

This standard specifies the cassettes, the tape, the track configuration, the data structure, the recording method and the MPEG TS format for D-VHS.

D-VHS is formatted on the basis of the VHS system, aiming for digital data recording. D-VHS basic format, which is commonly used in various types of data recording application, is specified.

D-VHS records the MPEG TS packets directly compatible with digital broadcast systems.

D-VHS has three recording modes according to the data-rate: STD (Standard), HS (High Speed) and LS (Low Speed/four types), to accommodate various digital broadcasts and needs. MPEG2 recording format for D-VHS including trick play is specified.

The pack format is specified for recording ancillary data, except the MPEG TS packets.

D-VHS MPEG format (ID: MTRM) is specified. D-VHS MPEG format is used for pre-recorded software tape for D-VHS, and self-encoding of analog video and audio signal (for example conventional TV) to MPEG TS packet.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60774-1:1994, *Helical-scan video tape cassette system using 12,65 mm (0,5 in) magnetic tape on type VHS – Part 1: VHS and compact VHS video cassette system*

IEC 60774-3:1993, *Helical-scan video tape cassette system using 12,65 mm (0,5 in) magnetic tape on type VHS – Part 3: S-VHS*

ISO/IEC 11172-3:1993, *Information technology – Coding of moving pictures and associated audio for digital storage media at up to about 1,5 Mbit/s – Part 3: Audio*

ISO/IEC 13818-1, *Information technology – Generic coding of moving pictures and associated audio information – Part 1: Systems*

ISO/IEC 13818-2, *Information technology – Generic coding of moving pictures and associated audio information – Part 2: Video*

ISO/IEC 13818-3:1998, *Information technology – Generic coding of moving pictures and associated audio information– Part 3: Audio*

ISO/IEC 13818-7:1998, *Information technology – Generic coding of moving pictures and associated audio information – Part 7: Advanced Audio Coding (AAC)*

ISO 639-2:1998, *Codes for the representation of names of languages – Part 2: Alpha-3 code*

ISO 2015:1976, *Numbering of weeks*

ISO 3166-1:1997, *Codes for the representation of names of countries and their subdivisions – Part 1: Country codes*

ISO 3901:2001, *Information and documentation – International Standard Recording Code (ISRC)*

ISO 8859-1:1997, *Information technology – 8-bit single-byte coded graphic character sets - Part 1: Latin alphabet No. 1*

ISO 13818-9:1996, *Information technology – Generic coding of moving pictures and associated audio information – Part 9: Extension for real-time interface for system decoders*

ARIB STD-B5:1996, *Ver 1.0, Data multiplex broadcasting system for the conventional television using the vertical blanking interval*

ARIB STD-B10:2004, *Ver 3.8, Service Information for Digital Broadcasting System*

ARIB STD-B24:2004, *Ver 4*, *Data coding system and transmission specification for digital broadcasting*

ARIB TR-B15:2004, Ver 2.9, *Operational guidelines for digital satellite broadcasting services using broadcasting satellites*

ATSC A/52B:2001, *Digital audio Compression (AC-3)*

ATSC.A/53B:2001, *ATSC Digital Television Standard*

ATSC A/70:2000, *Conditional Access System for Terrestrial Broadcast*

EIA 708 B:1999, *Digital Television (DTV) closed captioning*

EN 300 468:1998, *V1.3.1, Digital Video Broadcasting (DVB) - Specification for Service Information (SI) in DVB systems*

ETR 211:1997, *Digital Video Broadcasting (DVB); Guidelines on implementation and usage of Service Information (SI)*

ETS 300 743:1997, *Digital Video Broadcasting (DVB) - Subtitling systems*

JIS X 208:1997, *7-bit and 8-bit double byte coded KANJI sets for information interchange*

PNG (Portable Network Graphic) Specification Ver1.0, W3C Rec.Oct.1996, available at <http://www.w3.org/Graphics/PNG/>

SMPTE 302M:2002, *Television – Mapping of AES3 Data into MPEG-2 Transport Stream*

3 Terms, definitions and abbreviations

3.1 Terms and definitions

For the purposes of this part of IEC 60774-5, the following terms and definitions apply.

3.1.1
absolute track number
22-bit code recorded in the ID area of the Subcode sync block. The absolute track number is counted up every track and indicates the tape absolute position

3.1.2
absolute track number support flag
SF
2-bit code recorded in the ID area of the subcode sync block. SF indicates the condition of absolute track number

3.1.3
application detail
in the case of MPEG2 mode or CTP STD mode 1, the application detail is recorded in the main header. Application detail indicates the contents of main data

3.1.4
application ID
the application ID is recorded in the format ID. The application ID provides the definitions of subcode data area and main data area

3.1.5
code word
code word in ECC

3.1.6
data-AUX
in the case of MPEG2 transport stream recording or CTP stream recording, data-AUX is the next 1 byte after the main header. 6 data-AUX from 6 subsequent sync blocks constitute the pack data

3.1.7
data detail
the data detail is recorded in the sync block information area. The contents of Data detail depend on the contents of data type

3.1.8
data type
DT
the data type is recorded in the sync block information area in the main header. The data type indicates the contents of the sync block data

3.1.9
ECC block size
indicates the size of data which constitute the product codes of main code

3.1.10
ECC block number per track
number of ECC blocks per track

3.1.11
format ID
block of ID data in the subcode sync block or main header which contains ECC block size, ECC block number per track, program mode, scanner rotation speed, recording

3.1.12
format information area
first 4 bits of the main header. The format information area contains format ID, application details, application information, etc.

3.1.13
group of pictures
GOP
layer defined in MPEG

3.1.14
ID parity
IDP
error detecting code for ID

3.1.15

index flag
indicates the start point of the index

3.1.16

inter block gap
IBG
edit gap between two data areas

3.1.17
local time stamp generator
in the case of MPEG2 mode or CTP STD mode 1, the local time stamp generator is the drum reference generator in recording and playback. Local time stamp generator provides the value of the time stamp to be recorded

3.1.18
main data sync block
sync block in main code area. One main data sync block is 112 bytes

3.1.19
main header
in the case of MPEG2 mode or CTP STD mode, the main header is the first 2 bytes of the main data area in each main data sync block and it consists of format information area and sync block information area

3.1.20
marker flag
indicates the remarkable data area

3.1.21
outer interleave
number of tracks which constitute one interleave block for the outer error correcting codes

3.1.22
pack data
contains ancillary data as specified in the pack data format

3.1.23
packet header
4-byte information area associated with the transport packet. The packet header is recorded in front of each transport packet

3.1.24
pack header
first byte of pack data

3.1.25
precoder
device for the first process of interleaved NRZI

3.1.26
program mode
indicates the structure of the main code area

3.1.27
sequence number
the sequence number is recorded in the ID area of the main data sync block. The sequence number indicates the number of interleave sequences

3.1.28
skip flag
indicates the skip start point

3.1.29
start flag
indicates the start point of the program

3.1.30
subcode sync block
sync block in the subcode area. One subcode sync block is 28 bytes

3.1.31
sync block
SB
indicates the main data sync block or the subcode sync block or a 112 byte (896 bits) data length

3.1.32
sync block count
SBC
the sync block count is recorded in the sync block information area. Sync block count indicates the relation between D-VHS and the transport packet

3.1.33
sync block information area
12-bit data area in the main header. The sync block information area consists of data type, sync block count and data detail

3.1.34
sync block number
the sync block number is recorded in the ID area of the main data sync block and the subcode sync block. The sync block number indicates the position of the sync block

3.1.35
time stamp
input time of the transport packet. The time stamp is recorded in packer header as the sampled value of the local time stamp generator

3.1.36
track pair number
the track pair number is recorded in the ID area of the main data sync block. The track pair number represents interleave sequence

3.2 Abbreviations

AAC	Advanced Audio Coding
ATSC	Advanced Television Systems Committee
ARIB	Association of Radio Industries and Businesses
bslbf	bit string, left bit first
C/N	Carrier to Noise ratio
CGMS	Copy Generation Management System
CRC	Cyclic Redundancy Check Code
CTP	Compact Transport Packet (different format from MPEG2 TS packet)
DIT	Discontinuity Information Table
DTS	Digital Theater System
DVB	Digital Video Broadcasting
ES	Elementary Stream
ETR	ETSI Technical Report
ETS	European Telecommunication Standard
ETSI	European Telecommunication Standards Institute
IEC	International Electrotechnical Commission
I-picture	Intra-coded Frame or Field
IRD	Integrated Receiver-Decoder
ISO	International Organisation for Standardisation
ITU	International Telecommunication Union
HDTV	High Definition Television
HL	High Level
MJD	Modified Julian Date
ML	Main Level
MP	Main Profile
MPEG	Moving Pictures Expert Group
MTRM	MPEG Transport stream for Recording Media
NIT	Network Information Table
NRZI	Non-Return to Zero Inverse
PAT	Program Association Table
PCR	Program Clock Reference
PES	Packetized Elementary Stream
PID	Packet Identifier
PLTE	Palette
PMT	Program Map Table

PNG	Potable Network Graphics
PSI	Program Specific Information
rpchof	remainder polynomial coefficients, highest order first
RS code	Reed-Solomon Code
SI	Service Information
SIT	Selection Information Table
SDTV	Standard Definition Television
STD	System Target Decoder
TS	Transport Stream
uimsbf	unsigned integer most significant bit first
UTC	Universal Time, Co-ordinated

4 Fundamental specifications

4.1 Cassette

4.1.1 General

The cassettes described here shall be used for recording and playback of signals conforming to the D-VHS system.

4.1.2 Name

The following type name shall be used for D-VHS cassettes in order to indicate that they conform to the D-VHS system.

DF – *xxx* (*xxx* indicates tape length in minutes in STD mode)

for 160 min: DF – 160	for 300 min: DF – 300
for 180 min: DF – 180	for 360 min: DF – 360
for 240 min: DF – 240	for 420 min: DF – 420

4.1.3 Magnetic tape

4.1.3.1 General

a) The magnetic tape used in this cassette shall conform to 4.2.

b) After incorporation into the cassette, performance shall fulfil all items of the above-mentioned “magnetic tape” specifications. However, dropout measurements shall exclude the initial and final 1 min segments at the start and the end of tape length.

c) The magnetic tape wound shall be a single length and shall not include interim splices.

4.1.3.2 Length

The relationship between tape length and time for recording and playback is defined by the following formula.

$$L = \left[1{,}01 \; t + 3{,}0 \right]^{+3}_{0} \quad \text{(m)}$$

where

L is the tape length (m)

t is the recording or playback time (min)

NOTE *L* shall be an integer obtained by rounding up decimal fractions.

4.1.3.3 Thickness and strength

$$\begin{matrix} \text{DF - 160, DF - 180} \\ \text{DF - 240, DF - 300} \end{matrix} : Th = 17{,}5 \begin{matrix} +0{,}4 \\ -0{,}5 \end{matrix} \ \mu\text{m}$$

$$\text{DF - 360} \quad : Th = 14{,}4 \ \pm 0{,}5 \ \mu\text{m}: Sd = 0{,}64 \text{ N}: E = 1{,}03 \times 10^4 \text{ N/mm}^2$$

$$\text{DF - 420} \quad : Th = 12{,}4 \ \pm 0{,}4 \ \mu\text{m}: Sd = 0{,}64 \text{ N}: E = 1{,}37 \times 10^4 \text{ N/mm}^2$$

where

Th: Tape thickness (µm)

Sd: Dynamic collapse strength (N)

The *Sd* is measured with the “VG-5 dynamic collapse strength measuring instrument” manufactured by JVC or its equivalent.

E: Young’s modulus (N/mm^2)

$E = El + Et$

El: Young’s modulus of longitudinal direction (N/mm^2)

Et: Young’s modulus of transverse direction (N/mm^2)

4.1.3.4 Others

Other items not specified herein shall conform to the "S-VHS Video Tape Cassette" specified in IEC 60774-3:1993.

4.1.4 Leader tape and trailer tape

Leader tape and trailer tape shall conform to the "VHS Video Tape Cassette" specified in IEC 60774-1:1994.

4.1.5 Reel and cassette

For identification, D-VHS cassettes shall have identification holes (ID holes) defined in Figure 1. Other items, shape and dimensions shall conform to the "S-VHS Video Tape Cassette" specified in IEC 60774-3:1993.

Colors of cassettes shall be defined as follows.

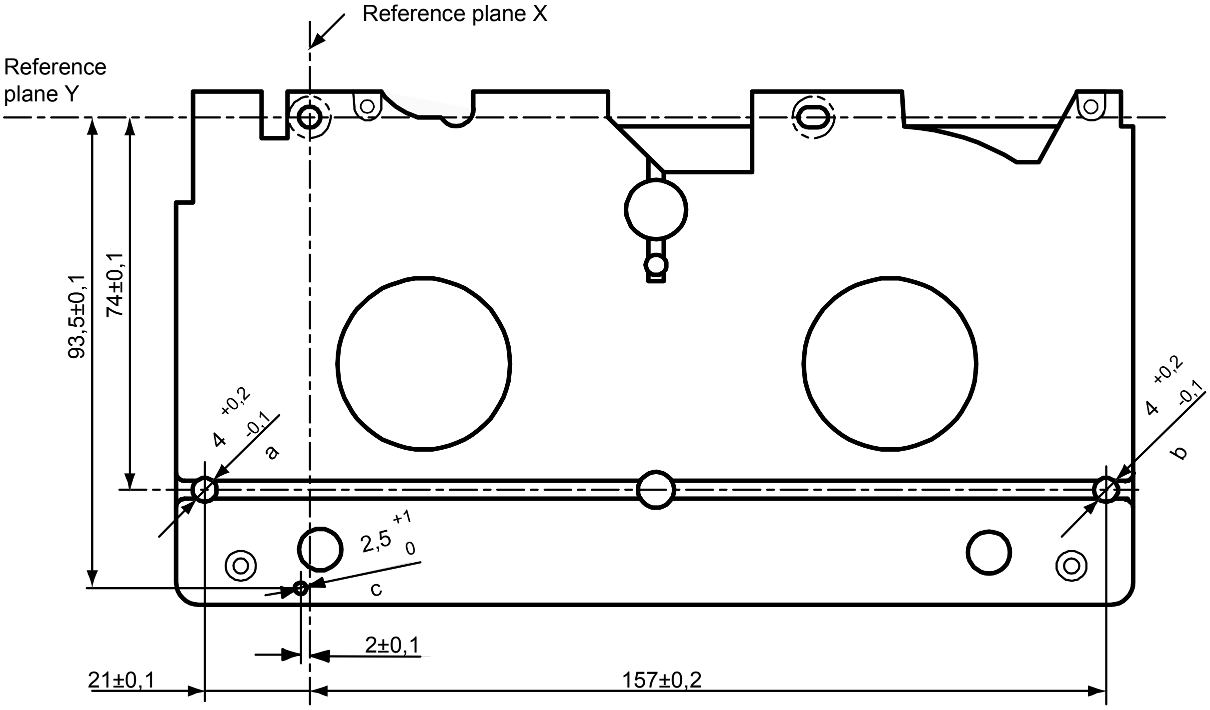
a) Windows and upper flanges of reels:

   Clear to indicate the remaining tape.

b) Cassette case:

   The colour of the cassette case, except the windows and upper flanges, shall be selected so that the light transmission is 0,45 % or less when measured with the cassette optical tester VT-2M manufactured by JVC or equivalent measuring instrument. In this case, the judgement shall be made with the higher value of either supply reel side or take-up reel side.

All units in mm



a Identification hole A: the depth shall be more than 3,8 mm from the reference plane Z.

b Identification hole B: the depth shall be more than 3,8 mm from the reference plane Z.

c Identification hole C: the depth shall be more than 3,0 mm from the reference plane Z.

Figure 1 – D-VHS cassette ID holes

4.2 Magnetic tape

4.2.1 General

This clause covers the technical details pertaining to the magnetic tape used in the D-VHS system, including technical descriptions other than those specified in 4.1. The recording characteristics are specified based on the standard tape “SRT-1 (S-VHS standard tape)”.

4.2.2 Test conditions

4.2.2.1 Environment

All tests shall be performed at room temperature of (20 ± 2) °C and relative humidity of (65 ± 5) % (hereinafter referred to as normal temperature and humidity), unless otherwise specified.

However, if there is no doubt about judgment, all tests may be performed at room temperature from 5 °C to 30 °C and relative humidity from 30 % to 80 %.

4.2.2.2 Test tape recorder

A cassette recorder conforming to D-VHS standards and S-VHS standards or equivalent measuring instruments shall be used.

4.2.3 Data recording properties

4.2.3.1 Reference RF recording current

The reference RF recording current is defined as the recording current which is necessary to obtain the maximum playback output level from the standard tape on which a 9,6 MHz square wave signal has been recorded.

4.2.3.2 Playback output

When a 9,6 MHz square wave signal is recorded with the reference RF recording current, then the output signal of 9,6 MHz is measured, it should be as follows with respect to the RF playback output of the standard tape.

9,6 MHz: + 1,0 dB or more

4.2.3.3 Carrier to noise ratio

When a 9,6 MHz square wave signal is recorded using the reference RF recording current, and the ratio between the playback output level and the noise level is measured, it should be as follows with respect to that of the standard tape.

Noise level: the average of 8,6 MHz and 10,6 MHz noise level

(Measuring condition of the noise level: resolution bandwidth 30 kHz)

C/N: + 1,0 dB or more

4.2.3.4 Dropout

Measuring method: When a 4,8 MHz square wave signal is recorded with the reference RF recording current, then the playback output level is measured, using the dropout counter VH06AZ manufactured by Shibasoku or its equivalent. A dropout is defined as a decline in the RF envelope output below –6 dB for a period exceeding 0,5 μs.

Dropout: there should be 200 dropouts or less per minute over the entire tape length.

4.2.4 Conformance with S-VHS specifications

Every item described in the IEC 60774-3: 1993, shall be fulfilled.

4.3 Basic format

4.3.1 Track configuration and dimensions

See Table 1.

Table 1 – Track configuration and dimensions

Item				Value	Figure number
1.	(*A*)	Tape width	mm	12,65 ± 0,01	2
2.	(*Vt*)	Tape speed	mm/s	(See [a])	
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01	
4.	(*Vh*)	Relative writing speed	m/s	(See [a])	
5.	(*P*)	Data track pitch	mm	(See [a])	2
6.	(*B*)	Total data width	mm	10,60	2
7.	(*W*)	Effective data width	mm	10,07	2
8.	(*L*)	Data track center (measures from reference edge)	mm	6,198	2
9.	(*T*)	Data track width	mm	(See [a])	2
10.	(*C*)	Control track width ([b])	mm	0,75 ± 0,1	2, 4
11.	(*R*)	Linear track width (mono track)	mm	1,0 ± 0,03	2, 4
12.	(*D*)	Linear track width (channel 2)	mm	0,35 ± 0,03	2, 4
13.	(*E*)	Linear track width (channel 1)	mm	0,35 ± 0,03	2, 4
14.	(*F*)	Linear track reference line (measured from reference edge)	mm	11,65 ± 0,03	2
15.	(*h*)	Linear-to-linear guardband width	mm	0,3	2, 4
16.	(θ_0)	Data track angle (tape stationary)		5° 56′ 7,4″	
17.	(θ)	Data track angle (tape running)		(See [a])	
18.	(α)	Data head gap azimuth angle		(See [c])	
19.	(*X*)	Position of linear and control head	mm	(See [a])	2
20.		Position of subcode sync (inside the bottom edge of *W*)		3 850 to 5 110 bits	2, 3
21.		Position of main code sync (inside the bottom edge of *W*)		13 706 to 14 966 bits	2, 3
22.		Tape back-tension (at the tape beginning and the entrance to drum)		30 to 45 g	
23.		Track shift error (from logical value)	µm	(See [a])	
24.		Track bend (peak-to-peak value)	µm	(See [a])	
25.		Number of tracks	tracks/s	(See [a])	
26.		Transmission rate	bits/s	(See [a])	

[a] The value of the parameter depends on Application ID.

[b] The control signal shall be recorded from the tape reference edge.

[c] The value of the parameter depends on Format ID.

Remarks

1) Where tolerances are not given, the quoted values are nominal.

2) Tests and measurements shall be carried out under the following condition. Temperature: 20 ± 2 °C, Relative humidity: (65 ± 5) %. However, provided no discrepancy in judgment is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, Relative humidity: 30 % to 80 %.

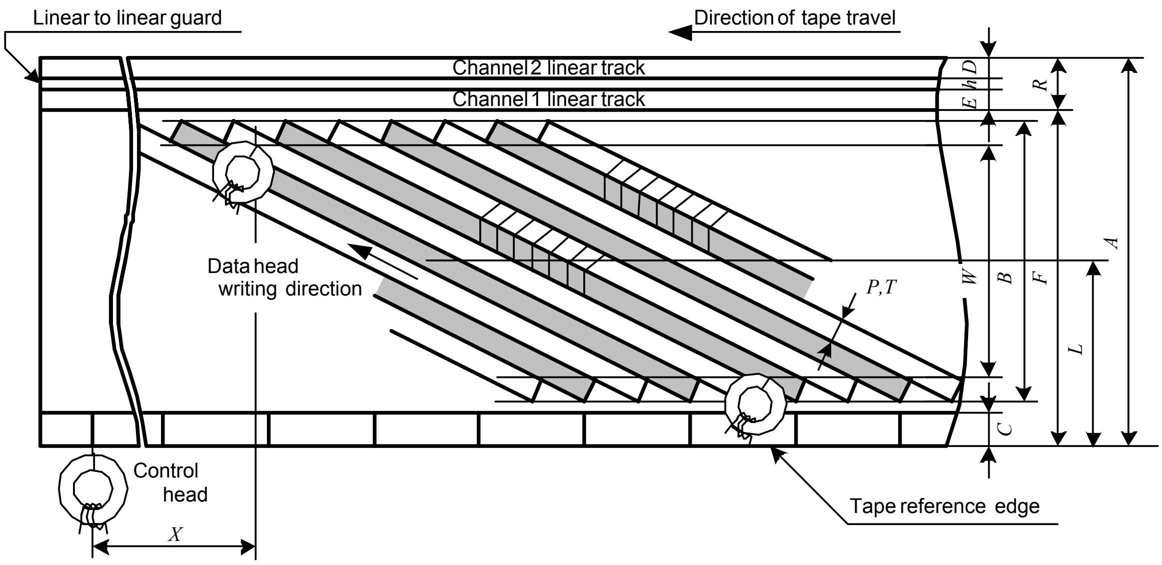
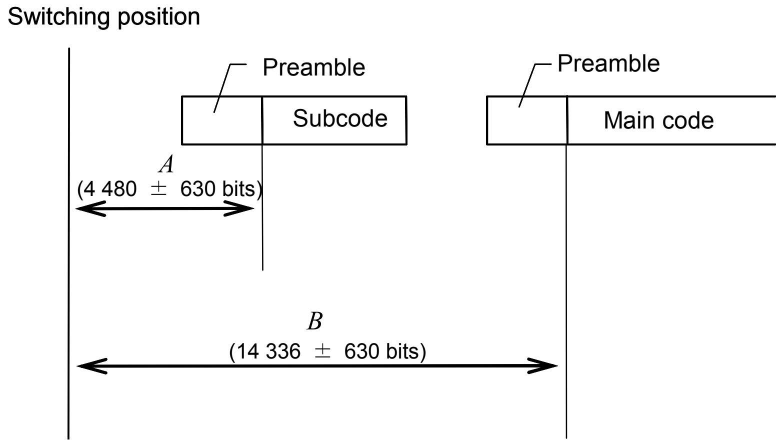


Figure 2– Magnetic tape position



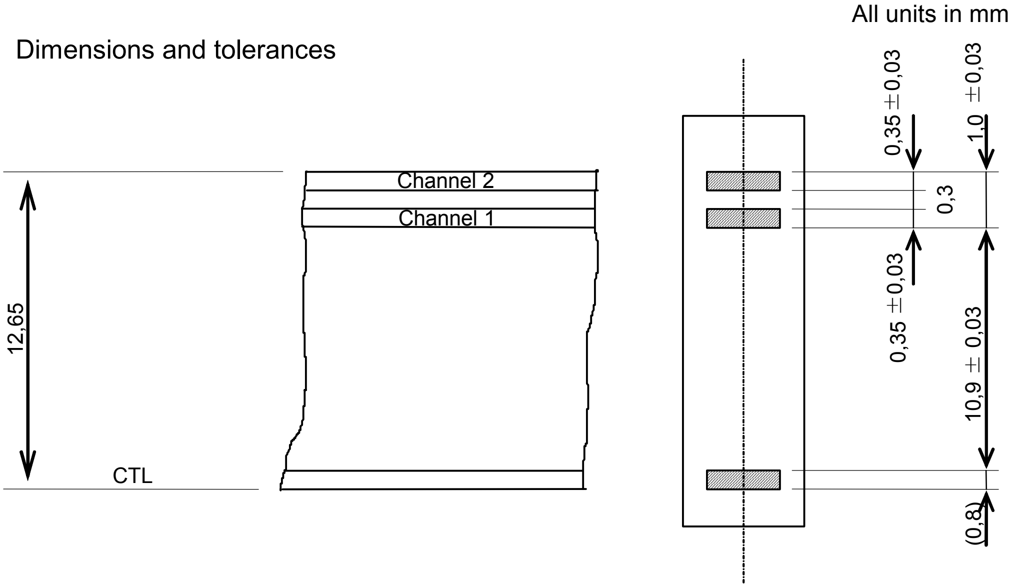
NOTE Specification of signal allocation is shown in 4.3.2.4. Signal allocation (Table 10)

In recording, the following rule shall apply.

$$3\,850 \text{ bits} \leq A \leq 5\,110 \text{ bits}$$

$$13\,706 \text{ bits} \leq B \leq 14\,966 \text{ bits}$$

Figure 3 – Switching position



Remark: linear and control head gap shall be in line.

Figure 4 – Linear and control head (reference)

4.3.2 D-VHS basic format

4.3.2.1 Recording data structure

Unless otherwise specified, MSB shall be written from the left and recorded first.

4.3.2.1.1 Track structure

A track consists of 356 sync blocks. However, when 1,001 flag in Format ID (see 4.3.2.1.3.3) is set to 1, a track consists of 356,356 sync blocks. Each of sync blocks is composed of 896 bits.

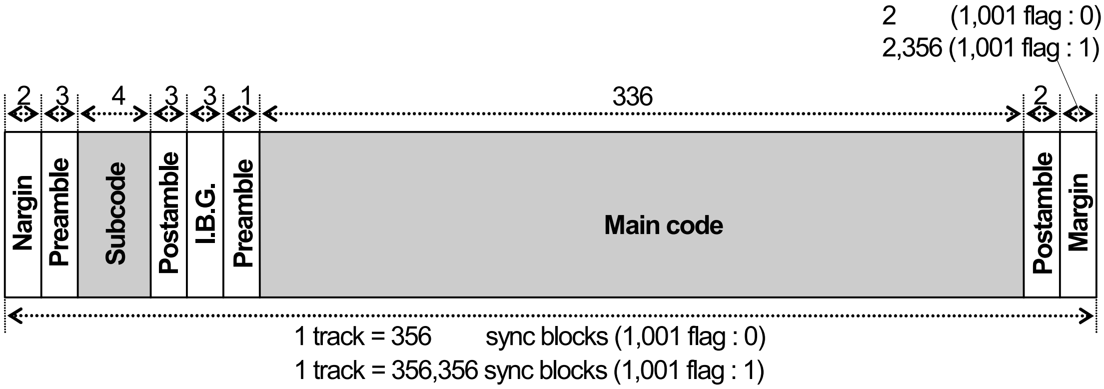


Figure 5 – Track structure

a) Main code area

Main code area consists of main data sync blocks.

b) Subcode area

Subcode area consists of subcode sync blocks. One sync block consists of 4 subcode sync blocks.

4.3.2.1.2 Main data sync block

Main data sync block consists of sync, ID, main data and inner parity.

One main data sync block has 99 symbols of main data and 8 symbols of inner parity.

One symbol is composed of 8 bits.

See Figure 6.

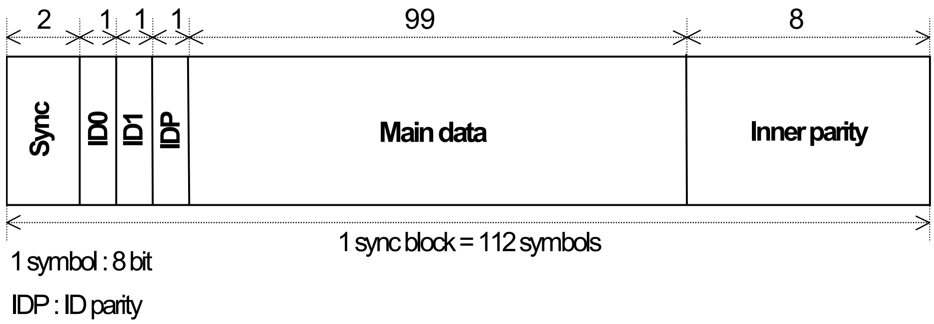


Figure 6 – Main data sync block

4.3.2.1.2.1 Sync

Synchronization pattern of main data sync block is defined before pre-coding.

Synchronization pattern has 2 undefined bits and 14 defined bits.

Synchronization pattern is defined as follows.

Synchronization pattern of main data sync block is “**01001011100011”.

4.3.2.1.2.2 ID

a) Configuration

See Table 2.

Table 2 – Configuration of ID (maindata sync block)

ID0								ID1							
MSB							LSB	MSB							LSB
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
Sequence number				Track pair number			Sync block number								

b) Sequence number

Sequence number is a 4-bit code.

The system to count the sequence number is defined in accordance with the application ID (see 4.3.2.1.3.3).

c) Track pair number

Track pair number is a 3-bit code.

Track pair number is counted up every track pair (plus and minus azimuth track).

The system to count track pair number is defined in accordance with the application ID (see 4.3.2.1.3.3).

d) Sync block number

Sync block number is a 9-bit code.

Sync block number is counted up every sync block and counted from 000000000 to 101001111.

The sync block number 000000000 shall be assigned to the first sync block of main code area.

e) ID parity

ID parity is an error detection code for ID0 and ID1.

ID parity is generated as follows.

Symbol definition

ID0 bit 7 to bit 4 $\rightarrow$ symbol 0, ID0 bit 3 to bit 0 $\rightarrow$ symbol 1, ID1 bit 7 to bit 4 $\rightarrow$ symbol 2,

ID1 bit 3 to bit 0 $\rightarrow$ symbol 3, ID parity bit 7 to bit 4 $\rightarrow$ symbol 4, ID parity bit 3 to bit 0 $\rightarrow$ symbol 5

See Table 3.

Table 3 – Symbol definition of ID parity (maindata sync block)

	MSB				LSB			
	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
ID0	symbol 0				symbol 1			
ID1	symbol 2				symbol 3			
IDP	symbol 4				symbol 5			

RS code over GF(2^4)

Primitive polynomial : $P(X) = X^4 + X + 1$

Generator polynomial : $G(X) = (X + \alpha)(X + 1)$ $\alpha = 0010$

$$Codeword = \sum_{i=0}^{5} Symbol_i \times X^{5-i}$$

The last 2 symbols are parity symbols.

4.3.2.1.2.3 Main data

One main data sync block has 99 symbols of main data.

4.3.2.1.2.4 Inner parity

See 4.3.2.2.

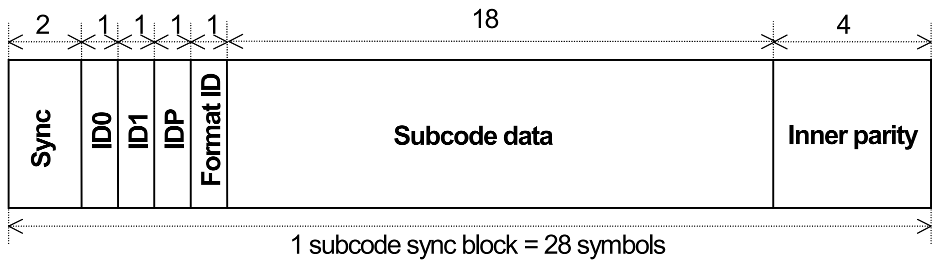
4.3.2.1.3 Subcode sync block

Subcode sync block consists of sync, ID, format ID, subcode data and inner parity.

One subcode sync block has 18 symbols of subcode data and 4 symbols of inner parity.

One symbol is composed of 8 bits.

See Figure 7.



1 symbol : 8 bit
IDP : ID parity

Figure 7 – Subcode sync block

4.3.2.1.3.1 Sync

Synchronization pattern of subcode sync block is defined before pre-coding.

Synchronization pattern has 2 undefined bits and 14 defined bits.

Synchronization pattern is defined as follows.

Synchronization pattern of subcode sync block is “**10110100011100”.

4.3.2.1.3.2 ID

a) Configuration

See Table 4.

Table 4 – Configuration of ID (subcode sync block)

SB#	ID0								ID1							
	MSB				LSB				MSB				LSB			
	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
4n	Reserved								Tag				Sync block number			
4n + 1	SF		Absolute track number (Upper byte)						Tag				Sync block number			
4n + 2	Absolute track number (Middle byte)								Tag				Sync block number			
4n + 3	Absolute track number (Lower byte)								Tag				Sync block number			

Reserved bits shall be set to “0”.

b) Absolute track number

Absolute track numbers are counted up every track and recorded as a 22 bit binary word.

The beginning of the tape is provided for the lead-in area.

The system to count the absolute track number is defined for each application with reference to the following formula. Only the lower 2 bits are counted differently for each data track pitch.

$$ATNo = \text{int}\left(\frac{n \times Tp}{14{,}5}\right)$$

where

$ATNo$: Absolute track number

Tp: data track pitch (µm)

$n = 0, 1, 2, 3, \ldots$

Also, the starting point of the absolute track number (000000H) shall be located on a point within 30 s from the beginning of the tape without regard to the program.

For pre-recorded tapes however, the starting point of absolute track number can be located at the starting point of the program.

c) SF: Absolute track number support flag

SF is used for the absolute track number support flag.

SF shall be recorded according to the definition shown in Table 5.

Table 5 – Absolute track number support flag

SF	Contents
00	Absolute track number is not supported.
01	Absolute track number is the estimated value and it is not necessarily unique.
10	Absolute track number is the estimated value and it is unique from the beginning of the tape to the present position, but a discontinuity may exist at the beginning of the present recording.
11	Absolute track number is continuous from the beginning of the tape.

If the recording of the absolute track number begins in the midst of the tape, an estimated value by the recorder can be recorded. In that case, the support flag shall be 01 during the double of the error (the difference between the estimated value and the value to be recorded if the absolute track number is kept recording continuously from the beginning). After that, the support flag shall be recorded as 10.

d) Tag

The tag is defined as shown in Table 6.

Table 6 – Tag

ID1			
Bit 7	Bit 6	Bit 5	Bit4
Start flag	Index flag	Skip flag	Marker flag

When the tag is in use, the following rules shall apply.

1) Start flag

The start flag shall be set to "1" from the start point of the program for 300 tracks.

2) Index flag

The index flag shall be set to "1" from the start point of index for 300 tracks.

3) Skip flag

The skip flag shall be set to "1" from the skip start point for 120 tracks. All the data from the start point of Skip flag to the next start point of start or index flag shall be skipped. If both skip flag and start flag or index flag are set to "1", skip flag shall have priority over the other flag.

4) Marker flag

Marker flag shall be set to "1" from the beginning of the data area to be marked for at least 120 tracks.

e) Sync block number

Sync block number is 4 bit code.

Sync block number is counted up every subcode sync block and counted from 0000 to 1111. Sync block number 0000 shall be assigned to the first subcode sync block of subcode area.

f) ID parity

ID parity is an error detecting code for ID0 and ID1.

ID parity is generated as follows.

Symbol definition

ID0 bit 7 to bit 4 $\rightarrow$ symbol 0, ID0 bit 3 to bit 0 $\rightarrow$ symbol 1, ID1 bit 7 to bit 4 $\rightarrow$ symbol 2,

ID1 bit 3 to bit 0 $\rightarrow$ symbol 3, ID parity bit 7 to bit 4 $\rightarrow$ symbol 4, ID parity bit 3 to bit 0 $\rightarrow$ symbol 5

See Table 7.

Table 7 – Symbol definition of ID parity (subcode sync block)

	MSB				LSB			
	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
ID0	symbol 0				symbol 1			
ID1	symbol 2				symbol 3			
IDP	symbol 4				symbol 5			

RS code over GF(2^4)

Primitive polynomial : $P(X) = X^4 + X + 1$

Generator polynomial : $G(X) = (X + \alpha)(X + 1) \quad \alpha = 0010$

$$Codeword = \sum_{i=0}^{5} Symbol_i \times X^{5-i}$$

Last 2 symbols are parity symbols.

4.3.2.1.3.3 Format ID

Format ID is defined as follows according to the sync block number. One set of format ID data is recorded in four consecutive sync blocks. See Table 8.

Table 8 – Format ID

SB#	MSB							LSB
	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
$4n$	ECC block size	ECC block number per track		Program mode			Scanner rotation speed	
$4n+1$	1,001 flag	Outer interleave			Recording mode			
$4n+2$	Reserved							
$4n+3$	Application ID					Reserved		

Reserved bits shall be set to “0”.

Application ID is defined as shown in Table 9.

Table 9 – Application ID

0xxxx	Consumer use
1xxxx	Professional or special use

The D-VHS recorders that are designed to record or playback the signal having the application ID value of 1xxxx shall not have the playback capability of the recorded signal having the application ID value of 0xxxx.

4.3.2.1.3.4 Subcode data

One subcode sync block has 18 symbols of subcode data. The definition of subcode data area varies according to the application ID in format ID.

4.3.2.1.3.5 Inner parity

See 4.3.2.2.

4.3.2.2 Error correcting code (inner parity)

4.3.2.2.1 Main code area

4.3.2.2.1.1 Data position definition

One main code area is composed of 336 main data sync blocks.

One main data sync block is composed of 107 symbols (without Sync, ID0, ID1, and ID parity). 99 symbols are main data and 8 symbols are inner parity.

The symbols in one main data sync block are numbered from 0 to 106 along recording direction.

The main data sync blocks in one main code area (one track) are numbered from 0 to 335 in the recording direction.

See Figure 8.

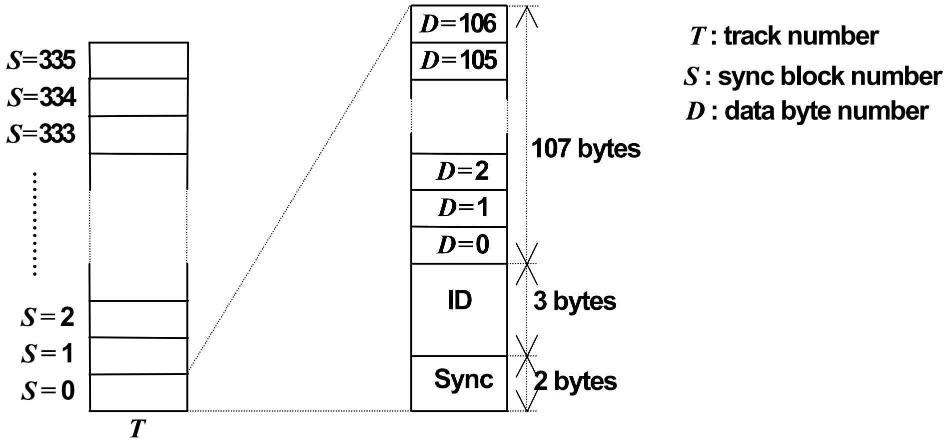


Figure 8 – Data position definition (main code)

4.3.2.2.1.2 ECC definition

Inner error correcting code: (107,99,9) RS code over GF(2^8).

The calculation is defined on GF(2^8) driven by the following polynomial.

$$\mathrm{P}(X) = X^8 + X^4 + X^3 + X^2 + 1$$

A primitive element α is defined as follows.

$$\alpha = 00000010$$

One inner code word is made of following symbols.

$$\mathrm{Codeword}(S) = \sum_{D=0}^{106} \mathrm{Data}_{S,D} \times X^{(106-D)}$$

$$(S = 0 \text{ to } 335)$$

where $\mathrm{Data}_{S,D}$ means a symbol defined by the sync block number: S, and data byte number: D.

Last 8 symbols (D = 99 to 106) are parity.

This equation indicates the inner codeword defined by the sync block number: S.

The generator polynomial for the inner error correcting code is as follows.

$$\mathrm{G}(X) = \prod_{i=0}^{7} (X + \alpha^i)$$

4.3.2.2.2 Subcode area

4.3.2.2.2.1 Data position definition

One subcode area is composed of 16 subcode sync blocks.

One subcode sync block is composed of 23 symbols (without Sync, ID0, ID1, and ID parity).

19 symbols are data and 4 symbols are inner parity.

The symbols in one sync block are numbered from 0 to 22 along the recording direction.

The subcode sync blocks in one subcode area (one track) are numbered from 0 to 15 along the recording direction.

See Figure 9.

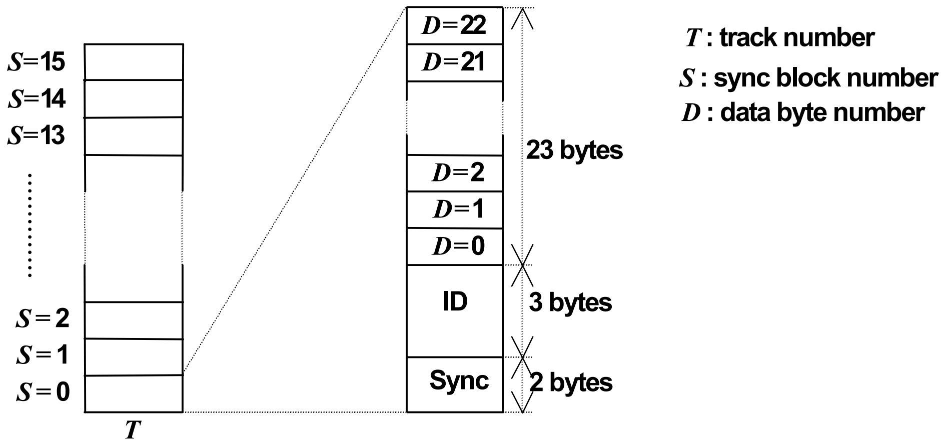


Figure 9 – Data position definition (subcode)

4.3.2.2.2.2 ECC definition

Inner error correcting code: (23,19,5) RS code over GF(2^8).

The calculation is defined on GF(2^8) driven by the following polynomial.

$$\mathrm{P}(X) = X^8 + X^4 + X^3 + X^2 + 1$$

A primitive element α is defined as follows.

$$\alpha = 00000010$$

One inner code word is made of following symbols.

$$\mathrm{Codeword}(S) = \sum_{D=0}^{22} \mathrm{Data}_{S,D} \times X^{(22-D)}$$

$$(S = 0 \text{ to } 15)$$

where $\mathrm{Data}_{S,D}$ means a symbol defined by the sync block number: S and data byte number: D.

Last 4 symbols (D = 19 to 22) are parity.

This equation indicates the inner codeword defined by the sync block number: S.

The generator polynomial for inner error correcting code is as follows.

$$G(X) = \prod_{i=0}^{3} (X + \alpha^{i})$$

4.3.2.3 Channel coding

Scrambled interleaved NRZI is used for this system.

4.3.2.3.1 Scrambler, descrambler

Prior to recording, the data in the main code area and subcode area shall be processed by the scramblers specified for the respective areas. In playback, respective descrambler processes shall be applied.

4.3.2.3.2 Precoder

After scrambling, all data (sync, ID, data and parity, but not including preamble, postamble, margin and IBG) is inverted and precoded. See Figure 10.

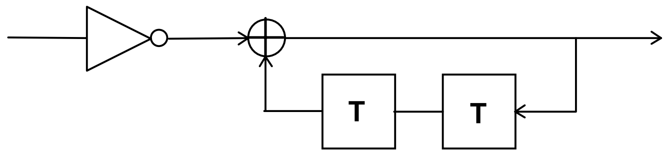


Figure 10 – Precoder

4.3.2.3.3 Margin, preamble, postamble, IBG

The recording pattern of margin, preamble, postamble, and IBG areas is as follows.

.....111000111000111000111000..... on the tape

This recording pattern shall be continuous for each 1 sync block length (112×8 bits).

At sync block boundaries the recording pattern need not be continuous.

Discontinuities in the tape pattern due to editing are allowed only in the IBG and margin areas.

4.3.2.4 Signal allocation

See Table 10.

Table 10 – Signal allocation

		number of SB	number of bits
margin		2	1 792
preamble		3	2 688
subcode		4	3 584
postamble		3	2 688
IBG		3	2 688
preamble		1	896
main code		336	301 056
postamble		2	1 792
margin	1,001 flag: 0	2	1 792
	1,001 flag: 1	2,356	2 111
total	1,001 flag: 0	356	318 976
	1,001 flag: 1	356,356	319 295

If all data in a subcode area or a main code area are rewritten, the tolerance for the errors of IBG shall be ± 440 bits as far as the provisions of switching position (see 4.3.1) are satisfied.

5 MPEG2 format

5.1 Track configuration and dimensions

Table 11 – Track configuration and dimensions (STD mode)

Item				Value
1.	(A)	Tape width	mm	12,65 ± 0,01
2.	**(Vt)**	**Tape speed**	**mm/s**	**16,67 ± 0,5 %**
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01
4.	**(Vh)**	**Relative writing speed**	**m/s**	**5,827 (5,821)**
5.	**(P)**	**Data track pitch**	**mm**	**0,029**
6.	(B)	Total data width	mm	10,60
7.	(W)	Effective data width	mm	10,07
8.	(L)	Data track center (measures from reference edge)	mm	6,198
9.	**(T)**	**Data track width**	**mm**	**0,029**
10.	(C)	Control track width [a)]	mm	0,75 ± 0,1
11.	(R)	Linear track width (mono track)	mm	1,0 ± 0,03
12.	(D)	Linear track width (channel 2)	mm	0,35 ± 0,03
13.	(E)	Linear track width (channel 1)	mm	0,35 ± 0,03
14.	(F)	Linear track reference line		
		(measured from reference edge)	mm	11,65 ± 0,03
15.	(h)	Linear-to-linear guardband width	mm	0,3
16.	(θ_0)	Data track angle (tape stationary)		5°56'7,4"
17.	**(θ)**	**Data track angle (tape running)**		**5°57'8,4" (5°57'8,5")**
18.	**(α_1, α_2)**	**Data head gap azimuth angle**		**30° ± 9'**
19.	**(X)**	**Position of linear and control head**	**mm**	**79,248**

Item			Value
20.	Position of subcode sync (inside the bottom edge of W)		3 850 bits to 5 110 bits
21.	Position of main code sync (inside the bottom edge of W)		13 706 bits to 14 966 bits
22.	Tape back-tension (at the tape beginning and the entrance to drum)		30 g to 45 g
23.	**Track shift error (from logical value)**	**µm**	**±7**
24.	**Track bend (peak-to-peak value)**	**µm**	**7 or less**
25.	**Number of tracks**	**tracks/s**	**60 (59,94)**
26.	**Transmission rate per channel**	**bits/s**	**(19 138 560 ± 500) ppm**

[a] The control signal shall be recorded from the tape reference edge.

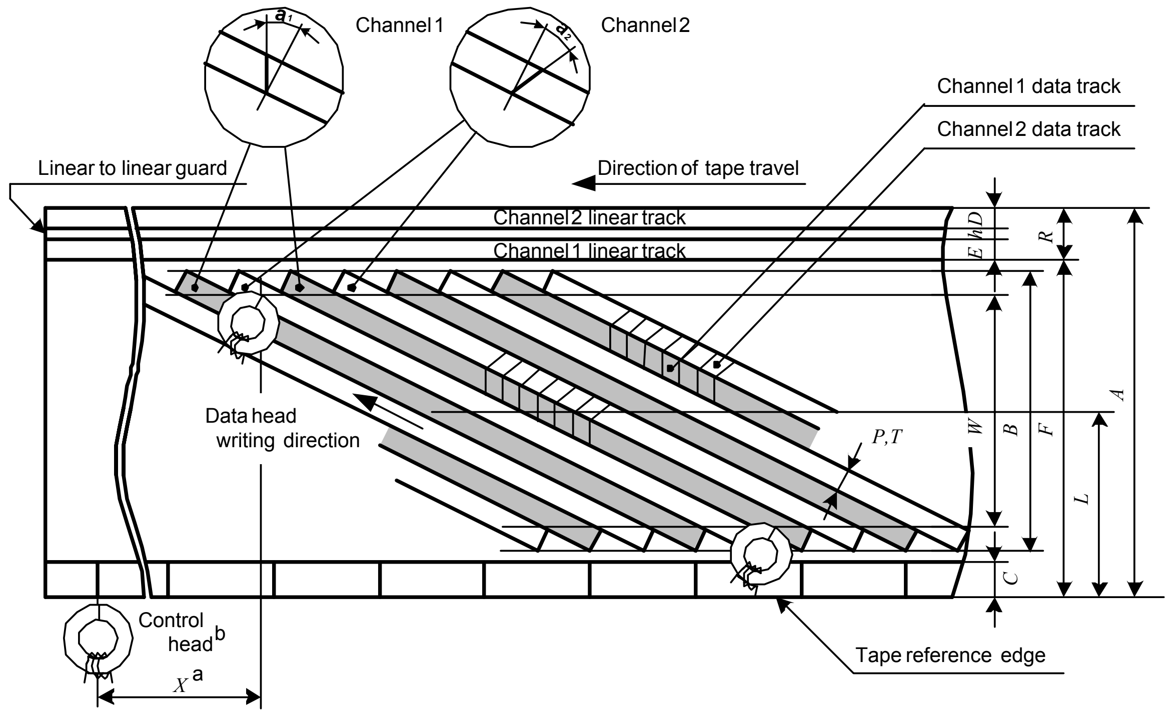
Remarks

1) When number of tracks is 59,94 tracks/s, relative writing speed and data track angle (tape running) are given in parentheses.

2) Where tolerances are not given, the quoted values are nominal.

3) Tests and measurements shall be carried out under the following conditions. Temperature: (20 ± 2) °C, relative humidity: (65 ± 5) %. However, provided no discrepancy in judgment is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, relative humidity: 30 % to 80 %.

4) The items in bold text are basic values and the items in normal text are calculated values.



Key

a X shall be measured from the end of the 180° scan of channel 2 to the recorded control signal on the tape.

The reference point on the control track is the transition point of the polarity of remanent magnetization on the tape, from south poles representing the direction of tape travel to its opposite.

b The control signal shall be recorded on the control track with sufficient current for saturation and the rise time of recording current shall be less than 200 µs.

A positive going edge of the recorded control signal shall be in coincidence with the start of channel 1 track scanning.

The control signal shall be recorded so that the rotating drum side of the control head poles would be north polarized when the pulse signal is positive.

Figure 11 – Magnetic tape position (STD mode)

Table 12 – Track configuration and dimensions (LS2 mode)

Item				Value
1.	(*A*)	Tape width	mm	12,65 ± 0,01
2.	**(*Vt*)**	**Tape speed**	**mm/s**	**8,33 ± 0,5 %**
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01
4.	**(*Vh*)**	**Relative writing speed**	**m/s**	**5,835 (5,829)**
5.	**(*P*)**	**Data track pitch**	**mm**	**0,029**
6.	(*B*)	Total data width	mm	10,60
7.	(*W*)	Effective data width	mm	10,07
8.	(*L*)	Data track center (measures from reference edge)	mm	6,198
9.	**(*T*)**	**Data track width**	**mm**	**0,029**
10.	(*C*)	Control track width [a]	mm	0,75 ± 0,1
11.	(*R*)	Linear track width (mono track)	mm	1,0 ± 0,03
12.	(*D*)	Linear track width (channel 2)	mm	0,35 ± 0,03
13.	(*E*)	Linear track width (channel 1)	mm	0,35 ± 0,03
14.	(*F*)	Linear track reference line		
		(measured from reference edge)	mm	11,65 ± 0,03
15.	(*h*)	Linear-to-linear guardband width	mm	0,3
16.	(θ_0)	Data track angle (tape stationary)		5°56'7,4"
17.	**(θ)**	**Data track angle (tape running)**		**5°56'37,9" (5°56'37,9")**
18.	**(α_1, α_2)**	**Data head gap azimuth angle**		**30° ± 9'**
19.	**(*X*)**	**Position of linear and control head**	**mm**	**79,248**
20.		Position of subcode sync (inside the bottom edge of *W*)		3 850 bits to 5 110 bits
21.		Position of main code sync (inside the bottom edge of *W*)		13 706 bits to 14 966 bits
22.		Tape back-tension		
		(at the tape beginning and the entrance to drum)		30 g to 45 g
23.		**Track shift error (from logical value)**	**µm**	**±7**
24.		**Track bend (peak-to-peak value)**	**µm**	**7 or less**
25.		**Number of tracks**	**tracks/s**	**30 (29,97)**
26.		**Transmission rate per channel**	**bits/s**	**(19 138 560) ± 500 ppm**

[a] The control signal shall be recorded from the tape reference edge.

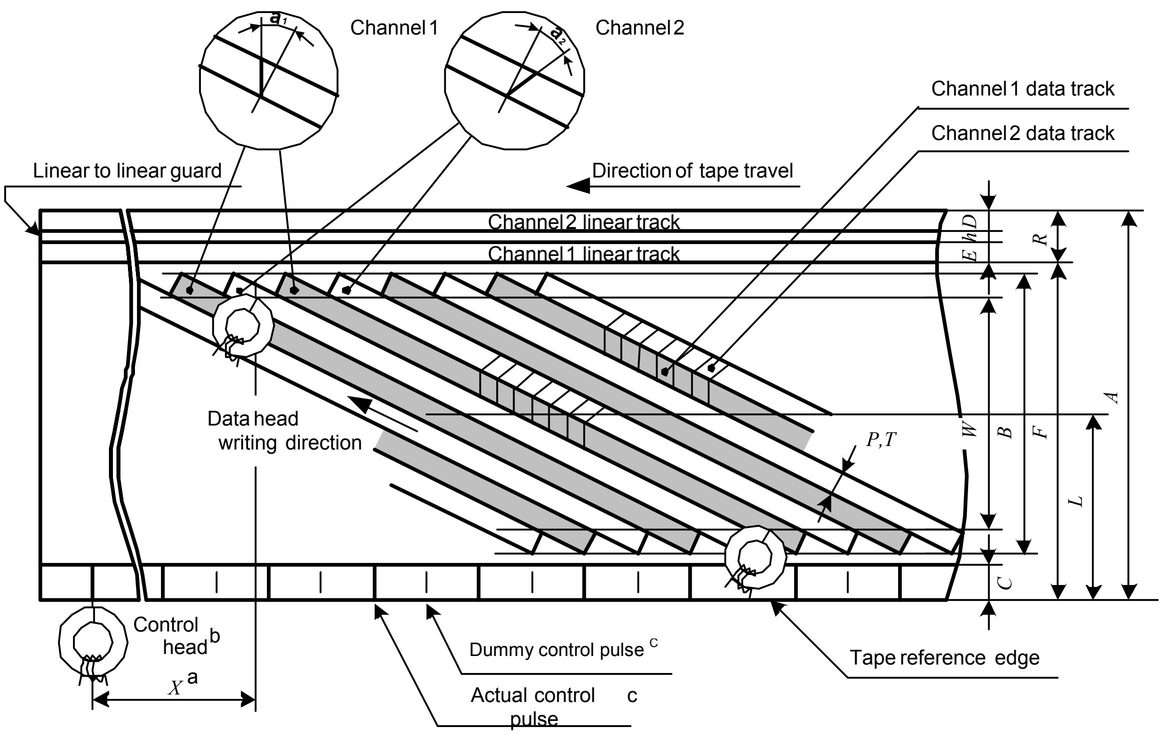
Remarks

1) When number of tracks is 29,97 tracks/s, relative writing speed and data track angle (tape running) are given in parentheses.

2) Where tolerances are not given, the quoted values are nominal.

3) Tests and measurements shall be carried out under the following conditions. Temperature: (20 ± 2) °C, relative humidity: (65 ± 5) %. However, provided no discrepancy in judgment is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, relative humidity: 30 % to 80 %.

4) The items in bold text are basic values and the items in normal text are calculated values.



a	X shall be measured from the end of the 180° scan of channel 2 to the recorded control signal on the tape.

The reference point on the control track is the transition point of the polarity of remanent magnetization on the tape, from south poles representing the direction of tape travel to its opposite.

b	The control signal shall be recorded on the control track with sufficient current for saturation and the rise time of recording current shall be less than 200 µs.

A positive going edge of the recorded control signal shall be in coincidence with the start of channel 1 track scanning.

The control signal shall be recorded so that the rotating drum side of the control head poles would be north polarized when the pulse signal is positive.

c	Control pulse frequency during recording shall be 30 Hz.

When the number of tracks is 29,97 tracks/s, the frequency shall be 29,97 Hz.

Figure 12 – Magnetic tape position (LS2 mode)

Table 13 – Track configuration and dimensions (LS3 mode)

Item				Value
1.	(*A*)	Tape width	mm	12,65 ± 0,01
2.	**(*Vt*)**	**Tape speed**	**mm/s**	**5,55 ± 0,5%**
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01
4.	**(*Vh*)**	**Relative writing speed**	**m/s**	**5,838 (5,832)**
5.	**(*P*)**	**Data track pitch**	**mm**	**0,029**
6.	(*B*)	Total data width	mm	10,60
7.	(*W*)	Effective data width	mm	10,07
8.	(*L*)	Data track center (measures from reference edge)	mm	6,198
9.	**(*T*)**	**Data track width**	**mm**	**0,029**
10.	(*C*)	Control track width [a]	mm	0,75 ± 0,1
11.	(*R*)	Linear track width (mono track)	mm	1,0 ± 0,03
12.	(*D*)	Linear track width (channel 2)	mm	0,35 ± 0,03
13.	(*E*)	Linear track width (channel 1)	mm	0,35 ± 0,03
14.	(*F*)	Linear track reference line (measured from reference edge)	mm	11,65 ± 0,03
15.	(*h*)	Linear-to-linear guardband width	mm	0,3
16.	(θ_0)	Data track angle (tape stationary)		5°56'7,4"
17.	**(θ)**	**Data track angle (tape running)**		**5°56'27,7" (5°56'27,7")**
18.	**(α_1, α_2)**	**Data head gap azimuth angle**		**30° ± 9'**
19.	**(*X*)**	**Position of linear and control head**	**mm**	**79,248**
20.		Position of subcode sync (inside the bottom edge of *W*)		3 850 bits to 5 110 bits
21.		Position of main code sync (inside the bottom edge of *W*)		13 706 bits to 14 966 bits
22.		Tape back-tension (at the tape beginning and the entrance to drum)		30 g to 45 g
23.		**Track shift error (from logical value)**	**µm**	**±7**
24.		**Track bend (peak-to-peak value)**	**µm**	**7 or less**
25.		**Number of tracks**	**tracks/s**	**20 (19,98)**
26.		**Transmission rate per channel**	**bits/s**	**(19 138 560) ± 500 ppm**

[a] The control signal shall be recorded from the tape reference edge.

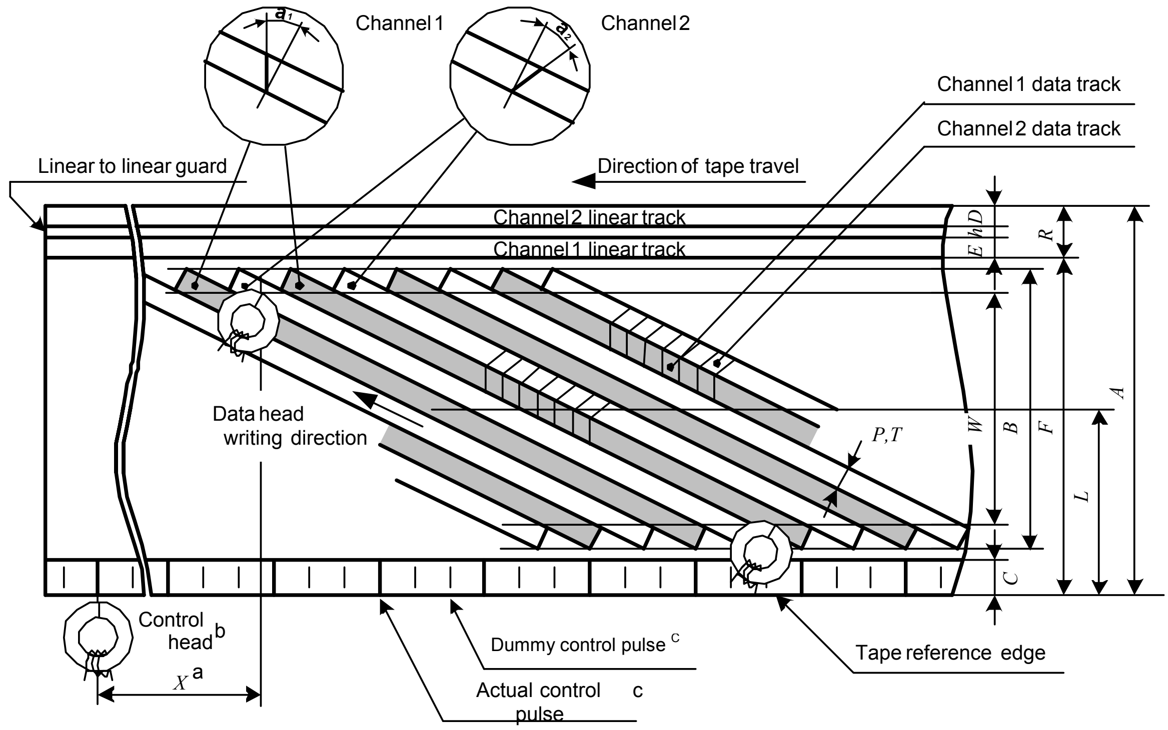
Remarks

1) When the number of tracks is 19,98 tracks/s, relative writing speed and data track angle (tape running) are given in parentheses.

2) Where tolerances are not given, the quoted values are nominal.

3) Tests and measurements shall be carried out under the following conditions. Temperature: (20 ± 2) °C, relative humidity: (65 ± 5) %. However, provided no discrepancy in judgement is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, relative humidity: 30 % to 80 %.

4) The items in bold text are basic values and the items in normal text are calculated values.



A X shall be measured from the end of the 180° scan of channel 2 to the recorded control signal on the tape.

The reference point on the control track is the transition point of the polarity of remanent magnetization on the tape, from south poles representing the direction of tape travel to its opposite.

B The control signal shall be recorded on the control track with sufficient current for saturation and the rise time of recording current shall be less than 200 µs.

A positive going edge of the recorded control signal shall be in coincidence with the start of channel 1 track scanning.

The control signal shall be recorded so that the rotating drum side of the control head poles would be north polarized when the pulse signal is positive.

C Control pulse frequency during recording shall be 30 Hz.

When the number of tracks is 19,98 tracks/s, the frequency shall be 29,97 Hz.

Figure 13 – Magnetic tape position (LS3 mode)

Table 14 – Track configuration and dimensions (LS5 mode)

Item				Value
1.	(A)	Tape width	mm	12,65 ± 0,01
2.	**(Vt)**	**Tape speed**	**mm/s**	**3,33 ± 0,5%**
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01
4.	**(Vh)**	**Relative writing speed**	**m/s**	**5,840 (5,834)**
5.	**(P)**	**Data track pitch**	**mm**	**0,029**
6.	(B)	Total data width	mm	10,60
7.	(W)	Effective data width	mm	10,07
8.	(L)	Data track center (measures from reference edge)	mm	6,198
9.	**(T)**	**Data track width**	**mm**	**0,029**
10.	(C)	Control track width [a)]	mm	0,75 ± 0,1
11.	(R)	Linear track width (mono track)	mm	1,0 ± 0,03
12.	(D)	Linear track width (channel 2)	mm	0,35 ± 0,03
13.	(E)	Linear track width (channel 1)	mm	0,35 ± 0,03
14.	(F)	Linear track reference line		
		(measured from reference edge)	mm	11,65 ± 0,03
15.	(h)	Linear-to-linear guardband width	mm	0,3
16.	(θ_0)	Data track angle (tape stationary)		5°56'7,4"
17.	**(θ)**	**Data track angle (tape running)**		**5°56'19,6" (5°56'19,6")**
18.	**(α_1, α_2)**	**Data head gap azimuth angle**		**30° ± 9'**
19.	**(X)**	**Position of linear and control head**	**mm**	**79,248**
20.		Position of subcode sync (inside the bottom edge of W)		3 850 bits to 5 110 bits
21.		Position of main code sync (inside the bottom edge of W)		13 706 bits to 14 966 bits
22.		Tape back-tension		
		(at the tape beginning and the entrance to drum)		30 to 45 g
23.		**Track shift error (from logical value)**	**µm**	**±7**
24.		**Track bend (peak-to-peak value)**	**µm**	**7 or less**
25.		**Number of tracks**	**tracks/s**	**12 (11,99)**
26.		**Transmission rate per channel**	**bits/s**	**(19 138 560 ± 500) ppm**

[a)] The control signal shall be recorded from the tape reference edge.

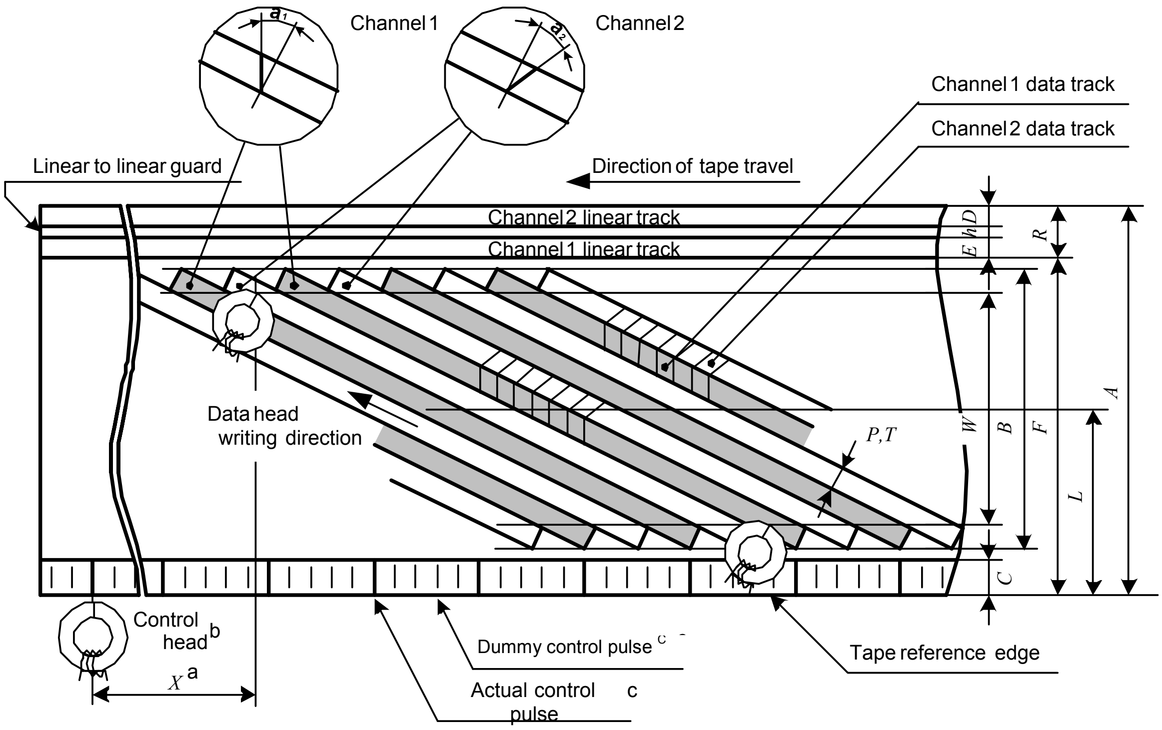
Remarks

1) When the number of tracks is 11,99 tracks/s, relative writing speed and data track angle (tape running) are given in parentheses.

2) Where tolerances are not given, the quoted values are nominal.

3) Tests and measurements shall be carried out under the following conditions. Temperature: (20 ± 2) °C, relative humidity: (65 ± 5) %. However, provided no discrepancy in judgment is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, relative humidity: 30 % to 80%.

4) The items in bold text are basic values and the items in normal text are calculated values.



a *X* shall be measured from the end of the 180° scan of channel 2 to the recorded control signal on the tape.

The reference point on the control track is the transition point of the polarity of remanent magnetization on the tape, from south poles representing the direction of tape travel to its opposite.

b The control signal shall be recorded on the control track with sufficient current for saturation and the rise time of recording current shall be less than 200 µs.

A positive going edge of the recorded control signal shall be in coincidence with the start of channel 1 track scanning.

The control signal shall be recorded so that the rotating drum side of the control head poles would be north polarized when the pulse signal is positive.

C Control pulse frequency during recording shall be 30 Hz.

When the number of tracks is 11,99 tracks/s, the frequency shall be 29,97 Hz.

Figure 14 – Magnetic tape position (LS5 mode)

Table 15 – Track configuration and dimensions (LS7 mode)

Item				Value
1.	(A)	Tape width	mm	12,65 ± 0,01
2.	**(Vt)**	**Tape speed**	**mm/s**	**2,38 ± 0,5 %**
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01
4.	**(Vh)**	**Relative writing speed**	**m/s**	**5,841 (5,835)**
5.	**(P)**	**Data track pitch**	**mm**	**0,029**
6.	(B)	Total data width	mm	10,60
7.	(W)	Effective data width	mm	10,07
8.	(L)	Data track center (measures from reference edge)	mm	6,198
9.	**(T)**	**Data track width**	**mm**	**0,029**
10.	(C)	Control track width [a)]	mm	0,75 ± 0,1
11.	(R)	Linear track width (mono track)	mm	1,0 ± 0,03
12.	(D)	Linear track width (channel 2)	mm	0,35 ± 0,03
13.	(E)	Linear track width (channel 1)	mm	0,35 ± 0,03
14.	(F)	Linear track reference line		
		(measured from reference edge)	mm	11,65 ± 0,03
15.	(h)	Linear-to-linear guardband width	mm	0,3
16.	(θ_0)	Data track angle (tape stationary)		5°56'7,4"
17.	**(θ)**	**Data track angle (tape running)**		**5°56'16,1" (5°56'16,1")**
18.	**(α_1, α_2)**	**Data head gap azimuth angle**		**30° ± 9'**
19.	**(X)**	**Position of linear and control head**	**mm**	**79,248**
20.		Position of subcode sync (inside the bottom edge of W)		3 850 bits to 5 110 bits
21.		Position of main code sync (inside the bottom edge of W)		13 706 bits to 14 966 bits
22.		Tape back-tension		
		(at the tape beginning and the entrance to drum)		30 g to 45 g
23.		**Track shift error (from logical value)**	**µm**	**±7**
24.		**Track bend (peak-to-peak value)**	**µm**	**7 or less**
25.		**Number of tracks**	**tracks/s**	**8,57 (8,56)**
26.		**Transmission rate per channel**	**bits/s**	**(19 138 560 ± 500) ppm**

[a)] The control signal shall be recorded from the tape reference edge.

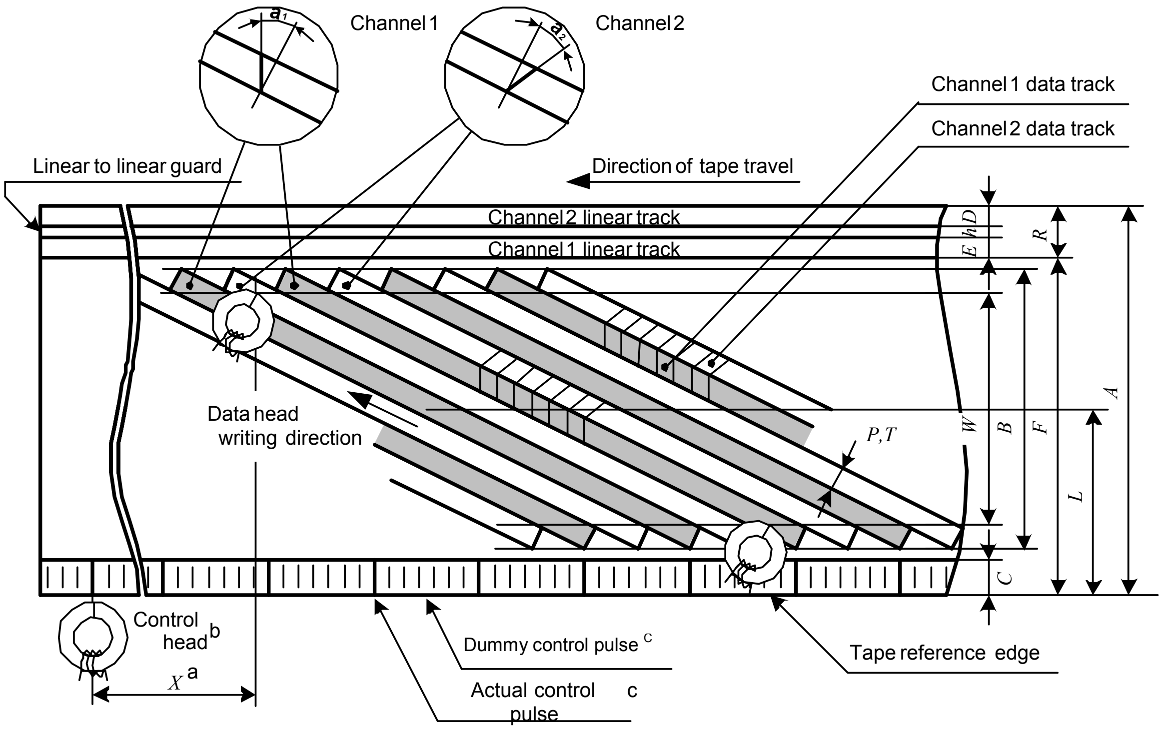
Remarks

1) When number of tracks is 8,56 tracks/s, relative writing speed and data track angle (tape running) are given in parentheses.

2) Where tolerances are not given, the quoted values are nominal.

3) Tests and measurements shall be carried out under the following conditions. Temperature: (20 ± 2) °C, relative humidity: (65 ± 5) %. However, provided no discrepancy in judgment is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, relative humidity: 30 % to 80 %.

4) The items in bold text are basic values and the items in normal text are calculated values.



a X shall be measured from the end of the 180° scan of channel 2 to the recorded control signal on the tape.

The reference point on the control track is the transition point of the polarity of remanent magnetization on the tape, from south poles representing the direction of tape travel to its opposite.

b The control signal shall be recorded on the control track with sufficient current for saturation and the rise time of recording current shall be less than 200 µs.

A positive going edge of the recorded control signal shall be in coincidence with the start of channel 1 track scanning.

The control signal shall be recorded so that the rotating drum side of the control head poles would be north polarized when the pulse signal is positive.

c Control pulse frequency during recording shall be 30 Hz.

When the number of tracks is 8,56 tracks/s, the frequency shall be 29,97 Hz.

Figure 15 – Magnetic tape position (LS7 mode)

Table 16 – Track configuration and dimensions (HS mode)

Item				Value
1.	(A)	Tape width	mm	12,65 ± 0,01
2.	**(Vt)**	**Tape speed**	**mm/s**	**33,35 ± 0,5%**
3.	(ϕ)	Drum diameter (upper drum)	mm	62 ± 0,01
4.	**(Vh)**	**Relative writing speed**	**m/s**	**5,810 (5,804)**
5.	**(P)**	**Data track pitch**	**mm**	**0,029**
6.	(B)	Total data width	mm	10,60
7.	(W)	Effective data width	mm	10,07
8.	(L)	Data track center (measures from reference edge)	mm	6,198
9.	**(T)**	**Data track width**	**mm**	**0,029**
10.	(C)	Control track width [a)]	mm	0,75 ± 0,1
11.	(R)	Linear track width (mono track)	mm	1,0 ± 0,03
12.	(D)	Linear track width (channel 2)	mm	0,35 ± 0,03
13.	(E)	Linear track width (channel 1)	mm	0,35 ± 0,03
14.	(F)	Linear track reference line		
		(measured from reference edge)	mm	11,65 ± 0,03
15.	(h)	Linear-to-linear guardband width	mm	0,3
16.	(θ_0)	Data track angle (tape stationary)		5°56'7,4"
17.	**(θ)**	**Data track angle (tape running)**		**5°58'9,8" (5°58'10,8")**
18.	**(α_1, α_2)**	**Data head gap azimuth angle**		**30° ± 9'**
19.	**(X)**	**Position of linear and control head**	**mm**	**79,248**
20.		Position of subcode sync (inside the bottom edge of W)		3 850 bits to 5 110 bits
21.		Position of main code sync (inside the bottom edge of W)		13 706 bits to 14 966 bits
22.		Tape back-tension		
		(at the tape beginning and the entrance to drum)		30 g to 45 g
23.		**Track shift error (from logical value)**	**µm**	**±7**
24.		**Track bend (peak-to-peak value)**	**µm**	**7 or less**
25.		**Number of tracks**	**tracks/s**	**120 (119,88)**
26.		**Transmission rate per channel**	**bits/s**	**(19 138 560 ± 500) ppm**

[a)] The control signal shall be recorded from the tape reference edge.

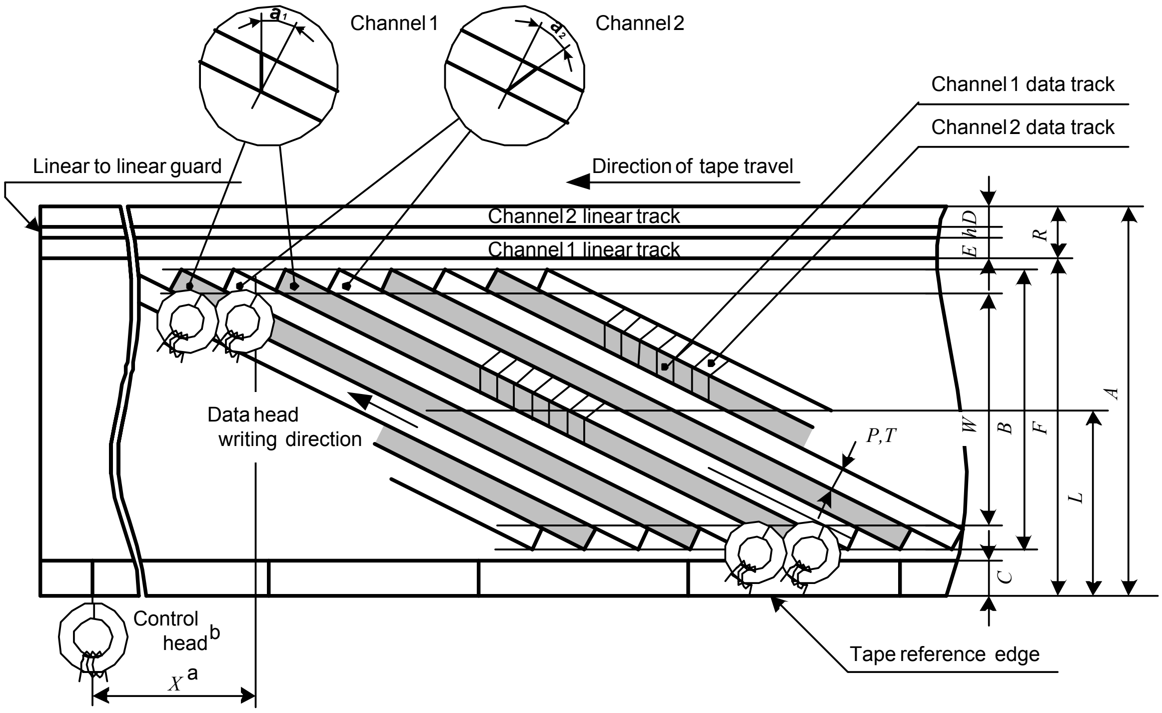
Remarks

1) When number of tracks is 119,88 tracks/s, relative writing speed and data track angle (tape running) are given in parentheses.

2) Where tolerances are not given, the quoted values are nominal.

3) Tests and measurements shall be carried out under the following condition. Temperature: (20 ± 2) °C, relative humidity: (65 ± 5) %. However, provided no discrepancy in judgment is caused, these may also be done under the following conditions. Temperature: 5 °C to 35 °C, relative humidity: 30 % to 80 %.

4) The items in bold text are basic values and the items in normal text are calculated values.



a X shall be measured from the end of the 180° scan of channel 2 to the recorded control signal on the tape.

The reference point on the control track is the transition point of the polarity of remanent magnetization on the tape, from south poles representing the direction of tape travel to its opposite.

b The control signal shall be recorded on the control track with sufficient current for saturation and the rise time of recording current shall be less than 200 µs.

A positive going edge of the recorded control signal shall be in coincidence with the start of channel 1 track scanning.

The control signal shall be recorded so that the rotating drum side of the control head poles would be north polarized when the pulse signal is positive.

Figure 16 – Magnetic tape position (HS mode)

5.2 MPEG2 recording format

5.2.1 General

When the application ID is set to 00000, recording data complies with MPEG2 recording mode format. In MPEG2 recording mode format, format ID data are defined in Tables 18 to 25.

Unless otherwise specified, MSB shall be written from the left and recorded first.

In this document, definition of k is as shown in Table 17.

Table 17 – Definition of k

Mode	k
STD	1
LS2	2
LS3	3
LS5	5
LS7	7
HS	—

1) ECC block size

Table 18 – ECC block size

0	112 sync blocks per 1 ECC block

2) ECC block number per track

Table 19 – ECC block number per track

00	3 ECC blocks per track

3) Program mode

Table 20 – Program mode

000	1 program per track (1 editable area per track)

4) Scanner rotation speed

Table 21 – Scanner rotation speed

00	1 800 rpm

5) 1,001 flag

Table 22 – 1,001 flag

0	Scanner rotation speed × 1
1	Scanner rotation speed × 1/1,001

6) Outer interleave

Table 23 – Outer interleave

000	6 tracks

7) Recording mode

Table 24 – Recording mode

0000	STD mode
0001	HS mode
0010	LS2 mode
0011	LS3 mode
0101	LS5 mode
0111	LS7 mode

8) Application ID

Table 25 – Application ID

00000	MPEG2

NOTE Tables 19 and 23 describe ECC structure for the main code area.

5.2.2 Number of tracks

5.2.2.1 Recording mode

5.2.2.1.1 NTSC

In case the function of D-VHS is added to the NTSC model of VHS, the number of tracks is as follows.

STD mode and LS mode: 60/k or 59,94/k tracks/s

HS mode: 120 or 119,88 tracks/s

5.2.2.1.2 PAL/SECAM

In case the function of D-VHS is added to the PAL/SECAM model of VHS, the number of tracks is as follows.

STD mode and LS mode: 60/k or 59,94/k tracks/s

HS mode: 120 or 119,88 tracks/s

5.2.2.2 Playback mode

In D-VHS playback mode, the number of tracks shall support both values (STD mode and LS mode: 60/k and 59,94/k tracks/s, HS mode: 120 and 119,88 tracks/s).

5.2.3 Recording data structure

5.2.3.1 Subcode data structure

5.2.3.1.1 General

In case of MPEG2 mode, subcode data area has 3 pack data as shown in Figure 17. One pack data is composed of 6 bytes. The first byte of each pack data is provided for pack header. Specification of the pack data is shown in 6.1.

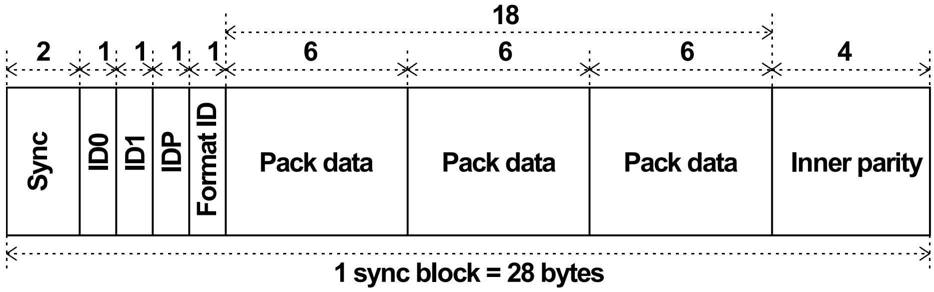


Figure 17 – Subcode sync block (MPEG2 mode)

5.2.3.1.2 ID

1) Absolute track number

The lower 2 bits of Absolute track number shall be increased as follows.

00 → 10 → 00 → 10 ···

5.2.3.2 Main data structure

5.2.3.2.1 General

In the case of MPEG2 mode, the first 2 bytes of the main data area in each sync block are provided for the main header and the next 1 byte of main data area is provided for data-AUX.

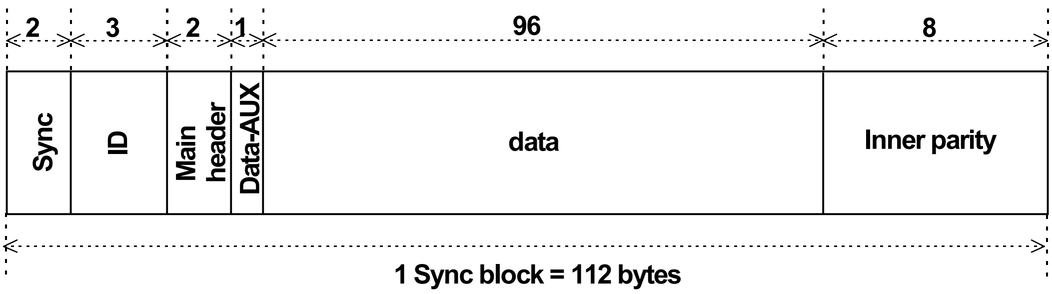


Figure 18 – Main data sync block (MPEG2 mode)

5.2.3.2.2 ID

1) Sequence number

The sequence number is counted up when track pair number returns to zero and sequence number repeats from 0000 to 1111.

2) Track pair number

The track pair number is counted up every track pair and repeats from 000 to 010 in MPEG2 recording mode. The first track of a track pair having the same track pair number is channel 1 and the second track is channel 2.

The track pair number also represents 6 track (3 track pair) interleave sequence.

5.2.3.2.3 Main header

The main header consists of the format information area and the sync block information area.

According to the sync block number, the format information area and the sync block information area are defined as shown in Table 26.

Table 26 – Configuration of main header

SB#	1st byte								2nd byte							
	MSB							LSB	MSB							LSB
	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0	bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
	Format info.				Sync block info.											
$26n$	Format ID				DT		SBC		Data detail							
$26n+1$					DT		SBC		Data detail							
$2(6n+1)$					DT		SBC		Data detail							
$2(6n+1)+1$					DT		SBC		Data detail							
$2(6n+2)$					DT		SBC		Data detail							
$2(6n+2)+1$					DT		SBC		Data detail							
$2(6n+3)$					DT		SBC		Data detail							
$2(6n+3)+1$					DT		SBC		Data detail							
$2(6n+4)$	Application detail				DT		SBC		Data detail							
$2(6n+4)+1$	App. detail extension				DT		SBC		Data detail							
$2(6n+5)$	$\overline{\text{CGMS}}$		Reserved		DT		SBC		Data detail							
$2(6n+5)+1$	Reserved				DT		SBC		Data detail							

Reserved bits shall be set to “0”.

Format ID is defined as shown in Table 27.

Table 27 – Format ID

SB#	$m = 0$				$m = 1$			
	bit 7	bit 6	bit 5	bit 4	bit 7	bit 6	bit 5	bit 4
$26n + m$	ECC block size	ECC block number per track		Program mode			Scanner rotation speed	
$2(6n+1)+m$	1,001 flag	Outer interleave			Recording mode			
$2(6n+2)+m$	Reserved							
$2(6n+3)+m$	Application ID					Reserved		
Reserved bits shall be set to “0”.								

Application ID is defined as shown in Table 28.

Table 28 – Application ID

0xxxx	Consumer use
1xxxx	Professional or special use

The D-VHS recorders that are designed to record or playback the signal having the Application ID value of 1xxxx shall not playback the recorded signal having the Application ID value of 0xxxx.

1) ECC block size **See 5.2.1, 1) ECC block size.**

2) ECC block number per track **See 5.2.1, 2) ECC block number per track.**

3) Program mode **See 5.2.1, 3) Program mode.**

4) Scanner rotation speed **See 5.2.1, 4) Scanner rotation speed.**

5) 1,001 flag **See 5.2.1, 5) 1,001 flag.**

6) Outer interleave **See 5.2.1, 6) Outer interleave.**

7) Recording mode **See 5.2.1, 7) Recording mode.**

8) Application ID **See 5.2.1, 8) Application ID.**

9) Application detail **See Table 29.**

Table 29 – Application detail

0000	MPEG2 transport stream
0001	CTP stream
1xxx	Encrypted MPEG2 transport stream

Other codes are reserved.

The D-VHS recorders that are not designed to process the recorded data having the aplication detail value of 1xxx shall not output the playback signal of such data.

When the application detail value is 1xxx, the three bits (xxx) represent the lower three bits of the region code. (Region code A)

10) Application detail extension

When the application detail value is 0000, the four bit application detail extension represents the Time compression ratio defined in Table 30.

The detail specification for the time compression function is to be determined.

Table 30 – Time compression ratio

Application detail Extension	Time compression ratio	lt [1]	Application detail Extension	Time compression ratio	lt [1]
0000	1: 1	1	0101	1: 5	5
0010	1: 2	2	0110	1: 6	6
0011	1: 3	3	0111	1: 7	7
0100	1: 4	4	Other codes are reserved		

[1] "lt" is the coefficient of Time compression ratio.

When the application detail value is 1xxx, the application detail extension represents the upper four bits of the region code (region code B).

The region code structure is defined as shown in Table 31.

Region Management Flag (RMF) N is the flag to indicate the permission of playback on players assigned to region number N.

The region management flag is defined as shown in Table 31.

0: may be played back in the corresponding region.

1: shall not be played back in the corresponding region.

Table 31 – Region management flag

Application detail				Application detail extension			
bit 7	bit 6	bit 5	bit 4	bit 7	bit 6	bit 5	bit 4
1	Region code A			Region code B			
	RMF 8	RMF 6	RMF 5	RMF 4	RMF 3	RMF 2	RMF 1

11) CGMS

See Table 32.

Table 32 – CGMS

00	Copy permitted
01	Reserved
10	One generation of copy permitted
11	Copy restricted

Two bits of CGMS shall be recorded as which is the inverse of the two bits.

CGMS information shall be the same within a track. In case of multi-program recording, the priority of CGMS information shall be 11,10,00. When input CGMS information is different among programs, the CGMS information of the highest priority shall be recorded.

12) Data Type (DT)

Data type indicates the contents of sync block data.

See Table 33.

Table 33 – Data type

Data type	Contents
00	Normal play data
01	Dummy data
10	Trick play data (for Track Select system)
11	Reserved

13) Sync Block Count (SBC)

Sync block count indicates the identification data to distinguish the contents of the main data (except main header and Data-AUX).

Definition of the sync block count varies according to the application detail and data type.

14) Data detail

The contents of data detail depend on the contents of data type.

See Table 34.

Table 34 – Data detail

Data type		Contents of data detail
Normal play data	(00)	Reserved
Dummy data	(01)	Reserved
Trick play data (for Track Select System)	(10)	See 5.2.6.2.2
Reserved	(11)	Reserved
Reserved bits shall be set to “0”.		

5.2.3.2.4 Data-AUX

Data-AUX consists of pack data.

One pack data consists of 6 data bytes from 6 sync blocks. The first data byte of pack data is provided for the pack header.

See Table 35.

Table 35 – Configuration of pack

SB#	Contents
$6n$	PC0 (Pack header)
$6n + 1$	PC1
$6n + 2$	PC2
$6n + 3$	PC3
$6n + 4$	PC4
$6n + 5$	PC5

5.2.4 Trick play data distribution

A trick play block consists of 48 consecutive tracks starting with the first track of sequence number 0 or 8. Trick play data are distributed in a macro unit of the trick play data block. A macro unit consists of several macro areas which consist of 2 or 3 basic areas.

5.2.4.1 Basic areas

5.2.4.1.1 STD mode and LS mode

There are 6 types of basic trick play areas Bi (i = 1..6). For i = 2, this is subdivided in types 2a and 2b.

1) For + 4k times normal speed:

The basic area for +4k times trick play speed consists of 56 sync blocks.

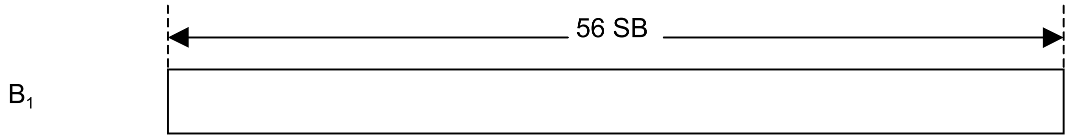


Figure 19 – + 4*k* times normal speed (STD and LS mode)

2) For –4*k* and + 6*k* times normal speed:

The basic area for –4*k* and +6*k* times trick play speed consists of 37 or 38 sync blocks.

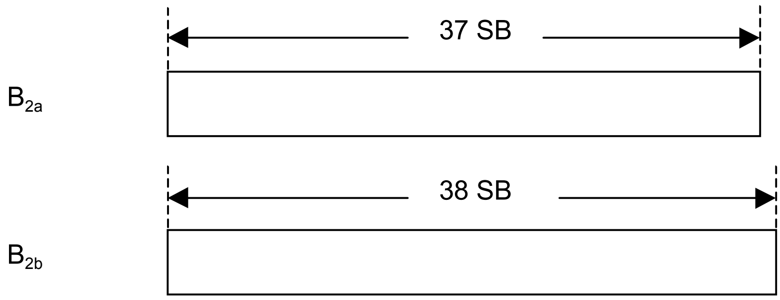
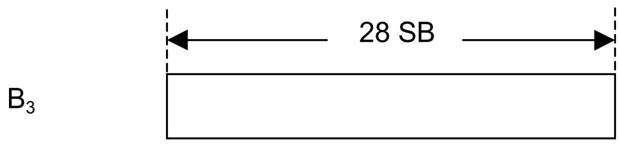


Figure 20 – –4*k* and + 6*k* times normal speed (STD and LS mode)

3) For –6*k* times normal speed:



The basic area for –6*k* times trick play speed consists of 28 sync blocks.

Figure 21 – –6*k* times normal speed (STD and LS mode)

4) For +12*k* times normal speed:

The basic area for +12*k* times trick play speed consists of 16 sync blocks.

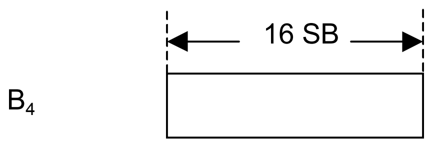


Figure 22 – +12*k* times normal speed (STD and LS mode)

5) For –12*k* times normal speed:

The basic area for –12*k* times trick play speed consists of 14 sync blocks.

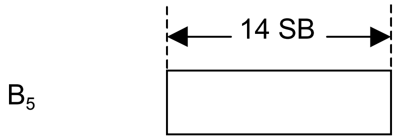


Figure 23 – –12k times normal speed (STD and LS mode)

6) For +24k and –24k times normal speed:

The basic area for +24k and –24k times trick play speed consists of 8 sync blocks.

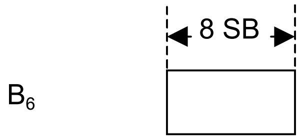


Figure 24 – –12k times normal speed (STD and LS mode)

5.2.4.1.2 HS mode

There are 5 types of basic trick play areas Bi (i = 1..5).

1) For + 3 times normal speed:

The basic area for +3 times trick play speed consists of 56 sync blocks.

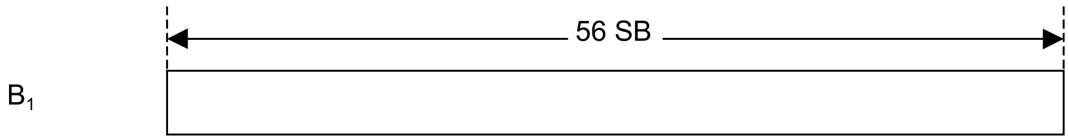


Figure 25 – + 3 times normal speed (HS mode)

2) For –3 times normal speed:

The basic area for –3 times trick play speed consists of 28 sync blocks.

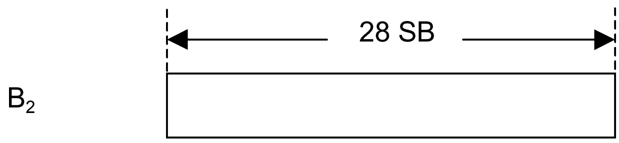


Figure 26 – – 3 times normal speed (HS mode)

3) For + 6 times normal speed:

The basic area for +6 times trick play speed consists of 16 sync blocks.

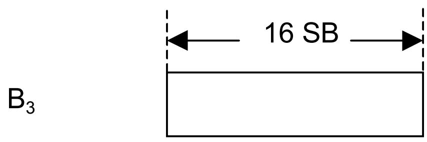


Figure 27 – + 6 times normal speed (HS mode)

4) For –6 times normal speed:

The basic area for –6 times trick play speed consists of 14 sync blocks.

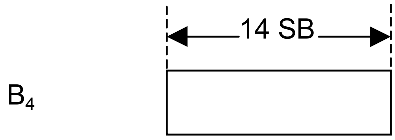


Figure 28 – – 6 times normal speed (HS mode)

5) For +12 and –12 times normal speed:

The basic area for +12 and -12 times trick play speed consists of 8 sync blocks.

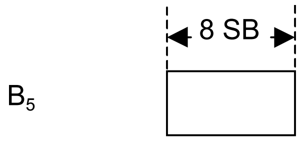


Figure 29 – +12 and –12 times normal speed (HS mode)

5.2.4.2 Basic units

5.2.4.2.1 STD mode and LS mode

A basic unit consists of a sequence of basic areas, and corresponds to one revolution of the drum at replay, starting with channel 1. See Figure 30.

The basic block corresponds to the shift of the tape in tracks during this one revolution and is therefore equal to 2 x (p/k) tracks in which p is the ratio of the absolute trick play speed to the normal play speed (p = 4k, 6k, 12k or 24k). The factor p is given by p = abs(n) in which n is the trick play speed factor.

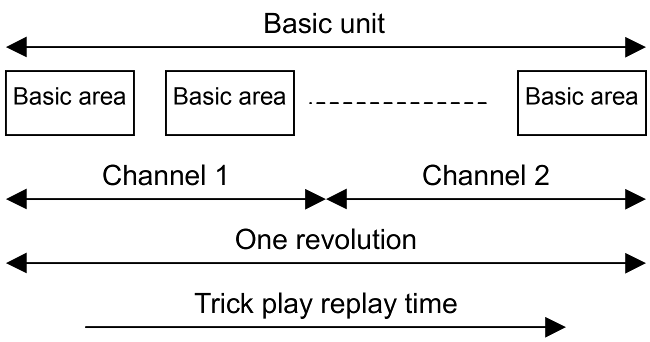


Figure 30 – Basic units (STD and LS mode)

5.2.4.2.2 HS mode

A basic unit consists of a sequence of basic areas, and corresponds to one revolution of the drum at replay. See Figure 31.

The basic block corresponds to the shift of the tape in tracks during this one revolution and is therefore equal to 4 $\times p$ tracks in which p is the ratio of the absolute trick play speed to the normal play speed (p = 3, 6 or 12). The factor p is given by p = abs(n) in which n is the trick play speed factor.

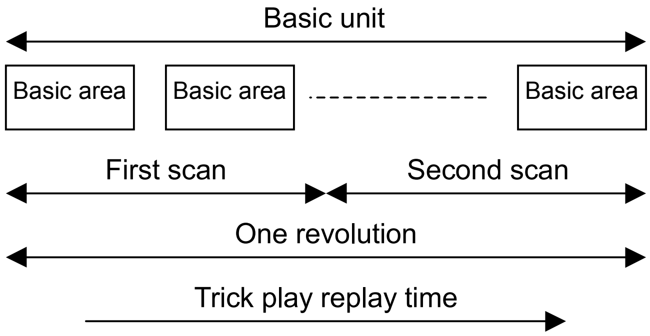
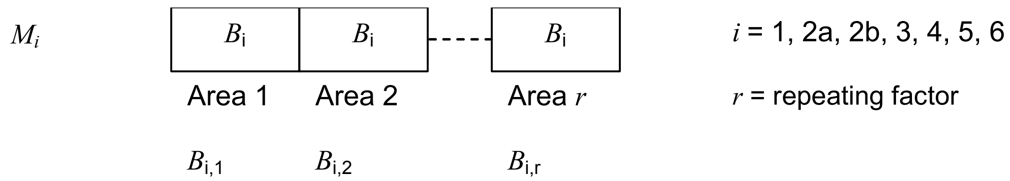


Figure 31 – Basic units (HS mode)

5.2.4.3 Macro areas

5.2.4.3.1 STD mode and LS mode

Basic areas described above are repeated and concatenated (back to back) to macro areas. Contents of each basic area in a macro area shall satisfy following rules. (This rule is applied except for ID, ID parity, 4 MSB's of format ID, Data-AUX and Inner parity area.) In the case of $-4k$ and $+6k$ times normal speed trick play, the basic area of type 2a can be extended to a length of 38 sync blocks by the addition of one normal play sync block before constructing the macro area. This normal play sync block can be in any position within the basic area. See Figure 32.



M_i = Macro area consisting of the concatenation of r basic areas B_i

$B_{i,j}$ = Area j of type B_i in the macro area M_i (j = 1..r) $\quad B_i = B_{i,j-1} \quad j = 2..r$

Figure 32 – Macro areas (STD and LS mode)

The repeating factor shall be as shown in Table 36.

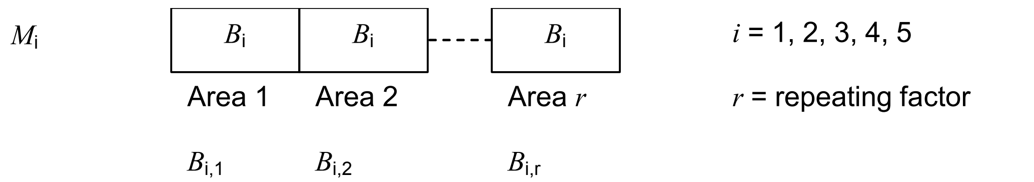
For $+4k$ or $-4k$, the repeating factor should not change within one program. It is strongly recommended to use repeating factor of 3 for +/-$4k$ trick play speed. A repeating factor of 2 can be used only if a repeating factor of 3 is not applicable.

Table 36 – Repeating factor (STD and LS mode)

Trick play speed	Repeating factor
$+4k/-4k$	2, 3
$+6k/-6k$	3
$+12k/-12k$	3
$+24k/-24k$	3

5.2.4.3.2 HS mode

The basic areas described above are repeated and concatenated (back to back) to macro areas. The contents of each basic area in a macro area shall satisfy the following rules. (This rule is applied except for ID, ID parity, 4 MSB's of format ID, Data-AUX and inner parity area.) See Figure 33.



M_i = Macro area consisting of the concatenation of r basic areas B_i

$B_{i,j}$ = Area j of type B_i in the macro area M_i (j = 1..r) $B_i = B_{i,j-1}$ j = 2..r

Figure 33 – Macro areas (HS mode)

The repeating factor shall be as shown in Table 37.

Table 37 – Repeating factor (HS mode)

Trick play speed	Repeating factor
+3/-3	3
+6/-6	3
+12/-12	3

5.2.4.4 Macro units

5.2.4.4.1 STD mode and LS mode

A macro unit consists of a sequence of macro areas, and corresponds to one revolution of the drum at replay, starting with channel 1. See Figure 34.

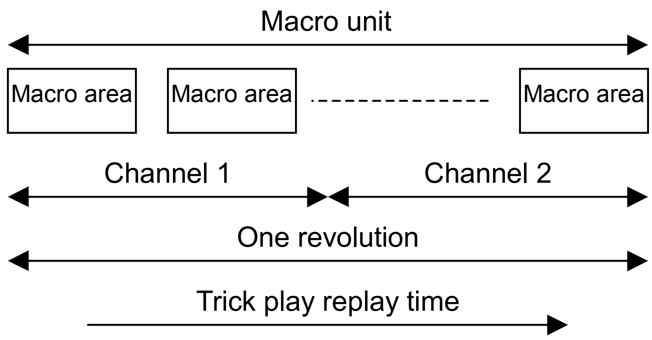


Figure 34 – Macro units (STD and LS mode)

5.2.4.4.2 HS mode

A macro unit consists of a sequence of macro areas, and corresponds to one revolution of the drum at replay. See Figure 35.

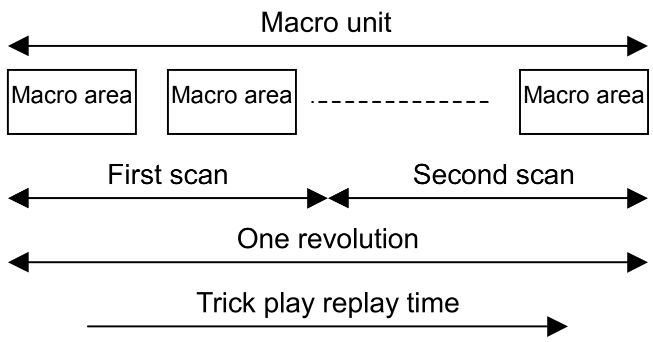


Figure 35 – Macro units (HS mode)

5.2.4.5 Macro area distribution in a trick play block

5.2.4.5.1 STD mode and LS mode

The macro block corresponds to the shift of the tape in tracks during one revolution as described above and is therefore equal to $2 \times p/k$ tracks in which p is the ratio of the absolute trick play speed to the normal play speed (p = 4k, 6k, 12k, or 24k). The factor p is given by p = abs(n) in which n is the trick play speed factor. One macro unit is recorded in one macro block.

The track numbering Tn within a macro block (2p/k tracks) runs from 0 to (2p/k)-1. The relation between track numbering Tn within a macro block for a certain speed n and track numbering T within the trick play block is given by the following formulas.

The track numbering within a trick play block (48 tracks) is defined as follows:

$$T = H + 2 \times TP\# + 6 \times (Seq\# \bmod 8)$$

In which H = 0 for the first track and H = 1 for the second track of a track pair with the same $TP\#$ (H = $Channel\#$ – 1). $TP\#$ and $Seq\#$ are the track pair number and sequence number in byte ID0 of the sync block ID.

The track numbering for a macro block for a certain speed n is defined as follows:

$$Tn = (48 + f \times T) \bmod (2p/k) \quad \text{with } f = \text{sign}(n)$$

For forward trick play speeds, this is equal to:

$$Tn = T \bmod (2p/k)$$

For reverse trick play speeds, this is equal to:

$$Tn = (48 - T) \bmod (2p/k)$$

The sequence of macro areas is placed within the macro unit that corresponds to the macro block given by Tn. The number of macro areas k_i in one macro unit and therefore one macro block depends on the speed and is shown in Table 38.

Table 38 – Macro area (STD and LS mode)

i	speed	k_i
1	$+4k$	2
2	$-4k$, $+6k$	3
3	$-6k$	4
4	$+12k$	7
5	$-12k$	8
6	$+24k$, $-24k$	14

Macro areas within a macro block are read at trick replay in the order of increasing Tn.

Corresponding subcode is read from tracks with the track number Ts in a trick play block.

Ts is given as follows.

Forward and reverse search: $Ts = (i \times p/k) \bmod 48$

5.2.4.5.2 HS mode

The macro block corresponds to the shift of the tape in tracks during one revolution as described above and is therefore equal to $4 \times p$ tracks in which p is the ratio of the absolute trick play speed to the normal play speed (p = 3, 6 or 12). The factor p is given by $p = \text{abs}(n)$ in which n is the trick play speed factor. One macro unit is recorded in one macro block.

The track numbering Tn within a macro block ($4p$ tracks) runs from 0 to $4p$-1. The relation between track numbering Tn within a macro block for a certain speed n and track numbering T within the trick play block is given by the following formulas.

The track numbering within a trick play block (48 tracks) is defined as follows:

$$T = H + 2 \times TP\# + 6 \times (Seq\# \bmod 8)$$

In which H = 0 for the first track and H = 1 for the second track of a track pair with the same $TP\#$ ($H = Channel\# - 1$). $TP\#$ and $Seq\#$ are the track pair number and sequence number in byte ID0 of the sync block ID.

Track numbering for a macro block for a certain speed n is defined as follows:

$$Tn = (47 + f + f \times T) \bmod (4p) \quad \text{with } f = \text{sign}(n)$$

For forward trick play speeds, this is equal to:

$$Tn = T \bmod (4p)$$

For reverse trick play speeds, this is equal to:

$$Tn = (46 - T) \bmod (4p)$$

The sequence of macro areas is placed within the macro unit that corresponds to the macro block given by Tn. The number of macro areas k_i in one macro unit and therefore one macro block depends on the speed and is shown in Table 39.

Table 39 – Macro area (HS mode)

i	speed	k_i
1	+3	2
2	-3	4
3	+6	7
4	-6	8
5	+12, -12	14

At forward trick play, the corresponding macro areas are read by the channel 1 heads.

At reverse trick play, the corresponding macro areas are read by channel 2 heads. Macro areas within a macro block are read at trick replay in the order of increasing Tn.

Corresponding subcode is read from tracks with the track number Ts in a trick play block. Ts is given as follows.

HS forward search: Ts = (i*2p) mod 48

HS reverse search: Ts = (i*2p-1) mod 48 i is an integer number

5.2.4.6 Ideal points of each macro area

5.2.4.6.1 STD mode and LS mode

The ideal recording points for trick play data (center of the macro area) in a trick play block are given in Table 42, Table 43, Table 44, Table 45 and Table 46. The ideal recording point is located at the start of the sync blocks indicated in the table. Numbers correspond to the $SB\#$ in the SB-ID. The distance in sync blocks between the center of the macro area and the ideal recording point shall not exceed the value given in the Table 40.

Table 40 – Ideal points (STD and LS mode)

Speed	Maximum allowance of the shift from the ideal position
+4k	6
-4k	2
+6k	6
-6k	4
+12k	4
-12k	3
+24k	1
-24k	1

5.2.4.6.2 HS mode

The ideal recording points for trick play data (center of the macro area) in a trick play block are given in the Table 47. The ideal recording point is located at the start of the sync blocks indicated in the table. Numbers correspond to the SB# in the SB-ID. The distance in sync blocks between the center of the macro area and the ideal recording point shall not exceed the value given in the Table 41.

Table 41 – Ideal points (HS mode)

Speed	Maximum allowance of the shift from the ideal position
+3	1
-3	0
+6	0
-6	0
+12	0
-12	0

Trick play data must satisfy following placement rules.

If within a macro block, more ideal points are available than there are macro areas in a macro block, the ideal points can be chosen freely.

Macro areas must always be placed in one piece. (No cut-off allowed due to overlap of macro areas or beginning or end of main data area.)

The positioning of the macro areas within the macro unit with respect to the ideal points shall not be changed during the recording of one program.

The combination of speeds shall not be changed during the recording of one program.

Table 42 – Ideal points of macro area (STD mode: k = 1)

Speed	-24		+24		-12		+12		-6		+6		-4		+4	
T	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*
0			0				0				0				0	
1	47		1		23		1		11	245	1		7	205	1	
2	46		2	22	22		2	56	10		2	133	6		2	228
3	45	290	3		21	237	3		9	144	3		5	62	3	
4	44		4	53	20		4	120	8		4		4		4	
5	43	262	5		19	183	5		7	42	5		3		5	110
6	42		6	84	18		6	185	6		6		2	133	6	
7	41	233	7		17	128	7		5		7	62	1		7	
8	40		8	115	16		8	250	4	194	8		0		0	
9	39	205	9		15	73	9		3		9	205	7	205	1	
10	38		10	146	14		10		2	93	10		6		2	228
11	37	176	11		13	18	11		1		11		5	62	3	
12	36		12	177	12		12		0		0		4		4	
13	35	148	13		11		13	23	11	245	1		3		5	110
14	34		14	208	10	265	14		10		2	133	2	133	6	
15	33	119	15		9		15	88	9	144	3		1		7	
16	32		16	239	8	210	16		8		4		0		0	
17	31	91	17		7		17	153	7	42	5		7	205	1	
18	30		18	270	6	155	18		6		6		6		2	228
19	29	62	19		5		19	218	5		7	62	5	62	3	
20	28		20		4	101	20		4	194	8		4		4	
21	27	34	21		3		21	282	3		9	205	3		5	110
22	26		22		2	46	22		2	93	10		2	133	6	
23	25		23		1		23		1		11		1		7	
24	24		24		0		0		0		0		0		0	
25	23		25		23		1		11	245	1		7	205	1	
26	22		26		22		2	56	10		2	133	6		2	228
27	21		27	37	21	237	3		9	144	3		5	62	3	
28	20	276	28		20		4	120	8		4		4		4	
29	19		29	68	19	183	5		7	42	5		3		5	110
30	18	247	30		18		6	185	6		6		2	133	6	
31	17		31	99	17	128	7		5		7	62	1		7	
32	16	219	32		16		8	250	4	194	8		0		0	
33	15		33	130	15	73	9		3		9	205	7	205	1	
34	14	190	34		14		10		2	93	10		6		2	228
35	13		35	161	13	18	11		1		11		5	62	3	
36	12	162	36		12		12		0		0		4		4	
37	11		37	192	11		13	23	11	245	1		3		5	110
38	10	133	38		10	265	14		10		2	133	2	133	6	
39	9		39	223	9		15	88	9	144	3		1		7	
40	8	105	40		8	210	16		8		4		0		0	
41	7		41	254	7		17	153	7	42	5		7	205	1	
42	6	76	42		6	155	18		6		6		6		2	228
43	5		43	285	5		19	218	5		7	62	5	62	3	
44	4	48	44		4	101	20		4	194	8		4		4	
45	3		45		3		21	282	3		9	205	3		5	110
46	2	19	46		2	46	22		2	93	10		2	133	6	
47	1		47		1		23		1		11		1		7	
0	0				0				0				0			

Boundaries of ECC blocks are indicated by thick lines. The gray area indicates the first scan.

Table 43 – Ideal points of macro area (LS2 mode: k = 2)

Speed	-24k		+24k		-12k		+12k		-6k		+6k		-4k		+4k	
T	Tn	SB	Tn	SB	Tn	SB	Tn	SB	Tn	SB	Tn	SB	Tn	SB	Tn	SB
0			0				0				0				0	
1	47		1		23		1		11	265	1		7	228	1	
2	46		2	21	22		2	53	10		2	120	6		2	194
3	45		3		21	247	3		9	155	3		5	70	3	
4	44		4	52	20		4	115	8		4	250	4		4	
5	43	267	5		19	190	5		7	46	5		3		5	93
6	42		6	82	18		6	177	6		6		2	149	6	
7	41	238	7		17	133	7		5		7	56	1		7	
8	40		8	112	16		8	239	4	210	8		0		0	
9	39	209	9		15	76	9		3		9	185	7	228	1	
10	38		10	142	14		10		2	101	10		6		2	194
11	37	180	11		13	19	11		1		11		5	70	3	
12	36		12	173	12		12		0		0		4		4	
13	35	151	13		11		13	22	11	265	1		3		5	93
14	34		14	203	10	276	14		10		2	120	2	149	6	
15	33	122	15		9		15	84	9	155	3		1		7	
16	32		16	233	8	219	16		8		4	250	0		0	
17	31	93	17		7		17	146	7	46	5		7	228	1	
18	30		18	264	6	162	18		6		6		6		2	194
19	29	64	19		5		19	208	5		7	56	5	70	3	
20	28		20	294	4	105	20		4	210	8		4		4	
21	27	35	21		3		21	270	3		9	185	3		5	93
22	26		22		2	48	22		2	101	10		2	149	6	
23	25		23		1		23		1		11		1		7	
24	24		24		0		0		0		0		0		0	
25	23		25		23		1		11	265	1		7	228	1	
26	22		26		22		2	53	10		2	120	6		2	194
27	21		27	36	21	247	3		9	155	3		5	70	3	
28	20	282	28		20		4	115	8		4	250	4		4	
29	19		29	67	19	190	5		7	46	5		3		5	93
30	18	253	30		18		6	177	6		6		2	149	6	
31	17		31	97	17	133	7		5		7	56	1		7	
32	16	223	32		16		8	239	4	210	8		0		0	
33	15		33	127	15	76	9		3		9	185	7	228	1	
34	14	194	34		14		10		2	101	10		6		2	194
35	13		35	158	13	19	11		1		11		5	70	3	
36	12	165	36		12		12		0		0		4		4	
37	11		37	188	11		13	22	11	265	1		3		5	93
38	10	136	38		10	276	14		10		2	120	2	149	6	
39	9		39	218	9		15	84	9	155	3		1		7	
40	8	107	40		8	219	16		8		4	250	0		0	
41	7		41	249	7		17	146	7	46	5		7	228	1	
42	6	78	42		6	162	18		6		6		6		2	194
43	5		43	279	5		19	208	5		7	56	5	70	3	
44	4	49	44		4	105	20		4	210	8		4		4	
45	3		45		3		21	270	3		9	185	3		5	93
46	2	20	46		2	48	22		2	101	10		2	149	6	
47	1		47		1		23		1		11		1		7	
0	0				0				0				0			

Boundaries of ECC blocks are indicated by thick lines. The gray area indicates the first scan.

Table 44 – Ideal points of macro area (LS3 mode: k = 3)

Speed	-24k		+24k		-12k		+12k		-6k		+6k		-4k		+4k	
T	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*
0			0				0				0				0	
1	47		1		23		1		11		1		7	237	1	
2	46		2	21	22		2	52	10		2	117	6		2	185
3	45		3		21	251	3		9	160	3		5	73	3	
4	44		4	51	20		4	113	8		4	242	4		4	
5	43	269	5		19	193	5		7	47	5		3		5	88
6	42		6	81	18		6	174	6		6		2	155	6	
7	41	240	7		17	135	7		5		7	54	1		7	
8	40		8	111	16		8	235	4	216	8		0		0	
9	39	210	9		15	78	9		3		9	179	7	237	1	
10	38		10	141	14		10		2	103	10		6		2	185
11	37	181	11		13	20	11		1		11		5	73	3	
12	36		12	172	12		12		0		0		4		4	
13	35	152	13		11		13	22	11		1		3		5	88
14	34		14	202	10	280	14		10		2	117	2	155	6	
15	33	123	15		9		15	83	9	160	3		1		7	
16	32		16	232	8	222	16		8		4	242	0		0	
17	31	93	17		7		17	144	7	47	5		7	237	1	
18	30		18	262	6	164	18		6		6		6		2	185
19	29	64	19		5		19	205	5		7	54	5	73	3	
20	28		20	292	4	106	20		4	216	8		4		4	
21	27	35	21		3		21	266	3		9	179	3		5	88
22	26		22		2	49	22		2	103	10		2	155	6	
23	25		23		1		23		1		11		1		7	
24	24		24		0		0		0		0		0		0	
25	23		25		23		1		11		1		7	237	1	
26	22		26		22		2	52	10		2	117	6		2	185
27	21		27	36	21	251	3		9	160	3		5	73	3	
28	20	284	28		20		4	113	8		4	242	4		4	
29	19		29	66	19	193	5		7	47	5		3		5	88
30	18	254	30		18		6	174	6		6		2	155	6	
31	17		31	96	17	135	7		5		7	54	1		7	
32	16	225	32		16		8	235	4	216	8		0		0	
33	15		33	126	15	78	9		3		9	179	7	237	1	
34	14	196	34		14		10		2	103	10		6		2	185
35	13		35	156	13	20	11		1		11		5	73	3	
36	12	167	36		12		12		0		0		4		4	
37	11		37	187	11		13	22	11		1		3		5	88
38	10	137	38		10	280	14		10		2	117	2	155	6	
39	9		39	217	9		15	83	9	160	3		1		7	
40	8	108	40		8	222	16		8		4	242	0		0	
41	7		41	247	7		17	144	7	47	5		7	237	1	
42	6	79	42		6	164	18		6		6		6		2	185
43	5		43	277	5		19	205	5		7	54	5	73	3	
44	4	50	44		4	106	20		4	216	8		4		4	
45	3		45		3		21	266	3		9	179	3		5	88
46	2	20	46		2	49	22		2	103	10		2	155	6	
47	1		47		1		23		1		11		1		7	
0	0				0				0				0			

Boundaries of ECC blocks are indicated by thick lines. The gray area indicates the first scan.

Table 45 – Ideal points of macro area (LS5 mode: k = 5)

Speed	-24k		+24k		-12k		+12k		-6k		+6k		-4k		+4k	
T	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*
0			0				0				0				0	
1	47		1		23		1		11		1		7	245	1	
2	46		2	21	22		2	51	10		2	114	6		2	178
3	45		3		21	254	3		9	163	3		5	76	3	
4	44		4	51	20		4	112	8		4	237	4		4	
5	43	271	5		19	195	5		7	48	5		3		5	85
6	42		6	81	18		6	172	6		6		2	161	6	
7	41	241	7		17	137	7		5		7	52	1		7	
8	40		8	111	16		8	232	4	221	8		0		0	
9	39	212	9		15	79	9		3		9	175	7	245	1	
10	38		10	141	14		10		2	106	10		6		2	178
11	37	182	11		13	20	11		1		11		5	76	3	
12	36		12	170	12		12		0		0		4		4	
13	35	153	13		11		13	21	11		1		3		5	85
14	34		14	200	10	283	14		10		2	114	2	161	6	
15	33	123	15		9		15	82	9	163	3		1		7	
16	32		16	230	8	224	16		8		4	237	0		0	
17	31	94	17		7		17	142	7	48	5		7	245	1	
18	30		18	260	6	166	18		6		6		6		2	178
19	29	65	19		5		19	202	5		7	52	5	76	3	
20	28		20	290	4	108	20		4	221	8		4		4	
21	27	35	21		3		21	263	3		9	175	3		5	85
22	26		22		2	49	22		2	106	10		2	161	6	
23	25		23		1		23		1		11		1		7	
24	24		24		0		0		0		0		0		0	
25	23		25		23		1		11		1		7	245	1	
26	22		26		22		2	51	10		2	114	6		2	178
27	21		27	36	21	254	3		9	163	3		5	76	3	
28	20	285	28		20		4	112	8		4	237	4		4	
29	19		29	66	19	195	5		7	48	5		3		5	85
30	18	256	30		18		6	172	6		6		2	161	6	
31	17		31	96	17	137	7		5		7	52	1		7	
32	16	226	32		16		8	232	4	221	8		0		0	
33	15		33	126	15	79	9		3		9	175	7	245	1	
34	14	197	34		14		10		2	106	10		6		2	178
35	13		35	156	13	20	11		1		11		5	76	3	
36	12	168	36		12		12		0		0		4		4	
37	11		37	185	11		13	21	11		1		3		5	85
38	10	138	38		10	283	14		10		2	114	2	161	6	
39	9		39	215	9		15	82	9	163	3		1		7	
40	8	109	40		8	224	16		8		4	237	0		0	
41	7		41	245	7		17	142	7	48	5		7	245	1	
42	6	79	42		6	166	18		6		6		6		2	178
43	5		43	275	5		19	202	5		7	52	5	76	3	
44	4	50	44		4	108	20		4	221	8		4		4	
45	3		45		3		21	263	3		9	175	3		5	85
46	2	20	46		2	49	22		2	106	10		2	161	6	
47	1		47		1		23		1		11		1		7	
0	0				0				0				0			

Boundaries of ECC blocks are indicated by thick lines. The gray area indicates the first scan.

Table 46 – Ideal points of macro area (LS7 mode: k = 7)

Speed	-24k		+24k		-12k		+12k		-6k		+6k		-4k		+4k	
T	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*
0			0				0				0				0	
1	47		1		23		1		11		1		7	249	1	
2	46		2	21	22		2	51	10		2	113	6		2	176
3	45		3		21	255	3		9	165	3		5	77	3	
4	44		4	51	20		4	111	8		4	234	4		4	
5	43	271	5		19	196	5		7	49	5		3		5	83
6	42		6	81	18		6	171	6		6		2	163	6	
7	41	242	7		17	138	7		5		7	52	1		7	
8	40		8	110	16		8	231	4	223	8		0		0	
9	39	212	9		15	79	9		3		9	173	7	249	1	
10	38		10	140	14		10		2	107	10		6		2	176
11	37	183	11		13	20	11		1		11		5	77	3	
12	36		12	170	12		12		0		0		4		4	
13	35	153	13		11		13	21	11		1		3		5	83
14	34		14	200	10	284	14		10		2	113	2	163	6	
15	33	124	15		9		15	81	9	165	3		1		7	
16	32		16	230	8	226	16		8		4	234	0		0	
17	31	94	17		7		17	141	7	49	5		7	249	1	
18	30		18	260	6	167	18		6		6		6		2	176
19	29	65	19		5		19	201	5		7	52	5	77	3	
20	28		20	289	4	108	20		4	223	8		4		4	
21	27	35	21		3		21	261	3		9	173	3		5	83
22	26		22		2	50	22		2	107	10		2	163	6	
23	25		23		1		23		1		11		1		7	
24	24		24		0		0		0		0		0		0	
25	23		25		23		1		11		1		7	249	1	
26	22		26		22		2	51	10		2	113	6		2	176
27	21		27	36	21	255	3		9	165	3		5	77	3	
28	20	286	28		20		4	111	8		4	234	4		4	
29	19		29	66	19	196	5		7	49	5		3		5	83
30	18	256	30		18		6	171	6		6		2	163	6	
31	17		31	95	17	138	7		5		7	52	1		7	
32	16	227	32		16		8	231	4	223	8		0		0	
33	15		33	125	15	79	9		3		9	173	7	249	1	
34	14	197	34		14		10		2	107	10		6		2	176
35	13		35	155	13	20	11		1		11		5	77	3	
36	12	168	36		12		12		0		0		4		4	
37	11		37	185	11		13	21	11		1		3		5	83
38	10	138	38		10	284	14		10		2	113	2	163	6	
39	9		39	215	9		15	81	9	165	3		1		7	
40	8	109	40		8	226	16		8		4	234	0		0	
41	7		41	245	7		17	141	7	49	5		7	249	1	
42	6	79	42		6	167	18		6		6		6		2	176
43	5		43	275	5		19	201	5		7	52	5	77	3	
44	4	50	44		4	108	20		4	223	8		4		4	
45	3		45		3		21	261	3		9	173	3		5	83
46	2	20	46		2	50	22		2	107	10		2	163	6	
47	1		47		1		23		1		11		1		7	
0	0				0				0				0			

Boundaries of ECC blocks are indicated by thick lines. The gray area indicates the first scan.

Table 47 – Ideal points of macro area (HS mode)

Speed	-12		+12		-6		+6		-3		+3	
T	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*	*Tn*	*SB*
47	47				23				11	258		
0	46		0		22		0		10		0	
1	45	292	1		21	245	1		9	169	1	
2	44		2	23	20		2	62	8		2	169
3	43	265	3		19	194	3		7	80	3	
4	42		4	56	18		4	133	6		4	
5	41	237	5		17	144	5		5	258	5	
6	40		6	88	16		6	205	4		6	
7	39	210	7		15	93	7		3	169	7	
8	38		8	120	14		8	276	2		8	169
9	37	183	9		13	42	9		1	80	9	
10	36		10	153	12		10		0		10	
11	35	155	11		11		11		11	258	11	
12	34		12	185	10		12		10		0	
13	33	128	13		9	245	13		9	169	1	
14	32		14	218	8		14	62	8		2	169
15	31	101	15		7	194	15		7	80	3	
16	30		16	250	6		16	133	6		4	
17	29	73	17		5	144	17		5	258	5	
18	28		18	282	4		18	205	4		6	
19	27	46	19		3	93	19		3	169	7	
20	26		20		2		20	276	2		8	169
21	25	18	21		1	42	21		1	80	9	
22	24		22		0		22		0		10	
23	23		23		23		23		11	258	11	
24	22		24		22		0		10		0	
25	21	292	25		21	245	1		9	169	1	
26	20		26	23	20		2	62	8		2	169
27	19	265	27		19	194	3		7	80	3	
28	18		28	56	18		4	133	6		4	
29	17	237	29		17	144	5		5	258	5	
30	16		30	88	16		6	205	4		6	
31	15	210	31		15	93	7		3	169	7	
32	14		32	120	14		8	276	2		8	169
33	13	183	33		13	42	9		1	80	9	
34	12		34	153	12		10		0		10	
35	11	155	35		11		11		11	258	11	
36	10		36	185	10		12		10		0	
37	9	128	37		9	245	13		9	169	1	
38	8		38	218	8		14	62	8		2	169
39	7	101	39		7	194	15		7	80	3	
40	6		40	250	6		16	133	6		4	
41	5	73	41		5	144	17		5	258	5	
42	4		42	282	4		18	205	4		6	
43	3	46	43		3	93	19		3	169	7	
44	2		44		2		20	276	2		8	169
45	1	18	45		1	42	21		1	80	9	
46	0		46		0		22		0		10	
47			47				23				11	

Boundaries of ECC blocks are indicated by thick lines. The gray area indicates the first scan.

5.2.5 Error correcting code (outer parity)

5.2.5.1 Error correcting code for the main data area

The outer error correcting code is applied only to the main code area except for sync and ID, ID parity and inner parity.

5.2.5.1.1 ECC structure

One ECC block consists of 102 data sync blocks and 10 outer parity sync blocks. See Figure 36.

The main code area has 336 main data sync blocks. There are 306 data sync blocks and 30 outer parity sync blocks. The combined main code areas of 6 tracks consist of 18 ECC blocks.

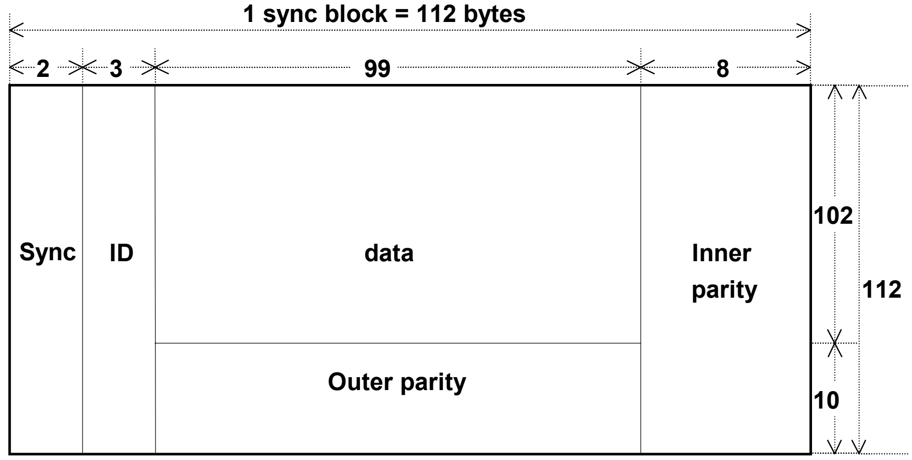


Figure 36 – ECC structure (main data area)

5.2.5.1.2 Data position definition

One ECC block is composed of 6 main code areas whose track pair numbers are 0 to 2 (6 tracks: the first track is channel 1). They are numbered from 0 to 5 along recording direction. See Figure 37.

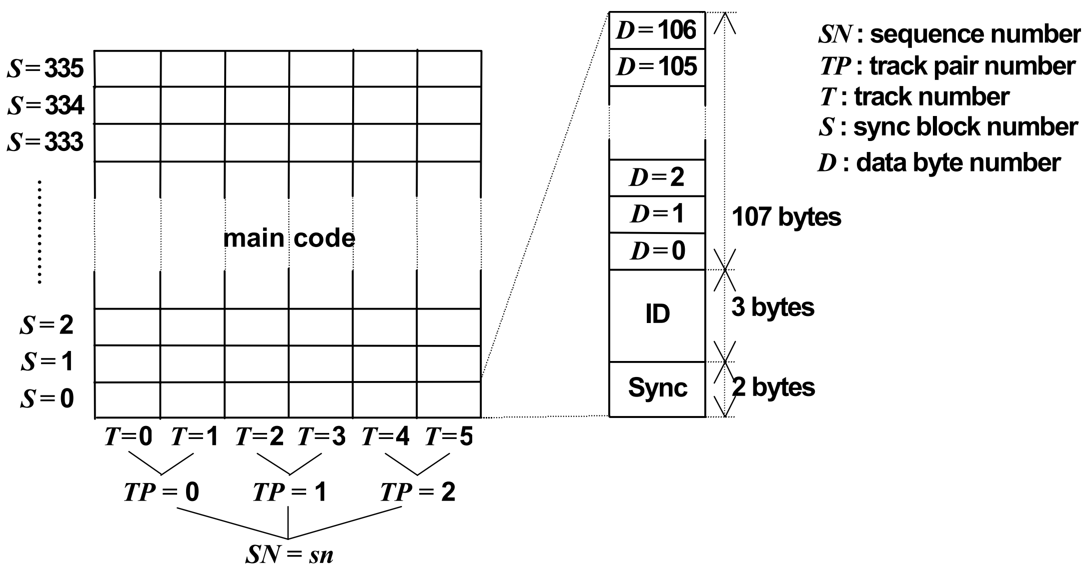


Figure 37 – Data position definition

5.2.5.1.3 ECC definition

The outer error correcting code for the main code area: (112,102,11) RS code over GF(2^8).

The calculation is defined on GF(2^8) driven by the following polynomial.

$$\mathrm{P}(X) = X^8 + X^4 + X^3 + X^2 + 1$$

A primitive element α is defined as follows.

$$\alpha = 00000010$$

The outer error correcting codes are interleaved among 6 tracks respectively.

One outer codeword is made of following symbols.

$$\mathrm{Codeword}(t,s,d) = \sum_{i=0}^{111} \mathrm{Data}_{T,S,D} \times X^{(111-i)}$$
$$(t = 0 \text{ to } 5,\ s = 0 \text{ to } 2,\ d = 0 \text{ to } 98)$$

where the variable i means the order of symbols in the codeword.

$$T = (t + 5 \times i) \bmod 6$$
$$S = s + 3 \times i$$
$$D = d$$

$\mathrm{Data}_{T,S,D}$ means a symbol defined by the track number: T, sync block number: S, and data byte number: D.

The last 10 symbols (i = 102 to 111) are parity.

This equation indicates the outer codeword whose first symbol is defined by track number t, sync block number s, and data byte number d.

The generator polynomial for outer error correcting code is as follows.

$$\mathrm{G}(X) = \prod_{i=0}^{9} (X + \alpha^i)$$

5.2.5.2 Error correcting code for trick play data

The outer error correcting code for the trick play data is applied only to MPEG-data area (packet header and MPEG2 transport stream). The use of an outer error correcting code at record and/or replay is optional. Use or no use of ECC for trick play data shall not be changed within one recording of a program.

5.2.5.2.1 ECC structure

One trick play ECC block consists of 102 data sync blocks and 10 outer parity sync blocks. When no outer error correcting code is used during the recording, the parity sync blocks are assigned to normal play data and marked with DT = 00 (normal play data) or DT = 01 (dummy data). The trick play sync block numbers St = 102 to 111 are non-existent in this case.

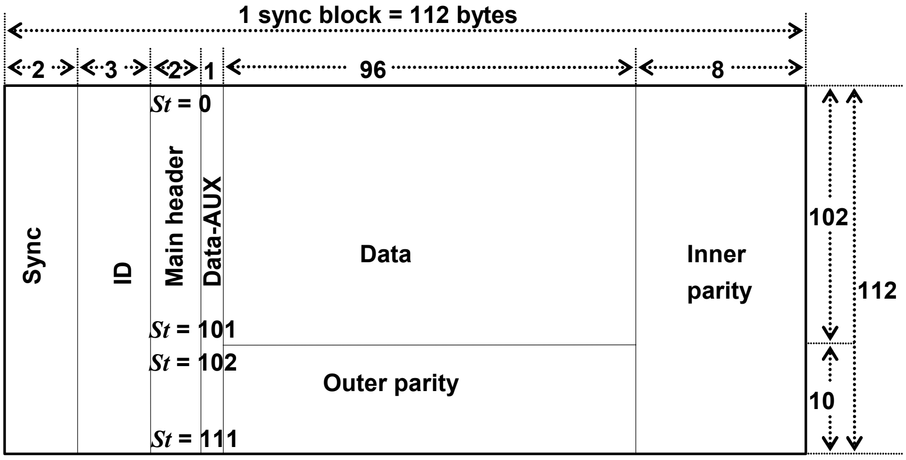


Figure 38 – ECC Structure (trick play)

5.2.5.2.2 Data position definition

The trick play ECC block is aligned within a macro block. A trick play macro block has one macro unit which is composed of 2 or 3 times repeated basic units. One trick play ECC block is composed of one basic unit and is divided into 102 data and 10 outer parity sync blocks. These sync blocks are numbered with special trick play sync block numbers that are stored in the second byte of the main header (data detail).

The data sync blocks are numbered 0 to 101. The outer parity sync blocks are numbered 102 to 111. Within one basic unit, sync blocks of an ECC block can be placed on the tape in any order. The outer ECC block shall be reconstructed according to the trick play sync block number.

5.2.5.2.3 ECC definition

The outer error correcting code for the trick play data: (112,102,11)RS code over GF(2^8).

The calculation is defined on GF(2^8) driven by the following polynomial.

$$\mathrm{P}(X) = X^8 + X^4 + X^3 + X^2 + 1$$

A primitive element α is defined as follows.

$$\alpha = 00000010$$

One outer codeword is made of following symbols.

$$\mathrm{Codeword}(d) = \sum_{i=0}^{111} \mathrm{Data}_{St,d} \times X^{(111-i)}$$

$$(St = i,\ d = 3 \text{ to } 98)$$

where: the variable i means the order of symbols in the codeword. $\mathrm{Data}_{St,d}$ means a symbol defined by the trick play sync block number: St, and data byte number d.

The last 10 symbols (i = 102 to 111) are parity.

This equation indicates the outer codeword defined by trick play sync block number St, and data byte number d.

The generator polynomial for outer error correcting code is as follows.

$$G(X) = \prod_{i=0}^{9} (X + \alpha^i)$$

5.2.6 Transport stream recording structure

5.2.6.1 MPEG2 TS recording for the normal play data

When the application detail in the main header (see 5.2.3.2.3) is set to 0000, the MPEG2 transport stream is recorded (MPEG2 transport stream recording).

MPEG2 transport packet (188 bytes) and associated packet header information (4 bytes) are recorded onto two sync blocks. The allocation of normal play data is shown in Figure 39. No dummy data is allowed to be put in between. Only trick play data should be allowed to be present between the former and latter half of an MPEG packet.

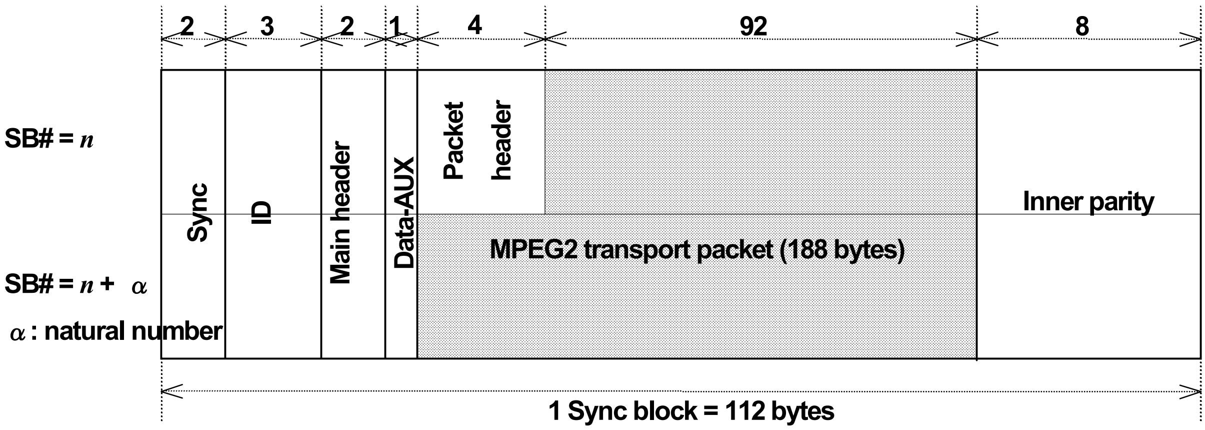


Figure 39 – MPEG2 TS recording (normal play)

5.2.6.1.1 Sync block count

When the application detail is set to 0000 (MPEG2 transport stream), the sync block count is defined as follows.

1) When the data type is set to “normal play data”.

See Table 48.

Table 48 – Sync block count (normal play)

Sync block count	Contents
00	Former half of MPEG2 transport packet
01	Latter half of MPEG2 transport packet
10, 11	Forbidden

“Dummy data” sync block cannot be inserted between the sync blocks which have the serial sync block count.

The serial number of the sync block count shall be completed in one track.

2) When data type is set to “dummy data”.

See Table 49.

Table 49 – Sync block count (dummy data)

Sync block count	Contents
00	Dummy normal play data
01 to 11	Reserved

5.2.6.1.2 Data detail

See Table 50.

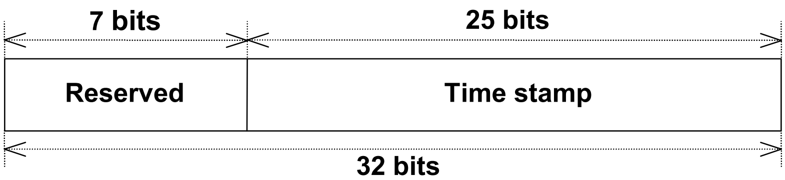
Table 50 – Data detail (normal play)

Data Detail							
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
Reserved							

5.2.6.1.3 Packet header

5.2.6.1.3.1 STD mode and LS mode

The lower 25 bits of the packet header are used for the time stamp which indicates the input time of the transport packet. See Figure 40.

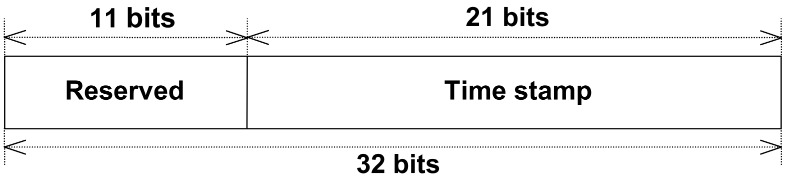


Reserved bits shall be set to “0”.

Figure 40 – Packet header (STD and LS mode/normal play)

5.2.6.1.3.2 HS mode

The lower 21 bits of packet header are used for time stamp which indicates the input time of the transport packet. See Figure 41.



Reserved bits shall be set to “0”.

Figure 41 – Packet header (HS mode/normal play)

5.2.6.1.4 Time stamp for MPEG2 TS recording

The input time of the transport packet is recorded as the value sampled from the local time stamp generator in order to make the time interval of playback exactly the same as that of input.

The clock frequency of the local time stamp generator (*Fts*) shall be 27 MHz locked to the input PCR data at recording. At playback the clock frequency shall be 27 MHz (±20 ppm).

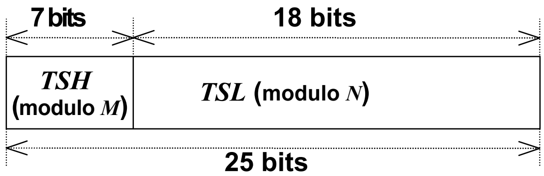
The drum shall be synchronized at recording and playback to *Fts* divided by 900 000 (1 800 rpm) or *Fts* divided by 900 900 (1 800/1,001 rpm).

The initial value of the time stamp generated by the local time stamp generator is “0”. The drum rotation phase should be synchronized with the value of the local time stamp generator at recording and playback.

5.2.6.1.4.1 STD mode and LS mode

The lower 18 bits of the time stamp (*TSL*) indicate the binary value counted up by the local time stamp generator (*Fts*; 27 MHz) and it repeats from 0 to 224 999 (mod 225 000; 1 800 rpm) or from 0 to 225 224 (mod 225 225; 1 800/1,001 rpm). See Figure 42.

The upper 7 bits of the time stamp (*TSH*) indicate the binary value counted up every time *TSL* changes to 0 (every quarter rotation of the drum) and it repeats from 0 to $12 \times k$-1 (mod ($12 \times k$)).



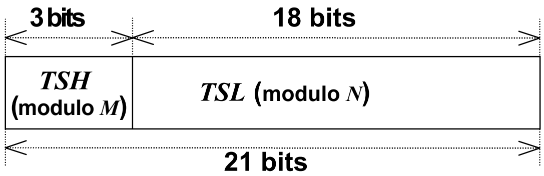
$$M = 12 \times k \qquad N = \begin{cases} 225\,000\ ;\ 1\,800\ \text{rpm} \\ 225\,225\ ;\ 1\,800\ /\ 1{,}001\ \text{rpm} \end{cases}$$

Figure 42 – Time stamp for MPEG2 TS recording (STD and LS mode/normal play)

5.2.6.1.4.2 HS mode

The lower 18 bits of the Time stamp (*TSL*) indicate the binary value counted up by the local time stamp generator (*Fts*; 27 MHz) and it repeats from 0 to 224 999 (mod 225 000; 1 800 rpm) or from 0 to 225 224 (mod 225 225; 1 800/1,001 rpm). See Figure 43.

The upper 3 bits of the time stamp (*TSH*) indicate the binary value counted up every time *TSL* changes to 0 (every quarter rotation of drum) and it repeats from 0 to 5 (mod 6).



$$M = 6 \qquad N = \begin{cases} 225\,000\ ;\ 1\,800\ \text{rpm} \\ 225\,225\ ;\ 1\,800\ /\ 1{,}001\ \text{rpm} \end{cases}$$

Figure 43 – Time stamp for MPEG2 TS recording (HS mode/normal play)

5.2.6.1.5 Transport packet recording position

Each transport packet shall be recorded on the tape in order of the time stamp value.

The 6 consecutive tracks which have the same sequence number sn are numbered from 0 to 5 as shown in Figure 37. The recorded transport packet of Time stamp $T.S[m,n]$ shall then satisfy the following rule, relating the actual recording position with track number t and sync block number s to the time stamp $T.S[m,n]$.

$$[X-1, Y] < [t, s] < [X+1, Y]$$

where

(STD mode and LS mode)

$$X = \mathrm{int}(m/(2\times k))$$
$$Y = \mathrm{int}[(n/N + m \bmod (2\times k))\times 306/(2\times k)]$$

(HS mode)

$$X = m$$
$$Y = \mathrm{int}[(n/N)\times 306]$$

$$N = \begin{cases} 225\,000;\ 1\,800\ \mathrm{rpm} \\ 225\,225;\ 1\,800/1{,}001\ \mathrm{rpm} \end{cases}$$

m (STD and LS mode): the upper 7 bits of the time stamp (TSH)

m (HS mode): the upper 3 bits of the time stamp (TSH)

n: the lower 18 bits of the time stamp (TSL)

X: track number of nominal recording position

Y: sync block number of nominal recording position

$[X, Y]$: the nominal recording position of a packet with the time stamp $T.S[m, n]$

t: track number of actual recording position

s: sync block number of actual recording position

$[t, s]$: the actual recording position of a packet with the time stamp $T.S[m, n]$

$X-1=-1$ means track number: 5 in the preceding sequence whose sequence number is $sn-1\,(sn \neq 0)$ or $snmax\,(sn = 0)$

$X+1=6$ means track number: 0 in the subsequent sequence whose sequence number is $(sn+1) \bmod (snmax+1)$

$sn_{\max}$: the maximum value of sn

5.2.6.2 MPEG2 TS recording for the trick play data

When the application detail in the main header (see 5.2.3.2.3) is set to 0000, the MPEG2 transport stream is recorded (MPEG2 transport stream recording).

The MPEG2 transport packet (188 bytes) and associated packet header information (4 bytes) are recorded onto two sync blocks. The allocation of trick play data is shown in Figure 44.

No dummy data are allowed in the trick play areas. Stuffing (padding) should be done with NULL packets in the MPEG transport stream. MPEG packets must be recorded on a fixed grid with respect to the ECC structure of Figure 38, which means that the former half of an MPEG transport packet is always recorded in a sync block with an even trick play sync block number, and the corresponding latter half of the same packet in the sync block with the next higher trick play sync block number.

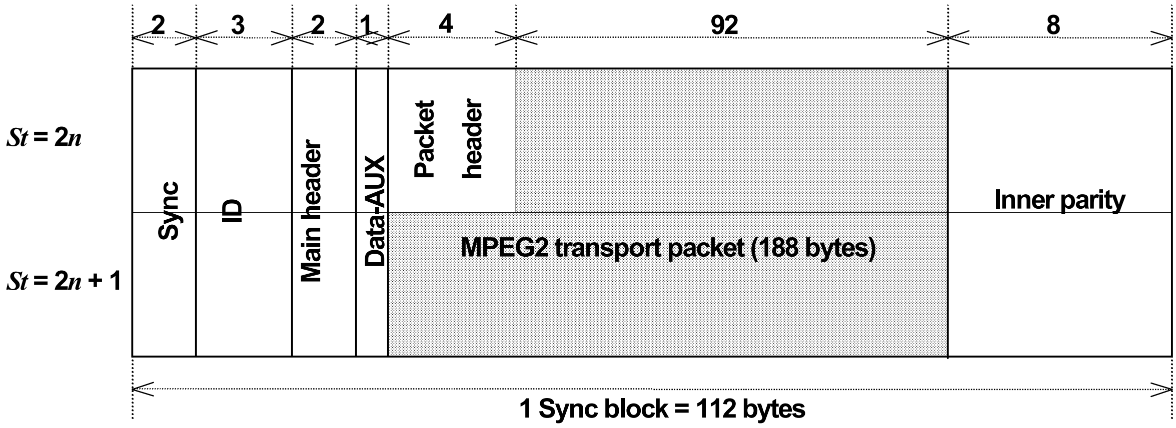


Figure 44 – MPEG2 TS recording (trick play)

5.2.6.2.1 Sync block count

When the application detail is set to 0000 (MPEG2 transport stream), the sync block count is defined as follows.

5.2.6.2.1.1 STD mode and LS mode

When the data type is set to "trick play data".

See Table 51.

Table 51 – Sync block count (STD and LS mode/ trick play)

Sync block count	Contents
00	Trick play data for $4 \times k$ times normal speed
01	Trick play data for $6 \times k$ times normal speed
10	Trick play data for $12 \times k$ times normal speed
11	Trick play data for $24 \times k$ times normal speed

5.2.6.2.1.2 HS mode

When the data type is set to "trick play data".

See Table 52.

Table 52 – Sync block count (HS mode/ trick play)

Sync block count	Contents
00	Reserved
01	Trick play data for 3 times normal speed
10	Trick play data for 6 times normal speed
11	Trick play data for 12 times normal speed

5.2.6.2.2 Data detail

See Table 53.

Table 53 – Data detail (trick play)

Data Detail							
bit 7	bit 6	bit 5	bit 4	bit 3	bit 2	bit 1	bit 0
F/R	Trick play sync block number						

1) F/R

This bit represents direction of the trick play data. See Table 54.

Table 54 – Direction of trick play

Direction	Contents
0	Trick play data for forward direction
1	Trick play data for reverse direction

2) Trick play sync block number

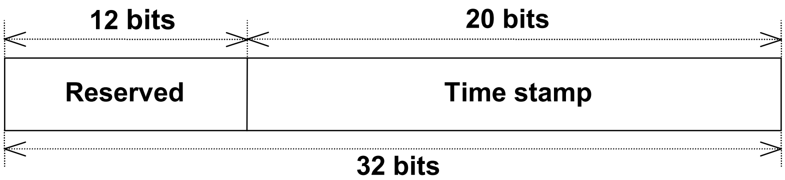
These seven bits represent the sync block number of the trick play data.

Sync blocks that contain trick play data are numbered from 0 to 101.

Sync blocks that contain outer parity for the trick play data are numbered from 102 to 111.

5.2.6.2.3 Packet header

The lower 20 bits of the packet header are used for the time stamp which indicates the input time of the transport packet. See Figure 45.



Reserved bits shall be set to “0”.

Figure 45 – Packet header (trick play)

5.2.6.2.4 Time stamp for MPEG2 TS recording

The time stamp shall be added to the trick play packets such that a valid MPEG-2 transport stream is outputted at playback of trick play data for a specific trick play speed.

5.2.6.2.4.1 STD mode and LS mode

The period of the time stamp generator for trick play is equal to one revolution of the drum at playback and starts with zero at the start of the scan of the channel 1 head. The time stamp generator for the playback of trick play data is identical to the time stamp generator for normal play data as described in 5.2.6.1.4 except that only the 2 LSB’s of TSH are used. This means that $TSHt = TSH \bmod 4$.

5.2.6.2.4.2 HS mode

The period of the time stamp generator for trick play is equal to one revolution of the drum and starts with zero at the start of the first scan.

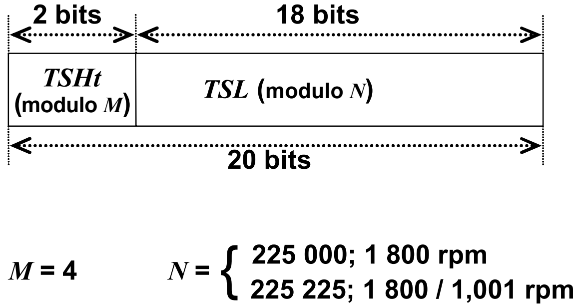


Figure 46 – Time stamp for MPEG2 TS recording (trick play)

5.2.6.2.5 Transport packet recording position

The trick play sync block number shall be added to each transport packet in the order of the playback. Trick play data sync blocks can be placed in any order within a basic unit. The output stream is divided in transport stream units with an average time duration equal to one revolution of the drum (33 ms). Subsequent transport stream units at the output correspond to subsequent basic units on tape. Subsequent means later in time during trick play replay. Within a basic unit (one revolution), packets are outputted in the order of an increasing trick play sync block number at an instant given by the time stamp value stored in the packet header. See Figure 47.

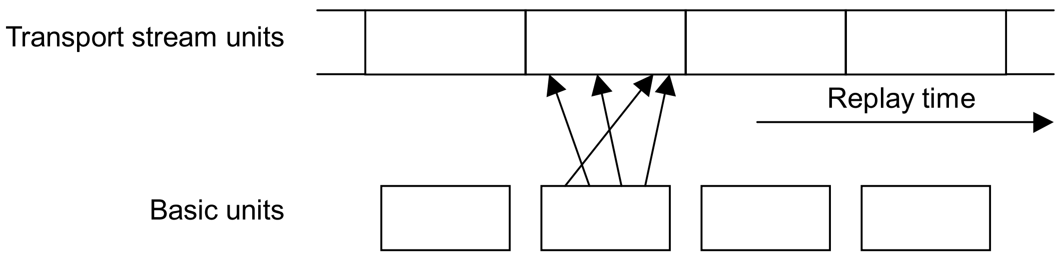


Figure 47 – TS recording position

The trick play transport packet of time stamp $T.S[m, n]$ shall satisfy the following rule, relating actual recording position with trick play sync block number S_t to the time stamp $T.S[m, n]$.

$$Y_t\text{-}51 < S_t < Y_t + 51$$

where

$$Y_t = \text{int}\left[(n/N + m_t) \times 102/4\right]$$

$$N = \begin{cases} 225\,000\,; 1\,800\ \text{rpm} \\ 225\,225\,; 1\,800/1{,}001\ \text{rpm} \end{cases}$$

m_t: the upper 2 bits of the time stamp (TSH_t) equal to the lower 2 bits of the normal play TSH(STD mode and LS mode)

n: the lower 18 bits of time stamp (TSL)

Y_t: trick play sync block number of nominal recording position of former half of the transport packet

S_t: trick play sync block number of actual recording position of former half of the transport packet.

The former half is stored in S_t. The latter half is stored in $S_t + 1$. In case of "Y_t - 51 < 0", " Y_t - 51" shall be interpreted as " Y_t - 51 + 102 in the preceding basic unit".

In case of " Y_t + 51 >101", " Y_t + 51" shall be interpreted as " Y_t + 51 - 102 in the subsequent basic unit".

Preceding and subsequent basic unit is based on the order that basic units are read from tape in trick play.

6 Pack

6.1 Pack format

6.1.1 C0: No information

This pack shall be recorded in the subcode or the main code area.

	MSB							LSB
PC0	0	0	0	0	0	0	0	0
PC1								
PC2								
PC3	Any data is available.							
PC4								
PC5								

Figure 48 – C0: No information

This pack is used for no information pack.

If there exists no data in the pack data, the pack header (PC0) shall be 00h.

6.1.2 C1: Cassette ID

This pack shall be recorded in the subcode area. See Figure 49.

	MSB							LSB
PC0	0	0	0	0	0	0	0	1
PC1	Tens of C number				Units of C number			
PC2	Thousands of C number				Hundreds of C number			
PC3	Maker code MSB					LSB	MSB	
PC4	Random number							
PC5								LSB

Figure 49 – C1: Cassette ID

C number: Cassette number

	001 to 999	FFFh = No information
Maker code:	0 to 55	56 to 61 = For pre-recorded tape 62 = Reserved 63 = No information

Random number: the method of numbering random number is not specified.

6.1.3 C2: Absolute time code

This pack shall be recorded in the subcode or the main code area. See Figure 50.

	MSB							LSB
PC0	0	0	0	1	1	0	0	1
PC1	BF	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	SGN	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 50 – C2: Absolute time code

This pack contains the absolute time code data which shows the elapsed time at the tape position where this pack is recorded.

This time code shall be set to the initial value within 90 s from beginning of the tape.

The absolute time code can be recorded as a minus value before the start point (ATC = initial value). If there exists discontinuity before the tape position where recording will be started, BF shall be 1.

Frames:

	Interlace scan	Progressive scan
525 / 60	00 to 29	00 to 59
625 / 50	00 to 24	00 to 49

00 = Initial value 7Fh = No information

Seconds:	00 to 59	00 = Initial value
Minutes:	00 to 59	00 = Initial value
Hours:	00 to 23	00 = Initial value
Days:	00 to 99	00 = Initial value

BF: Blank flag

0 = All time codes before and in this pack are increased according to the increment rule of time code. (n: a natural number).

1 = There exists a time code that does not conform to the definition of BF=0.

CF: Continuity flag

0 = All time code packs where the time code is continuous.

1 = All time code packs where the time code is discontinuous.

(“Continuous“ means increment is being made according to the number of frames per track in compliance with the increment rule of time code.)

DF: Drop frame flag

0 = Drop frame mode

1 = Non drop frame mode

Drop frame mode should be applicable for 1 800/1,001 rpm.

Drop frame sequence shall be based on the SMPTE/EBU format.

In the drop frame mode of the 525/60 interlace scan system, the first two frame numbers (00 and 01) shall be omitted from the count at the start of each minute except minutes 0,10, 20, 30, 40 and 50.

In the drop frame mode of the 525/60 progressive scan system, the first four frame numbers (00, 01, 02 and 03) shall be omitted from the count at the start of each minute except minutes 0, 10, 20, 30, 40 and 50.

SF: Scanning flag

0 = Interlace scan

1 = Progressive scan

SGN: Sign flag

0 = plus value

1 = minus value

Before the starting point, the minus value of the time code can be used.

6.1.4 C3: Optional time code

This pack shall be recorded in the subcode or the main code area. See Figure 51.

	MSB							LSB
PC0	0	0	0	1	1	0	0	0
PC1	BF	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	SGN	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 51 – C3: Optional time code

Frames: See C2: Absolute time code
Seconds: See C2: Absolute time code
Minutes: See C2: Absolute time code
Hours: See C2: Absolute time code
Days: See C2: Absolute time code

BF: See C2: Absolute time code
CF: See C2: Absolute time code
DF: See C2: Absolute time code
SF: See C2: Absolute time code
SGN: See C2: Absolute time code

6.1.5 C4: Total time

This pack shall be recorded in the subcode or the main code area. See Figure 52.

	MSB							LSB
PC0	0	0	0	1	0	0	0	1
PC1	0	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	0	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 52 – C4: Total time

This pack shows the total recording time of the real time.

When this pack is in use, all data (PC0 to PC5) shall be written.

Frames: See C2: Absolute time code
Seconds: See C2: Absolute time code
Minutes: See C2: Absolute time code
Hours: See C2: Absolute time code
Days: See C2: Absolute time code

CF: See C2: Absolute time code
DF: See C2: Absolute time code
SF: See C2: Absolute time code

6.1.6 C5: Rec date

This pack shall be recorded in the subcode or the main code area. See Figure 53.

<table>
<tr><td></td><td>MSB</td><td></td><td></td><td></td><td></td><td></td><td></td><td>LSB</td></tr>
<tr><td>PC0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>PC1</td><td colspan="2">Reserved</td><td colspan="2">Tens of day</td><td colspan="4">Units of day</td></tr>
<tr><td>PC2</td><td colspan="3">Week
MSB LSB</td><td>Tens of month</td><td colspan="4">Units of month</td></tr>
<tr><td>PC3</td><td colspan="4">Tens of year</td><td colspan="4">Units of year</td></tr>
<tr><td>PC4</td><td>VF</td><td>TM</td><td colspan="2">Tens of time zone</td><td colspan="4">Units of time zone</td></tr>
<tr><td>PC5</td><td colspan="8">Reserved</td></tr>
</table>

Reserved bits shall be set to '0'.

Figure 53 – C5: Rec date

This pack is used to memorize the date and time zone to indicate when and where the recording is being made.

Day: 01 to 31 3Fh = No information
Week: 0 = Sunday 4 = Thursday
1 = Monday 5 = Friday
2 = Tuesday 6 = Saturday
3 = Wednesday 7 = No information
Month: 01 to 12 = January to December 1Fh = No information
Year: Last two figures of A.D.: 00 to 99 FFh = No information

VF: Validity flag 0 = TM and Time zone data is invalid
1 = TM and Time zone data is valid
TM: Thirty minutes flag 30 min unit of the time differential from UTC
0 = 0 min
1 = 30 min
Time zone: 00 to 23 3Fh = No information

Example

For Tokyo Time zone = 001001b PC4 = 10001001b UTC plus 9:00

For New York with daylight saving time

Time zone = 011001b PC4 = 10011001b UTC plus 19:00

For New Delhi where 30 min unit of the time differential from UTC is adopted

Time zone = 000101b PC4 = 11000101b UTC plus 5:30

where UTC: Universal time

6.1.7 C6: Rec time

This pack shall be recorded in the subcode or the main code area. See Figure 54.

	MSB							LSB
PC0	0	0	0	1	0	0	0	0
PC1	0	0	0	0	0	0	0	0
PC2	0	Tens of seconds			Units of seconds			
PC3	0	Tens of minutes			Units of minutes			
PC4	0	0	Tens of hours		Units of hours			
PC5	0	0	0	0	0	0	0	0

Figure 54 – C6: Rec time

This pack is used to memorize the time of the day to indicate when the recording is being made.

Seconds:	00 to 59
Minutes:	00 to 59
Hours:	00 to 23

6.1.8 C7: Source

This pack shall be recorded in the subcode or the main code area. See Figure 55.

	MSB							LSB
PC0	0	0	0	0	0	0	1	1
PC1	Tens of TV channel				Units of TV channel			
PC2	Reserved				Hundreds of TV channel			
PC3	AF	Total number of additional packs (MSB … LSB)						
PC4	Reserved							
PC5	Reserved							

Reserved bits shall be set to '0'.

Figure 55 – C7: Source

This pack shows TV channel number and can be used to carry the additional information for multi-channel recording.

TV channel: the number of the television channel

001 to 999 = Television channel

EEEh = Pre-recorded tape

FFFh = No information

TV channel should indicate the channel number which is assigned to the broadcasting station, and it may indicate the channel number which is set by the user on the receiver.

AF: Additional information flag

0 = No additional pack exists

1 = An additional information pack or additional source pack exists

TAP number: Total number of additional packs:

In the case where AF = 0 — 0 = This pack is a single pack

n (≠0) = Additional source pack number

In the case where AF = 1 — 0 = End of additional information of source

n (≠0) = Total number of additional information packs or additional source packs

End of additional information pack:

PC1 and PC2 of "End of additional information pack" shall be the same as the first source pack.

6.1.9 C8: Text header

This pack shall be recorded in the subcode or the main code area. See Figure 56.

	MSB							LSB
PC0	0	0	1	0	0	X	X	X
PC1	TP number (MSB … LSB)							
PC2	Text code							
PC3	Tens of program number				Units of program number			
PC4	Tens of index number				Units of index number			
PC5	Reserved							

Reserved bits shall be set to '0'.

Figure 56 – C8: Text header

XXX: Application type 000 = Title 001 = Caption

TP number: Total number of text data pack 1 to 255

Text code: Text code designates the character set

00h = ISO 8859-1 (Latin-1)

01h = JIS X 208 (1997)

NOTE Two bytes are required for the coding of this character set.

For each byte, the code position shall be in the range of 02/1 to 07/14.

Program number: 01 to 99 FFh = No information

EEh = Same header as previous text

Index number: 01 to 99 FFh = No information

EEh = Same header as previous text

When the program number and the index number are both set to AAh, the text data pack contains the cassette information.

When the program number and the index number are both set to EEh, then the program number and the index number are the same as in the previous text header.

6.1.10 C9: Text data

This pack shall be recorded in the subcode or the main code area. See Figure 57.

	MSB							LSB
PC0	0	0	1	0	1	X	X	X
PC1	P number (MSB … LSB)							
PC2	Data							
PC3	Data							
PC4	Data							
PC5	Data							

Figure 57 – C9: Text data

This pack contains font data, graphic data, text data, according to the text code designated in the text header pack.

The text data pack data can be repeatedly sent by using the same pack page number.

XXX: Application type 000 = Title 001 = Caption

P number: Pack page number 1 to TP (designated in the text header pack)

6.1.11 C10: Transparent

See Figure 58.

	MSB							LSB
PC0	0	0	0	0	0	1	1	1
PC1								
PC2								
PC3	Any data is available.							
PC4								
PC5								

Figure 58 – C10: Transparent

6.1.12 C11, C 12: ISRC

These packs shall be recorded in the main code area. (International standard recording code). See Figure 59.

	MSB							LSB
PC0	0	0	0	0	0	1	0	0
PC1	Reserved		I1 (MSB … LSB)					
PC2	Reserved		I2 (MSB … LSB)					
PC3	Reserved		I3 (MSB … LSB)					
PC4	Reserved		I4 (MSB … LSB)					
PC5	Reserved		I5 (MSB … LSB)					

	MSB							LSB
PC0	0	0	0	0	0	1	0	1
PC1	I7 (MSB … LSB)				I6 (MSB … LSB)			
PC2	I9 (MSB … LSB)				I8 (MSB … LSB)			
PC3	I11 (MSB … LSB)				I10 (MSB … LSB)			
PC4	Reserved				I12 (MSB … LSB)			
PC5	Reserved							

Reserved bits shall be set to '0'.

Figure 59 – C11, C 12: ISRC

These packs are prepared for pre-recorded tapes.
ISRC is defined in ISO 3901:2001.
If ISRC is not used, ISRC packs shall not be recorded.

I1 to I2:	Country code
I3 to I5:	Owner code
I6 to I7:	The year of recording
I8 to I12:	The serial number of recording

6.1.13 C13: Marker

This pack shall be recorded in the subcode area. See Figure 60.

	MSB							LSB
PC0	0	0	0	0	0	1	1	0
PC1	Mode (MSB … LSB)							
PC2	Data detail							
PC3								
PC4								
PC5								

Figure 60 – C13: Marker

Mode: Marker mode 00h to FFh 00h = No information

10h = Marker of TOC start

11h = Marker of start point

Other codes are reserved.

Data detail: the data detail depends on PC1 (Mode). See Figure 61.

This pack shows the mode of the marker flag in the tag area.

This pack shall be written in the area where the marker flag is set to '1'.

Mode = 00h: No information

PC2	
PC3	Any data is available.
PC4	
PC5	

a) No information

Mode = 10h: Marker of TOC start

PC2	Reserved	
PC3	SF	MSB
PC4	Absolute track number	
PC5		LSB

SF and absolute track number are all “0”: No information
Reserved bits should be set to “0”

b) Marker of TOC start

Mode = 11h: Marker of start point

PC2	User number	Marker number
PC3	SF	MSB
PC4	Absolute track number	
PC5		LSB

User number: 0 to 7
Marker number: 01 to 1F (00: No information)
SF and absolute track number are all “0”: No information

c) Marker of start point

Figure 61 – Data detail of marker

6.1.14 C14: TOC of start point

This pack shall be recorded in the subcode or the main code area. See Figure 62.

	MSB							LSB
PC0	0	0	1	1	0	0	0	0
PC1	Tens of program number				Units of program number			
PC2	Tens of index number				Units of index number			
PC3	SF		MSB					
PC4	Absolute track number							
PC5								LSB

Figure 62 – C14: TOC of start point

This pack contains absolute track number which shows the start point of the program and index.

Program number: 01 to 99 AAh = TOC start
BBh = TOC end
FFh = No information

Index number: 01 to 99 AAh = TOC start
BBh = TOC end
FFh = No information

Absolute track number:

When the index number is set to FFh, the absolute track number shows the start point of the program whose program number is shown in program number (PC1).

When the index number is set to the value from 01 to 99, the absolute track number shows the start point of the index whose program number is shown in program number (PC1) and index number is shown in index number (PC2).

6.1.15 C15: TOC of program information

This pack shall be recorded in the subcode or the main code area. See Figure 63.

	MSB							LSB
PC0	0	0	1	1	0	0	0	1
PC1	Tens of Total number of programs				Units of Total number of programs			
PC2	Tens of program_A				Units of program_A			
PC3	Tens of total number of indexes in the program_A				Units of total number of indexes in the program_A			
PC4	Tens of program_B				Units of program_B			
PC5	Tens of total number of indexes in the program_B				Units of total number of indexes in the program_B			

Figure 63 – C15: TOC of program information

This pack contains the total number of programs in a cassette and total number of indexes in two of the programs.

program_B = program_A + 1

If program_B is not used, PC4 and PC5 must be set to FFh.

Program number:	01 to 99	FFh = No information
Total number of programs:	01 to 99	FFh = No information
Total number of indexes:	00 to 99	FFh = No information

6.1.16 C16: Trick play information

This pack shall be recorded in the subcode area when trick play stream is recorded. See Figure 64.

	MSB							LSB
PC0	0	0	0	0	1	0	0	0
PC1	Reserved			Application ID				
PC2	Recording mode				Application detail			
PC2	Reserved					TP mode		ECC
PC4	Trick play detail (1)							
PC5	Trick play detail (2)							

Reserved bits shall be set to '0'.

Figure 64 – C16: Trick play information

This pack shows the application ID, the application detail, the trick play mode, the trick play ECC and the trick play detail.

Application ID: 00000 = MPEG2

This part shall be the same as Application detail in the corresponding format ID area.

Recording mode: 0000 = STD mode

= HS mode

= LS2 mode

= LS3 mode

= LS5 mode

= LS7 mode

= Reserved

This part shall be the same as the application detail in the corresponding format ID area.

Application detail: 0000 = MPEG2 transport stream
1xxx = Pre-recorded tape
others = Reserved

This part shall be the same as the application detail in the corresponding format ID area.

TP Mode: trick play mode 00 = Track select, MPEG2 TS
(without using DSM trick mode flag)
01 = Track select, MPEG2 TS I-frame only
(using DSM trick mode flag)
others = Reserved

ECC: outer ECC for the trick play data

0	= No ECC parity
1	= with ECC parity

Trick play detail (1): trick play detail (1)

0xxxxxxx	= +24 $\times k$ data is not present
1xxxxxxx	= +24 $\times k$ data present
x0xxxxxx	= -24 $\times k$ data is not present
x1xxxxxx	= -24 $\times k$ data present
xx0xxxxx	= +12 $\times k$ data is not present
xx1xxxxx	= +12 $\times k$ data present
xxx0xxxx	= -12 $\times k$ data is not present
xxx1xxxx	= -12 $\times k$ data present
xxxx0xxx	= +6 $\times k$ data is not present
xxxx1xxx	= +6 $\times k$ data present
xxxxx0xx	= -6 $\times k$ data is not present
xxxxx1xx	= -6 $\times k$ data present
xxxxxx0x	= +4 $\times k$ data is not present
xxxxxx1x	= +4 $\times k$ data present
xxxxxxx0	= -4 $\times k$ data is not present
xxxxxxx1	= -4 $\times k$ data present

k = 1: STD mode

Trick play detail (2): trick play detail (2)

0xxxxxxx	= +24 $\times k$ data is trick play data derived from normal play data
1xxxxxxx	= +24 $\times k$ data is special data not derived from normal play data
x0xxxxxx	= -24 $\times k$ data is trick play data derived from normal play data
x1xxxxxx	= -24 $\times k$ data is special data not derived from normal play data
xx0xxxxx	= +12 $\times k$ data is trick play data derived from normal play data
xx1xxxxx	= +12 $\times k$ data is special data not derived from normal play data
xxx0xxxx	= -12 $\times k$ data is trick play data derived from normal play data
xxx1xxxx	= -12 $\times k$ data is special data not derived from normal play data
xxxx0xxx	= +6 $\times k$ data is trick play data derived from normal play data
xxxx1xxx	= +6 $\times k$ data is special data not derived from normal play data
xxxxx0xx	= -6 $\times k$ data is trick play data derived from normal play data
xxxxx1xx	= -6 $\times k$ data is special data not derived from normal play data
xxxxxx0x	= +4 $\times k$ data is trick play data derived from normal play data
xxxxxx1x	= +4 $\times k$ data is special data not derived from normal play data
xxxxxxx0	= -4 $\times k$ data is trick play data derived from normal play data
xxxxxxx1	= -4 $\times k$ data is special data not derived from normal play data

k = 1: STD mode

Special data shall be used in combination with the maker code and the random number in the cassette ID.

6.1.17 P1: Program information

This pack shall be recorded in the subcode or the main code area. See Figure 65.

	MSB							LSB
PC0	0	0	1	1	0	0	1	0
PC1	Tens of program number				Units of program number			
PC2	Tens of index number				Units of index number			
PC3	Tens of Total number of programs				Units of Total number of programs			
PC4	Tens of Total number of indexes				Units of Total number of indexes			
PC5	Reserved							

Reserved bits shall be set to '0'.

Figure 65 – P1: Program information

This pack shows the current program number and index number during playback, and total number of programs in a cassette and total number of indexes in the program.

Program number:	01 to 99	FFh = No information
Index number:	01 to 99	FFh = No information

Total number of programs: total number of programs in this tape.
01 to 99 FFh = No information

Total number of indexes: total number of indexes in this program.
00 to 99 FFh = No information

6.1.18 P2: Program total time

This pack shall be recorded in the subcode or the main code area. See Figure 66.

	MSB							LSB
PC0	0	0	0	1	0	0	1	0
PC1	0	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	0	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 66 – P2: Program total time

This pack shows the total recording time of the program.

Frames: See C2: Absolute time code
Seconds: See C2: Absolute time code
Minutes: See C2: Absolute time code
Hours: See C2: Absolute time code
Days: See C2: Absolute time code

CF: See C2: Absolute time code
DF: See C2: Absolute time code
SF: See C2: Absolute time code

6.1.19 P3: Program time code

This pack shall be recorded in the subcode or the main code area. See Figure 67.

	MSB							LSB
PC0	0	0	0	1	1	0	1	0
PC1	BF	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	SGN	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 67 – P3: Program time code

This pack contains the time code data to show the elapsed time in the program.

This time code shall be set to the initial value at the program start position and the minus value can be used before the start position.

Frames:	See C2: Absolute time code	
Seconds:	See C2: Absolute time code	00 = Initial value
Minutes:	See C2: Absolute time code	00 = Initial value
Hours:	See C2: Absolute time code	00 = Initial value
Days:	See C2: Absolute time code	00 = Initial value

BF:	See C2: Absolute time code
CF:	See C2: Absolute time code
DF:	See C2: Absolute time code
SF:	See C2: Absolute time code
SGN:	See C2: Absolute time code

6.1.20 I1: Index total time

This pack shall be recorded in the subcode or the main code area. See Figure 68.

	MSB							LSB
PC0	0	0	0	1	0	0	1	1
PC1	0	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	0	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 68 – I1: Index total time

This pack shows the total recording time of the current index.

Frames:	See C2: Absolute time code
Seconds:	See C2: Absolute time code
Minutes:	See C2: Absolute time code
Hours:	See C2: Absolute time code
Days:	See C2: Absolute time code
CF:	See C2: Absolute time code
DF:	See C2: Absolute time code
SF:	See C2: Absolute time code

6.1.21 I2: Index time code

This pack shall be recorded in the subcode or the main code area. See Figure 69.

	MSB							LSB
PC0	0	0	0	1	1	0	1	1
PC1	BF	Tens of frames			Units of frames			
PC2	CF	Tens of seconds			Units of seconds			
PC3	DF	Tens of minutes			Units of minutes			
PC4	SF	SGN	Tens of hours		Units of hours			
PC5	Tens of days				Units of days			

Figure 69 – I2: Index time code

This pack contains time code data to show the elapsed time in index.

This time code shall be set to the initial value at the index start position.

Frames:	See C2: Absolute time code	
Seconds:	See C2: Absolute time code	00 = Initial value
Minutes:	See C2: Absolute time code	00 = Initial value
Hours:	See C2: Absolute time code	00 = Initial value
Days:	See C2: Absolute time code	00 = Initial value

BF:	See C2: Absolute time code
CF:	See C2: Absolute time code
DF:	See C2: Absolute time code
SF:	See C2: Absolute time code
SGN:	See C2: Absolute time code

6.1.22 E1: G_ECM Pack

This pack is required for pre-recorded tape.

This pack is recorded only in the main code area.

This pack shall not be outputted from a D-VHS recorder/player.

See Figure 70.

	MSB							LSB
PC0	1	1	1	1	1	1	0	0
PC1								
PC2								
PC3	Data							
PC4								
PC5								

Figure 70 – E1: G_ECM Pack

6.1.23 E2: L_ECM Pack

This pack is required for pre-recorded tape.

This pack is recorded only in the main code area.

This pack shall not be outputted from a D-VHS recorder/player.

See Figure 71.

	MSB							LSB
PC0	1	1	1	1	1	1	0	0
PC1								
PC2								
PC3	Data							
PC4								
PC5								

a) One type of L_ECM Pack

	MSB							LSB
PC0	1	1	1	1	1	1	0	0
PC1								
PC2								
PC3	Data							
PC4								
PC5								

b) Another type of L_ECM Pack

Figure 71 – E2: L_ECM Pack

6.1.24 E3: A_ECM Pack

This pack is required for pre-recorded tape.

This pack is recorded only in the main code area.

This pack shall not be outputted from a D-VHS recorder/player.

See Figure 72.

	MSB							LSB
PC0	1	1	1	1	1	1	0	1
PC1								
PC2								
PC3	Data							
PC4								
PC5								

a) One type of A_ECM Pack

	MSB							LSB
PC0	1	1	1	1	1	1	0	1
PC1								
PC2								
PC3	Data							
PC4								
PC5								

b) Another type of A_ECM Pack

Figure 72 – E3: A_ECM Pack

6.1.25 E4: R_ECM Pack

This pack is required for pre-recorded tape

This pack is recorded only in the main code area.

This pack shall not be outputted from a D-VHS recorder/player.

See Figure 73.

	MSB							LSB
PC0	1	1	1	1	1	1	1	0
PC1								
PC2								
PC3				Data				
PC4								
PC5								

Figure 73 – E4: R_ECM Pack

6.2 Pack application

6.2.1 General

Pack is an optional function. The following is a guide to using pack.

Location

There are two pack recording areas, the main code area (51 packs/track) and the subcode area (48 packs/track).

Packs should be repeatedly recorded as much as possible.

As a general rule, pack application of pre-recorded tape should be recorded in the main code area.

Basically, pack application using single pack can be recorded anywhere in the pack data area.

In applications using two or more packs as well as multi-channel recording applications, there are cases where related Packs need to be recorded continuously.

The continuity of packs means that

1) there is continuity in the case of recording across tracks.
2) there is no continuity in the case of recording across the main code area and the subcode area.
3) time codes can be recorded anywhere in both areas without disturbance of continuity.

Numbers in the pack format

Decimal number:	‘*n*’	for example, 161
Binary number:	‘*n*’b	for example, 10100001b
Hex-decimal number:	‘*n*’h	for example, A1h

6.2.2 Pack application using single pack

No information

If there exists no data in the pack data, the pack header (PC0) shall be 00h.

Cassette ID
Cassette ID pack is used for identification of the cassette.
Cassette ID should be repeatedly recorded once or more than once per any 60 tracks.
The cassette number of the end-user’s recorded tape can be recorded by the end-user.
The reserved area (in PC2) is not determined yet, but some of the bits may be used for identification of the end-user’s tape and pre-recorded tape.

ISRC
ISRC pack should be repeatedly recorded once or more than once per any 600 tracks.

Trick play information
Trick play information pack shall be recorded once or more than once per track while trick play data is present.

Source
Source pack should be repeatedly recorded once or more than once per any 60 tracks.
When recording source packs without additional information, the TV channel number should be the same on the same track.

In addition, source pack can be used in conjunction with packs of information for multi-channel.

Marker
Marker pack should be recorded within the range where the marker flag in the tag area is set to "1" and shows the marker mode.
Marker pack should be repeatedly recorded once or more than once per any 6 tracks in the marker flag active area.

Program information
Program information pack should be repeatedly recorded once or more than once per any 60 tracks.

Total time, program total time and index total time
Total time, program total time and index total time packs should be repeatedly recorded once or more than once per any 60 tracks.

Rec. date and Rec. time
Rec. date and Rec. time packs should be repeatedly recorded once or more than once per any 60 tracks.

Time code
Time code (absolute, program, index and optional time code) packs should be repeatedly recorded once or more than once per pair of tracks. In addition, they should be repeatedly recorded on the other different azimuth track of adjacent pair of tracks.

Each time code value shall be the same on the same track.
However, at the transition of two programs/indexes, minus value time code before the start of the program/index and plus value time code before the end of the preceding program/index can be different.

	Program n-1			Program n				
Time code of Program n	-3	-2	-1	0	1	2	3	
Time code of Program n-1	91	92	93					

Figure 74 – An example of different time code on a track

In the case of recording data from two or more than two frames on a track, only the same time code of the first frame should be recorded.
In the case of recording time code packs in the sync block of the subcode area, the time code packs on continuous tracks shall be recorded at a shifted location. The shift between the tracks for the packs of each individual time code shall be as follows:

$N(T) = [N(T - 1) + 13] \text{ mod.}16$

$N(T)$: Sync block number of track T

$N(T - 1)$: Sync block number of preceding track $T - 1$

Absolute time code
Absolute time code shall be set to the initial value within 90 s from the beginning of the tape.
Absolute time code can be recorded as a minus value before the start point (*ATC* = initial value).

Program time code
Program time code shall be set to the initial value at the program start point and a minus value can be used before the start point.

Index time code
Index time code shall be set to the initial value at the index start point and a minus value can be used before the start point.

Optional time code
Optional time code can be applied only for professional use.
Optional time code can be used for any purpose in accordance with the rule of the time code pack.

6.2.3 Pack application using two or more than two packs

6.2.3.1 Text header and text data

The application rule of the text header and text data depends on the application type.

There are two kinds of text recording methods, text and text group.

a) Text

Text consists of the text header pack and text data packs.
Text header contains the total number of text data pack, text code, program number and index number
Characters are distributed sequentially to a series of text data packs with pack page number.

For example, number of 8 bit (ISO 8859-1) code in one pack(page): 4 characters/page

The text data packs shall be recorded in order of P number after the text header pack.
Other packs (except text header and text data packs) can be inserted among the text data packs.

	Text header pack TP number = n	Text data pack P number = 1	Text data pack P number = 2	Other packs	Text data pack P number = 3	...	Text data pack P number = n	

Figure 75 – An example of a text recording

The same Text Data Pack can be recorded repeatedly in order of P number by using the same pack number.

	Text header pack TP number = n	Text data pack P number = 1	Text data pack P number = 1	Text data pack P number = 2	Text data pack P number = 2	.....	Text data pack P number = n	

a) in rule

	Text header pack TP number = n	Text data pack P number = 1	Text data pack P number = 2	...	Text data pack P number = n	Text data pack P number = 1	...	Text data pack P number = n	

b) out of rule

Figure 76 – An example of same text data pack recording

b) Text group

Characters which need to use more than 255 packs can be recorded using two or more than two texts, which are called the text group.
For the text headers other than the one in the first text, program number and index number shall be set to EEh. In this case, the value EEh does not indicate any actual program number or index number.

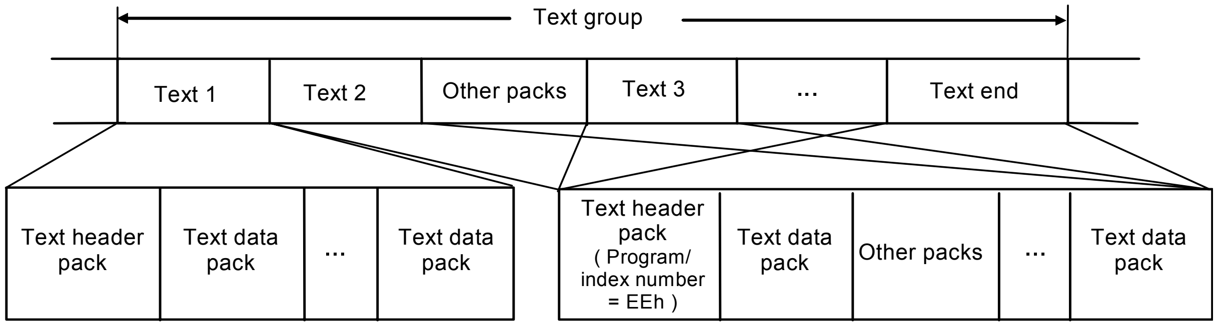


Figure 77 – An example of a text group recording

Title (application type = 0)

Text header can indicate three kinds of titles as follows.

Cassette:	Program number = AAh	Index number = AAh
Program n:	Program number = n	Index number = FFh
Index m of program n:	Program number = n	Index number = m

When recording the title in the subcode area, the title can use the *text header* pack and a maximum of eight *text data* packs.

6.2.3.2 Table of contents

There are a certain number of programs in a cassette, and program can consist of indexes.

TOC contents shall include *TOC of start point* pack which shows start point of the program or index.

Program and index numbers are indicated as follows.

Program n:	Program number = n	Index number = FFh
Index m of program n:	Program number = n	Index number = m

TOC Contents can include TOC of Program Information pack which show total number of Program in Cassette and total number of Index in Program, and can include additional information. The additional information about program number n or index number m can be recorded as “Additional information of TOC”. “Additional information of TOC” can consist of packs such as text (title), source, rec. date, rec. time, total time and so on except time code packs.

Time code packs are *absolute*, *program*, *index* and *optional time code*.

Also, “Additional information of TOC” shall be recorded continuously after the *TOC of start point* pack of program number m or Index number n without other packs, except the time code pack.

TOC of start point pack	Additional information of TOC

Figure 78 – Additional information of TOC

6.2.3.2.1 TOC format

There are two types of TOC.

a) Type A: TOC without TOC start and end pack

TOC contents

Figure 79 – TOC without TOC start and end pack

b) Type B: TOC with TOC start and end packs

In the case of type B, this start point should be indicated using a marker flag in the tag area with the *marker* pack of the TOC start mode.

The *marker* pack of the TOC start mode should be recorded once or more than once per any 6 tracks in the marker flag active area.

Marker of TOC start: marker flag in the tag area (set 1)
Marker pack (mode = 10h)
TOC start: TOC of the start point pack (Program number: AAh, Index number: AAh)
TOC end: TOC of the start point pack (Program number: BBh, Index number: BBh)

Pack :	Marker pack (TOC start)	TOC start pack	TOC contents	TOC end pack
Marker flag :	O N	OFF		

Figure 80 – TOC with TOC start and end pack

The marker pack shall be recorded in the subcode area. After the marker pack, the TOC (TOC start, TOC contents and TOC end) shall be recorded in the main code area or the subcode area.

6.2.3.2.2 TOC applications

There are three kinds of TOC applications.

a) TOC of the **previous programs in cassette**

The TOC of the previous programs should be recorded using type A.

This TOC shall include the TOC of the start point packs of all previous programs including the program. In other words, when recording new program n, this TOC should include all TOC from program 1 to program n.

TOC of the previous programs can include the TOC of the program information pack, but cannot include additional information.

The TOC of the previous programs should be recorded in the plus area of the program time code.

At least one TOC of the start point pack consisting of the TOC of previous programs should be recorded per track.

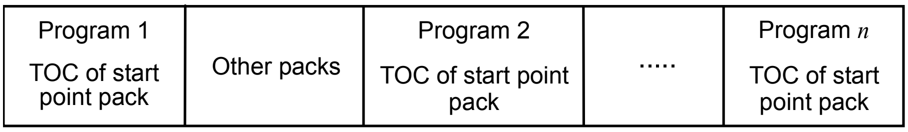


Figure 81 – An example of a TOC of previous programs in program *n*

b) TOC of each program

TOC of each program should be recorded using type B.

This TOC shall include the TOC of the start point pack of the program and can include the TOC of the program information packs and "additional information of TOC". The TOC of each program shall be recorded in the minus area of the program time code at the beginning of the program.

Packs of TOC start, TOC contents and TOC end should be recorded continuously without other packs apart from the time code pack.

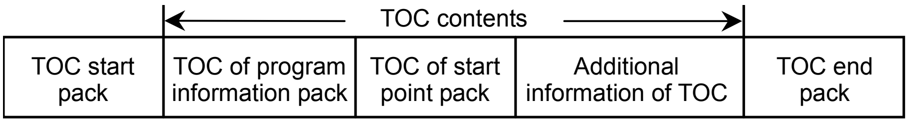


Figure 82 – An example of TOC of each program

c) TOC of all programs in the cassette

TOC of all programs should be recorded using type B.

This TOC shall include the TOC of the start point packs of all the programs in the cassette and can include the TOC of the program information packs and "additional information of TOC".

TOC of all the programs shall be recorded in the minus area of the absolute time code.

Packs of TOC start, TOC contents and TOC end should be recorded continuously without other packs except the time code pack.

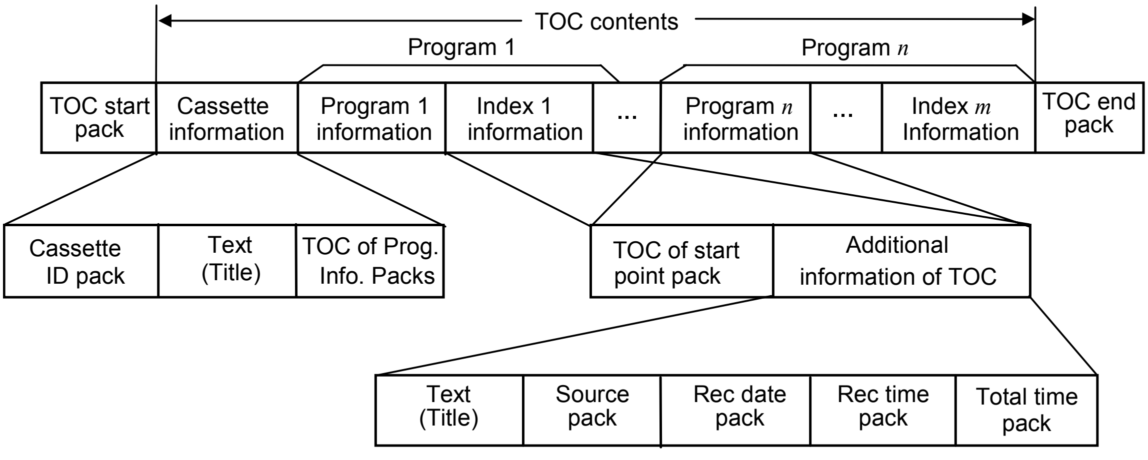


Figure 83 – An example of the TOC of all programs

6.2.4 Packs in multi-channel recording

Multi-channel information can be recorded in the subcode or the main code area appending additional information packs or additional source packs to the source pack.

When using the additional source pack, the AF flag of the leading source pack shall be set to 1 and the total number of additional source packs shall be written.

Additional source packs shall be recorded continuously after the first source pack without other packs, except the time code pack.

Indication of multi-channel recording

Source packs can be used for the TV channel indication of multi-channel recording.

These packs shall be repeatedly recorded once or more than once per any 60 tracks.

		← Additional source pack →		
	Source Pack (AF = 1)	Source pack (AF = 0)	.....	Source pack (AF = 0)
TAP number	*n*	1		*n*

TAP number of the first source pack indicates the number of additional source packs excluding time code packs.

Figure 84 – An example of indication of multi-channel recording

Information of each channel

The additional information of each channel can be recorded as “additional information of source”.

“Additional information of source” shall include the program information pack which shows the program number. “Additional information of source” can consist of packs such as text (title), marker, program time code, index time code, program total time, index total time packs and so on except absolute time code.

“End of additional information of source” shall be recorded at the end of additional information.

“End of additional information of source”: source pack (AF = 1, TAP number = 0)

		← Additional information of source →				
	Source pack (AF = 1)	Program info. pack	Text (Title) pack	.....	Program total time pack	End of additional information of source
TAP number	*n*	1	2		*n*-1	

Figure 85 – An example of Information in multi-channel recording

TAP number of the first source pack indicates the number of packs in “additional information of source” excluding absolute time code packs.

Program time code packs and index time code packs between the first source pack and “end of additional information of source” are regarded as a part of the additional information. In this case, the recording position of the program/index time code pack may not comply with the rule of time code shift (see “time code” in 6.2.2).

6.2.5 Time code application

6.2.5.1 625/50 system

When recording data from two or more frames on a track, only the same time code of the first frame shall be recorded.

a) An example of time code in the interlace scan system (STD mode)

Table 55 – Time code in the interlace scan system (STD mode)

Absolute track number	Time code (MM:SS:FF)
n	00: 00: 00
$n+2$	00: 00: 00
$n+4$	00: 00: 00
$n+6$	00: 00: 01
$n+8$	00: 00: 01
$n+10$	00: 00: 02
$n+12$	00: 00: 02
$n+14$	00: 00: 02
$n+16$	00: 00: 03
$n+18$	00: 00: 03
$n+20$	00: 00: 04
$n+22$	00: 00: 04
$n+24$	00: 00: 05
$n+26$	00: 00: 05
.	.
$n+116$	00: 00: 24
$n+118$	00: 00: 24
$n+120$	00: 01: 00
.	.
n: 0 or an arbitrary positive integer. MM:SS:FF: MM, SS and FF shows two digits of minute, second and frame respectively.	

b) An example of time code in the progressive scan system (STD mode)

Table 56 – Time code in the **progressive scan system (STD mode)**

Absolute track number	Time code (MM:SS:FF)
n	00: 00: 00
$n + 2$	00: 00: 00
$n + 4$	00: 00: 01
$n + 6$	00: 00: 02
$n + 8$	00: 00: 03
$n + 10$	00: 00: 04
$n + 12$	00: 00: 05
$n + 14$	00: 00: 05
$n + 16$	00: 00: 06
$n + 18$	00: 00: 07
$n + 20$	00: 00: 08
$n + 22$	00: 00: 09
$n + 24$	00: 00: 10
$n + 26$	00: 00: 10
.	.
$n + 116$	00: 00: 48
$n + 118$	00: 00: 49
$n + 120$	00: 01: 00
.	.
n: 0 or an arbitrary positive integer. MM:SS:FF: MM, SS and FF shows two digits of minute, second and frame respectively.	

6.2.5.2 525/60 system (1 800/1,001 rpm)

In the case of the drop frame mode, the drop frame sequence shall be based on the SMPTE format.

For the drop frame mode of the interlace scan system, the first two frame numbers (00 and 01) shall be omitted from the count at the start of each minute except minutes 0, 10, 20, 30, 40 and 50.

For the drop frame mode of the progressive scan system, the first four frame numbers (00, 01, 02 and 03) shall be omitted from the count at the start of each minute except minutes 0, 10, 20, 30, 40 and 50.

In case of recording at 1 800/1,001 rpm, non-drop frame time code is available.

See Table 57 for an example of the time code in the progressive scan system (STD mode)

Table 57 – Time code in the progressive scan system (STD mode)

Absolute track number	Time code (MM:SS:FF)
n	00: 00: 00
$n + 2$	00: 00: 01
$n + 4$	00: 00: 02
$n + 6$	00: 00: 03
.	.
$n + 116$	00: 00: 58
$n + 118$	00: 00: 59
$n + 120$	00: 01: 00
.	.
$n + 7\,196$	00: 59: 58
$n + 7\,198$	00: 59: 59
$n + 7\,200$	01: 00: 04
$n + 7\,202$	01: 00: 05
.	.
$n + 71\,996$	10: 00: 34
$n + 71\,998$	10: 00: 35
$n + 72\,000$	10: 00: 36
.	.
n: 0 or an arbitrary positive integer. MM:SS:FF: MM, SS and FF shows two digits of minute, second and frame respectively.	

Annex A
(normative)

D-VHS MPEG format

A.1 General

This D-VHS MPEG format defines MPEG stream and Service Information (SI) that are generated by the D-VHS system on a D-VHS tape. When the value of format_identifier in the Registration_descriptor is 0x4d54524d (MTRM in ASCII), the MPEG stream and the SI shall comply with this D-VHS MPEG format. The format_identifier is defined in this D-VHS MPEG format.

The purpose of this D-VHS MPEG format is to increase inter-exchangeability and interoperability. The data structure recorded on D-VHS tape complies basically with "transport stream" defined in ISO/IEC 13818-1. And the video elementary data defined in this D-VHS MPEG format shall comply with ISO/IEC 13818-2. The audio elementary data defined in this D-VHS MPEG format shall comply with ISO/IEC 11172-3 at least.

A.2 D-VHS MPEG transport stream specification

A.2.1 System layer

a) ISO/IEC 13818-1 compliant transport stream.

b) In the case of using a stuffing mechanism, null packet or adaptation field stuffing shall be used.

c) This stream shall be fully MPEG compliant with reference to ISO 13818-9.

d) It is recommended to limit peak data rate as close as possible to the average data rate.

A.2.2 Transport packet layer

a) As in ISO/IEC 13818-1. Null packets are intended for padding in the transport stream.

b) PID values for service information are given in EN 300 468. SIT and DIT are used in the D-VHS system. This service information is specified in A.3.

A.2.3 Adaptation field

a) The random_access_indicator bit shall be set whenever a random access point occurs in video streams (i.e. video sequence header immediately followed by an I-picture). Refer to A.2.7 a) constrained presentation period of GOP.

b) The time interval between two consecutive PCR values of the same program shall not exceed 100 ms. It is recommended that this interval should be no greater than 40 ms.

c) Other fields are optional.

A.2.4 PES packet

a) The PES_scrambling_control bit shall be set to "00".

b) Trick play data shall be fully MPEG-2 compliant. Use of the DSM_trick_mode flag in the header of the video PES packets is optional for the manufacturer.

c) The following trick mode fields shall not be transmitted in the normal playback stream.

– trick_mode_control;

– field_id;

– intra_slice_refresh;

– rep_cntrl.

d) For the trick play data, DSM-trick mode control is allowed. The trick play stream shall be fully compliant with the MPEG format. In the case of using DSM_trick_mode_flag, the video syntax as follows need not be set to the correct value.

– bit_rate;

– temporal_reference;

– vbv_delay;

– repeat_first_field;

– v_axis_positive;

– field_sequence;

– subcarrier;

– burst_amplitude;

– subcarrier_phase.

Other syntaxes shall be correct.

e) Other fields are optional.

A.2.5 PSI

a) PSI shall be coded using private_section structure defined in ISO/IEC 13818-1.

b) PAT and PMT shall be repeated with a maximum time interval of 120 ms between repetition. A time interval of 100 ms is recommended.

A.2.6 Program and program element descriptors

a) If profile_and_level_indication in Video_stream_descriptor is not present, then the video bit-stream shall comply with the constraints of the main profile at the main level.

b) When MPEG audio is recorded, Audio_stream_descriptor in the PMT may be used. If Audio_stream_descriptor is not present, then MPEG audio shall be recorded and all audio frames in the stream shall have the same bit rate.

c) The Hierarchy_descriptor shall not be used in the stream except when MPEG2 audio is recorded. Only when MPEG2 audio is recorded as more than one hierarchical layer, shall the Hierarchy_descriptor be used in the PMT.

d) The Maximum_bitrate_descriptor shall be in the PMT in the extended program information. The Smoothing_buffer_descriptor is recommended to be in the PMT in the extended program information.

A.2.7 Video

a) The video elementary stream shall have GOP structure. In the normal play stream, the video sequence header shall be immediately followed by an I-picture, and should be encoded at least once every 36 display fields in the case of the 525/60 TV system, and should be encoded at least once every 30 display fields in the case of TV system 625/50.

b) If quantiser matrices other than the default are used, the appropriate intra_ quantiser_matrix and/or non_intra_quantiser_matrix shall be included in every sequence header.

A.2.7.1 MP@ML

a) The frame rate shall be 24/1,001 Hz, 24 Hz, 25 Hz, 30/1,001 Hz, 30 Hz.

b) The aspect ratio shall be 4:3 or 16:9.

c) The encoded picture shall have a full screen luminance resolution of the values given in Table A.1, Table A.2, Table A.3 and Table A.4.

Table A.1 – MPEG2(MP@ML) Parameters constraints for NTSC

Vertical_size_ value	Horizontal_size_ value	Aspect_ratio_ information	Frame_rate_code	Progressive_ sequence
480	720	2 / 3	1 / 2 / 4 / 5	1
			4 / 5	0
	704		1 / 2 / 4 / 5	1
			4 / 5	0
	640		1 / 2 / 4 / 5	1
			4 / 5	0
	544		1 / 2 / 4 / 5	1
			4 / 5	0
	480		1 / 2 / 4 / 5	1
			4 / 5	0
	352		1 / 2 / 4 / 5	1
			4 / 5	0
240	352		1 / 2 / 4 / 5	1

Table A.2 – MPEG2(MP@ML) Parameters constraints for PAL/SECAM

Vertical_size_ value	Horizontal_size_ value	Aspect_ratio_ information	Frame_rate_code	Progressive_ sequence
576	720	2 / 3	3	0 / 1
	544			
	480			
	352			
288	352			1

[Aspect_ratio_information]
2 = 4:3 DAR 3 = 16:9 DAR
[Frame_rate_code]
1 = 24/1,001 Hz 2 = 24 Hz 3 = 25 Hz 4 = 30/1,001 Hz 5 = 30 Hz

Table A.3 – MPEG1 Parameters constraints for NTSC

Vertical_size	Horizontal_size	Pel_aspect_ratio	Picture_rate	Constrained_ parameters_flag
240	352	6 / 12	1 / 2 / 4 / 5	1

Table A.4 – MPEG1 Parameters constraints for PAL/SECAM

Vertical_size	Horizontal_size	Pel_aspect_ratio	Picture_rate	Constrained_ parameters_flag
288	352	3 / 8	3	1

[Pel_aspect_ratio]

3 = 16:9(625 line) 6 = 16:9(525 line) 8 = CCIR601 625 line 12 = CCIR601 525 line

[Picture_rate]

1 = 24/1,001 Hz 2 = 24 Hz 3 = 25 Hz 4 = 30/1,001 Hz 5 = 30 Hz

d) Picture_display_extension can be described in the stream. But description of this picture_display_extension is not necessarily required. When picture_display_extension is described, frame_center_horizontal_offset in Sequence_display_extension can be any value other than zero only in the case source aspect ratio is 16:9. Display_horizontal_size and horizontal_pan_scan_vector values depend on horizontal_size. See Table A.5.

Table A.5 – Permitted values for display_horizontal_size and pan scan vector

Horizontal size	Display horizontal size	Permitted horizontal pan scan vector
720	540	-1 440 – + 1 440
704	540	-1 312 – + 1 312
640	480	-1 280 – + 1 280
544	405	-1 080 – + 1 080
480	360	-960 – + 960
352	270	-656 – + 656

A value of 720 samples for horizontal_size is intended to correspond with an active line period of 53,33 µs. Value of 704 samples or 352 samples for horizontal_size is intended to correspond with an active line period of 52,15 µs. The center of 52,15 µs active line period shall coincide with that of 53,33 µs active line period for 720 sample case. In case horizontal_size is 544, the center 540 pixels in the source pictures (544) should be expanded. Frame_center_vertical_offset shall be zero.

A.2.7.2 MP@HL/MP@H14L

a) Frame rate shall be 24/1,001 Hz, 24 Hz, 25 Hz, 30/1,001 Hz, 30 Hz, 50 Hz, 60/1,001 Hz.

b) Aspect ratio shall be 4:3 or 16:9.

c) The encoded picture shall have a full screen luminance resolution of the following values in the Table A.6 and Table A.7.

Table A.6 – MPEG2(MP@HL/MP@H14L) parameters constraints for NTSC

Vertical_size_ value	Horizontal_size_ value	Aspect_ratio_ information	Frame_rate_code	Progressive_ sequence
1 080	1 920	3	1 / 2 / 4 / 5	1
			4 / 5	0
	1 440		4 / 5	0
720	1 280		1 / 2 / 4 / 5 / 7	1
480	720	2 / 3	7	1

Table A.7 – MPEG2(MP@HL/MP@H14L) parameters constraints for PAL/SECAM

Vertical_size_ value	Horizontal_size_ value	Aspect_ratio_ information	Frame_rate_code	Progressive_ sequence
1 152	1 440	3	3	0
1 080	1 920			1
				0
720	1 280		3 / 6	1
576	720	2 / 3	6	1

[Aspect_ratio_information]

2 = 4:3 DAR 3 = 16:9 DAR

[Frame_rate_code]

1 = 24/1,001 Hz 2 = 24 Hz 3 = 25 Hz 4 = 30/1,001 Hz 5 = 30 Hz 6 = 50 Hz

7 = 60/1,001 Hz

A.2.8 Audio

Audio data comprises some mandatory audio and some optional audio.

Data for multiple audio recording systems may be recorded.

MPEG1 audio or MPEG2 audio are mandatory, Dolby AC-3, AAC, DTS and Linear PCM audio are optional.

A.2.8.1 MPEG1 audio

a) MPEG1 audio shall be encoded in one of the following modes.
   - MPEG1 single channel;
   - MPEG1 dual channel;
   - MPEG1 joint stereo;
   - MPEG1 stereo.
b) The encoded stream shall use either Layer1 or Layer2 coding. Layer2 is recommended.
c) Bitrate shall be one of the following values.
   - 32, 64, 96, 128, 160, 192, 224, 256, 288, 320, 352, 384 kbps for Layer2;
   - 32, 64, 96, 128, 160, 192, 224, 256, 288, 320, 352, 384, 416, 448 kbps for Layer1.
d) Sampling frequency shall be one of the following values.
   - 32 kHz, 44,1 kHz, 48 kHz.
e) Encoding stream shall not have emphasis.
f) The CRC word shall be in the encoded stream.
g) The transmission of DRC is optional.

A.2.8.2 MPEG2 audio

See ISO/IEC 13818-3.

a) MPEG2 audio shall be encoded as either the MPEG2 multi-channel audio or the one backward compatible to MPEG1 (dematrix procedure = 0 or 1).

b) For MPEG2 encoded bit-streams with total bitrates greater than 448 kbit/s for Layer1 or 384 kbps for Layer2, an extension bit-stream shall be used. The bit rate of that extension may be in the range of 0 kbps to 682 kbps.

c) Sampling frequency shall be one of the following values.

   – 32 kHz, 44,1 kHz, 48 kHz

d) Encoding stream shall not have emphasis.

e) The CRC word shall be in the encoded stream.

f) The transmission of DRC is optional.

A.2.8.3 Dolby digital (AC-3)

a) The mode shall be selected from the coding modes specified in ATSC A/52B except the 1 + 1 mode.

b) The sampling frequency shall be 48 kHz.

c) The bitrate shall be encoded at bit rate as follows.

   – Main audio service or associated audio service containing all necessary program elements; equal to or less than 640 kbps.

   – Single channel associated service containing a single program element; equal to or less than 128 kbps.

   - Two channel dialogue associated service; equal to or less than 192 kbps.

   - Combined bit rate of a main and an associated service intended to be simultaneously decoded; equal to or less than 512 kbps.

d) The value of stream_id shall be 0xBD (private_stream_1).

e) The transmission of DRC is optional.

A.2.8.4 AAC

See ISO/IEC 13818-7.

a) The mode shall be one of the following modes.

   - mono, stereo, multichannel stereo (2/1, 2/2, 3/0, 3/1, 3/2, 3/2 + LFE), dual mono.

b) The sampling frequency shall be one of the following values.

   - 32 kHz, 44,1 kHz, 48 kHz.

c) The encoding stream shall not have emphasis.

d) The profile shall be Low Complexity (LC) profile.

e) The CRC shall be in the encoded stream.

f) The value of adts_buffer_fullness shall not be 0x7FF.

g) The bit rate of encoded data shall be equal to or less than 288 kbps for each channel.

h) The number of raw_data_block() per header in the ADTS structure shall be 1.

i) The transmission of DRC is optional.

A.2.8.5 DTS

See the paper "Delivering high quality multichannel sound to the consumer" in DTS Coherent Acoustics presented at the 100th Convention 1996 May 11-14, Copenhagen AES.

a) The mode shall be one of the following modes.

   - 1/0, 2/0, 3/0, 2/1, 2/2, 3/2 (0,1 low frequency effects channel is available in each audio mode).

b) Sampling frequency shall be 48 kHz.

c) Bitrate shall be encoded at the following value.

   – 188,25 kbps to 1 509,75 kbps in CBR.

d) Frame duration shall be 512 samples.

e) The value of stream_id shall be 0xBD (private_stream_1).

A.2.8.6 Linear PCM audio

This format is for the B-mode digital audio on analogue BS.

a) Linear PCM Audio shall be compliant to SMPTE 302M.

b) SMPTE linear PCM audio header shall be set as follows:

   number_channels: 00 (2 audio channels),

   channel_identification: 0,

   bits_per_sample: 00 (16bits per sample).

c) The encoding stream shall not have emphasis.

d) The sampling frequency shall be 48 kHz.

e) The value of stream_id shall be 0xBD (private_stream_1).

f) The input buffer size assumed for encoding shall not exceed 16 384 bytes.

   It is recommended that the input buffer for decoding be 65 536 bytes or more.

A.2.9 Closed caption

The closed caption data may be recorded in the picture user data of MPEG2 elementary video as specified in ATSC A/53B and EIA 708 B. Syntax and semantics of user data are as shown in Table A.8.

Table A.8 – Picture user data structure of closed caption

Syntax	Numberof bits	Identifier
user_data( ) {		
user_data_start_code	32	bslbf
ATSC_identifier	32	bslbf
user_data_type_code	8	uimsbf
if (user_data_type_code == ‘0x03’) {		
process_em_data_flag	1	bslbf
process_cc_data_flag	1	bslbf
additional_data_flag	1	bslbf
cc_count	5	uimsbf
em_data	8	bslbf
for (i=0 ; i < cc_count ; i++) {		
marker_bits	5	‘1111 1’
cc_valid	1	bslbf
cc_type	2	bslbf
cc_data_1	8	bslbf
cc_data_2	8	bslbf
}		
marker_bits	8	‘1111 1111’
if (additional_data_flag) {		
while(nextbits() != ‘0000 0000 0000 0000 0000 00 01’) {		
additional_user_data	8	
}		
}		
}		
next_start_code()		
}		

user_data_start_code – this is set to 0x0000 01B2.

ATSC_identifier – this is a 32 bit code that indicates that the video user data conforms to this specification. The value of the ATSC_identifier shall be 0x47413934.

user_data_type_code – the 8-bit code is set to 0x03.

process_em_data_flag – this flag is set to indicate whether it is necessary to process the em_data. If it is set to 1, the em_data shall be parsed and its meaning has to be processed. When it is set to 0, the em_data can be discarded.

process_cc_data_flag – this flag is set to indicate whether it is necessary to process the cc_data. If it is set to 1, the cc_data shall be parsed and its meaning has to be processed. When it is set to 0, the cc_data can be discarded.

additional_data_flag – this flag is set to 1 to indicate the presence of additional user data.

cc_count – this 5-bit integer indicates the number of closed caption constructs following this field. It can have values 0 through 31. The value of cc_count shall be set according to the frame rate and coded picture structure (field or frame) such that a fixed bandwidth of 9 600 bits/s is maintained for the closed caption payload data. 16 bits of closed caption payload data are carried in each pair of the fields cc_data_1 and cc_data_2.

em_data – 8 bits to representing the emergency message.

cc_valid – this flag is set to ‘1’ to indicate that the two closed caption data bytes that follow are valid. If set to ‘0’, the two data bytes are invalid.

cc_type – denotes the type of the two closed caption data bytes that follow.

cc_data_1 – the first byte of a closed caption data pair.

cc_data_2 – the second byte of a closed caption data pair.

additional_user_data – any further demand for picture user data could be met by defining this part of the bit stream.

A.2.10 Subtitling

The subtitling data may be recorded as private_stream_1. Two data formats are specified. One is the PNG format and the another is the ETS 300 743 format.

In the following regions, the PNG or the ETS 300 743 format may not be supported by decoding devices. Closed caption is used for subtitling purposes.

Regions: United States of America.

A.2.10.1 PNG

PNG shall be recorded by using the packet format of the subtitling data which complies with ARIB STD-B24.

a) Further constraints and clear indications to the format of PNG described in ARIB STD-B24 are as follows.

   - data_identifier in Synchronized_PES_data() for Subtitling shall be 0x80.

   - Private_stream_id shall be 0xFF.

The maximum size of PES shall be equal to or less than 32 768 bytes.

The input buffer size assumed for encoding shall not exceed 65 536 bytes.

The minimum time interval of the PES packet shall be equal to or more than 100 ms.

For accurate presentation, the complete PES packet should be delivered to the input buffer at least 500 ms prior to the time of PTS of the packet.

- Maximum number of languages per one ES shall be one.

Maximum value of PES rate shall be equal to or less than 256 kbps.

PES data bytes in a packet shall consist of one complete data_group() as specified in ARIB STD-B24. Within the data_group() construct, the following restrictions shall apply:

- data_group_id shall be 0x0 or 0x1.

- data_group_version may not necessarily be correct.

- data_group_link_number and last_data_group_link_number shall be 0.

Within the caption_management_data() and the caption_data() construct, the following restrictions shall apply:

- TMD shall be 00b.

- num_languages shall be 1, and language_tag shall be 0.

- DMF shall be 0010b or 1010b.

- Format shall be 1000b, 1001b, 1010b, or 1011b.

- TCS shall be 00b.

Frequency of caption_management_data() coding should be once per 300 ms at most, and once per 1 s at least.

- When colour map data (CLUT) are coded as a data_unit(), the unit shall be coded as a part of caption_management_data(). To create CLUT, following restrictions shall apply:

- The alpha value of the raster color entry shall be set to 0.

- The first color map address field in the CLUT shall be set to 128 or higher.

Within the data_unit() construct, the following restrictions shall apply:

- data_unit_parameter shall be 0x20, 0x34 or 0x35.

For data_unit() whose data_unit_parameter is 0x20, restrictions specified in ARIB TR-B15 shall apply. Furthermore, the data_unit() is only for the display control of PNG and shall not include printable characters.

- Graphics format of PNG shall comply with the PNG Specification.

Constraints of parameters in chunk are as follows.

Table A.9 – Constraints of parameters in chunk

Chunk	Constraints
IHDR	Bitdepth: 1, 2, 4, 8 Colortype: only 3(palette) Compression type: only 0 (equal or less than 32 768 bytes, deflate/inflate) Filtering method: only 0 Interlace method: only 0 (non interlace)
IDAT	Filter type: only 0 (no filter)

– When bitdepth is selected to 1, CLUT shall be selected between index 0 and index 1. When bitdepth is selected to 2, CLUT shall be selected between index 0 and index 3. When bitdepth is selected to 4, CLUT shall be selected between index 0 and index 15.

– PLTE chunk may or may not be valid. CLUT shall be used for the selection of the colour palette. (PLTE of PNG chunk shall not be used). For index value 0 to 127 of the image, predefined CLUT specified in ARIB TR-B15 shall be used for presentation. Index value 128 or higher refers to CLUT entries coded in data_unit() in the stream. Decoders incapable of handling these additional CLUT entries, may treat the index value as 7 bits, assuming bit 7 is always 0. This situation should be considered on creating additional CLUT.

– Constraints of display conditions are shown in Table A.10.

Table A.10 – Constraints of display conditions

Resolution	960 × 540 × 8, 16:9 (display size: 1 920 × 1 080 / 1 440 × 1 080 / 1 280 × 720)
	720 × 480 × 8, 16:9
	720 × 480 × 8, 4:3
Display position	x, y, arbitrary address
Size	x, y, arbitrary size

– In the case of display on the 1 920 x 1 080 plane, 960 x 540 PNG data shall be doubled its resolution of horizontal and vertical.

– In the case of display on the 1 440 x 1 080 plane, 960 x 540 PNG data shall be 3/2 times its resolution of horizontal.

– In the case of display on the 1 280 x 720 plane, 960 x 540 PNG data shall be doubled its resolution of horizontal and vertical, and after making 1 920 x 1 080 PNG data, it shall be 2/3 times its resolution of horizontal and vertical.

b) Subtitle coding regarding regions

– In the following regions, private_stream_1 compliant to PNG of ARIB STD-B24 shall be the mandatory presentation data to be displayed in these regions.

Regions: Japan.

– In the above-mentioned regions, the optional stream compliant to another format may be recorded.

A.2.10.2 ETS 300 743

Subtitling data shall be recorded by using the packet format of Subtitling data which comply with ETS 300 743.

a) Further constraints and clear indications to the format of ETS 300 743.

- The value of Data_identifier shall be 0x20.
- The value of subtitle_stream_id shall be 0x00.
- The maximum number of languages per 1 ES shall be 1.

b) Subtitle coding regarding regions

– In the following regions, private_stream_1 compliant to ETS 300 743 shall be recorded as the primary subtitling data. ETS 300 743 shall be the mandatory presentation data to be displayed in these regions.

Regions: European countries.

– In the above-mentioned regions, the optional stream compliant to another format may be recorded.

A.2.11 Playback

A.2.11.1 Video

a) In the case of the NTSC model, the MP@ML decoder shall support all the values of the full screen luminance resolution in Table A.1, and the MP@HL/MP@H14L decoder shall support all the values of the full screen luminance resolution in Table A.6 and Table A.1.

b) In the case of the PAL/SECAM model, the MP@ML decoder shall support all the values of full screen luminance resolution in Table A.2, and the MP@HL/MP@H14L decoder shall support all the values of the full screen luminance resolution in the Table A.6, and Table A.2.

A.2.11.2 Audio

The decoder shall decode the MPEG1 audio (including MPEG1 part of MPEG2 audio).

A.2.11.3 Closed caption

If the closed caption data is recorded, it is expected that the decoder decode the closed caption data and superimpose the closed caption data on video.

A.3 D-VHS MPEG transport stream service information specification

A.3.1 Definition of the D-VHS stream

The D-VHS stream shall not carry any SI data other than the Selection Information Table (SIT) described in Table A.11. The SIT may contain a summary of all relevant SI information contained in the stream. All relevant MPEG-2 PSI information shall be coded to correctly describe the partial MPEG-2 transport stream.

When the format_identifier whose value is 0x4d54524d (MTRM in ASCII) is recorded in the Registration_descriptor, it flags the bitstream as a D-VHS compliant stream. This allows the set-top box to know D-VHS compliant bitstream and to ignore the absence of any mandatory SI tables and only use information coded into the SIT.

In addition to the SIT table, a second table, called the Discontinuity Information Table (DIT), is defined. This table is to be inserted at transition points at which SI may be discontinuous.

A.3.2 Tables and descriptors

The D-VHS stream shall consist of PAT, PMT, SIT, DIT and Video/Audio elementary streams. Possible locations of the descriptor in PMT and SIT are shown in Table A.11.

Table A.11 – Possible locations of the descriptor

Descriptor	Tag value	PMT	SIT
Registration_descriptor	0x05	M1	
DTCP_descriptor	0x88	M1	
Video_stream_descriptor	0x02	O2(M*4)	
Audio_stream_descriptor	0x03	O2	
Maximum_bitrate_descriptor	0x0E	M1	
Smoothing_buffer_descriptor	0x10	O1	
Stream_identifier_descriptor	0x52	M2	
ISO_639_language_descriptor	0x0A	O2	
Hierarchy_descriptor	0x04	O2(M*3)	
Parental_rating_descriptor	0x55		O1

Descriptor	Tag value	PMT	SIT
Partial_transport_stream_descriptor	0x63		M1
Short_event_descriptor	0x4D		O2
Extended_event_descriptor	0x4E		O2
Multilingual_service_name_descriptor	0x5D		O2
Content_descriptor	0x54		O2
Partial_TS_time_descriptor	0xC3		O2
Component_descriptor	0x50		O2(M*1)
Caption_service_descriptor	0x86		O2(M*2)

Key

M: Mandatory; O: Option; 1: 1st loop; 2: 2nd loop.

*1: If subtitling data is recorded, this descriptor for subtitling data is mandatory.

*2: If closed caption data is recorded, this descriptor is mandatory. This descriptor shall not appear on events with no closed captioning service.

*3: If MPEG2 audio is recorded as more than one hierarchical layer, this descriptor is mandatory.

*4: If MPEG2 video comply with the constraints of profile and level except main profile at main level, this descriptor is mandatory.

A.3.2.1 PAT

All information in the D-VHS stream shall be described in the PAT, see Table A.12.

Table A.12 – PAT table

Syntax	Number of bits	Identifier
program_association_section(){		
table_id	8	uimsbf
section_syntax_indicator	1	bslbf
0	1	bslbf
Reserved	2	bslbf
section_length	12	uimsbf
transport_stream_id	16	bslbf
Reserved	2	bslbf
version_number	5	uimsbf
current_next_indicator	1	bslbf
section_number	8	uimsbf
last_section_number	8	uimsbf
for(i=0;i<N;i++){		
program_number	16	uimsbf
Reserved	3	bslbf
If(program_number=0){		
network_PID	13	uimsbf
}		
else{		
program_map_PID()	13	uimsbf
}		
}		
CRC_32	32	rpchof
}		

table_id – this is an 8 bit field, which shall be set to 0x00.

section_syntax_indicator – the section_syntax_indicator is a 1 bit field which shall be set to '1'.

section_length – this is a twelve bit field, the first two bits of which shall be '00'. It specifies the number of bytes of the section, starting immediately following the section_length field, and including the CRC. The value in this field shall not exceed 1 021.

transport_stream_id – this is a 16 bit field which serves as a label to identify this transport stream from any other multiplex within a network. Its value is defined by the user.

version_number – this 5 bit field is the version number of the whole program association table. The version number shall be incremented by 1 whenever the definition of the program association table changes. Upon reaching the value 31, it wraps around to 0. When the current_next_indicator is set to '1', then the version_number shall be that of the currently applicable program association table. When the current_next_indicator is set to '0', then the version_number shall be that of the next applicable program association table.

current_next_indicator – a 1 bit indicator, which when set to '1' indicates that the program association table sent is currently applicable. When the bit is set to '0', it indicates that the table sent is not yet applicable and shall be the next table to become valid.

section_number – this 8 bit field gives the number of this section. The section_number of the first section in the program association table shall be 0x00. It shall be incremented by 1 with each additional section in the program association table.

last_section_number – this 8 bit field specifies the number of the last section (that is, the section with the highest section_number) of the complete program association table.

program_number – program_number is a 16 bit field. It specifies the program to which the program_map_PID is applicable. If this is set to 0x0000 then the following PID reference shall be the network PID. For all other cases, the value of this field is user defined. This field shall not take any single value more than once within one version of the program association table. The program_number may be used as a designation for a broadcast channel, for example.

network_PID – network_PID is a 13 bit field specifying the PID of the transport stream packets which shall contain the selection information table. The value of the network_PID field is 0x001F. The presence of the network_PID is mandatory.

program_map_PID – program_map_PID is a 13 bit field specifying the PID of the transport stream packets which shall contain the program_map_section applicable for the program as specified by the program_number. No program_number shall have more than one program_map_PID assignment. The value of the program_map_PID is defined by the user.

CRC_32 – this is a 32 bit field that contains the CRC value that gives a zero output of the registers in the decoder defined in Annex B of ISO/IEC 13818-1 after processing the entire program association section.

A.3.2.2 PMT

All elementary streams in the D-VHS stream shall be described in the PMT, see Table A.13.

Table A.13 – PMT table

Syntax	Number of bits	Identifier
TS_program_map_section(){		
table_id	8	uimsbf
section_syntax_indicator	1	bslbf
0	1	bslbf
Reserved	2	bslbf
section_length	12	uimsbf
program_number	16	bslbf
Reserved	2	bslbf
version_number	5	uimsbf
current_next_indicator	1	bslbf
section_number	8	uimsbf
last_section_number	8	uimsbf
Reserved	3	bslbf
PCR_PID	13	uimsbf
Reserved	4	bslbf
program_info_length	12	uimsbf
for(i=0; i<N1; i++) {		
descriptor()		
}		
for(i=0;i<N2;i++){		
stream_type	8	uimsbf
Reserved	3	bslbf
elementary_PID	13	uimsbf
Reserved	4	bslbf
ES_info_length	12	bslbf
for(j=0;j<M;j++){		
descriptor()		
}		
}		
CRC_32	32	rpchof
}		

table_id – this is an 8 bit field, which in the case of a TS_program_map_section shall be always set to 0x02.

section_syntax_indicator – the section_syntax_indicator is a 1 bit field which shall be set to '1'.

section_length – this is a 12 bit field, the first two bits of which shall be '00'. It specifies the number of bytes of the section starting immediately following the section_length field, and including the CRC. The value in this field shall not exceed 1 021.

program_number – program_number is a 16 bit field. It specifies the program to which the program_map_PID is applicable. One program definition shall be carried within only one TS_program_map_section. This implies that a program definition is never longer than 1 016 bytes. See Annex C of ISO/IEC 13818-1 for ways to deal with the cases when that length is not sufficient. The program_number may be used as a designation for a broadcast channel, for example. By describing the different program elements belonging to a program, data from different sources (for example sequential events) can be concatenated together to form a continuous set of streams using a program_number. For examples of applications refer to Annex C of ISO/IEC 13818-1.

version_number – this 5 bit field is the version number of the TS_program_map_section. The version number shall be incremented by 1 modulo 32 when a change in the information carried within the section occurs. Version number refers to the definition of a single program, and therefore to a single section. When the current_next_indicator is set to '1', then the version_number shall be that of the currently applicable TS_program_map_section. When the current_next_indicator is set to '0', then the version_number shall be that of the next applicable TS_program_map_section.

current_next_indicator – a 1 bit field, which when set to '1' indicates that the TS_program_map_section sent is currently applicable. When the bit is set to '0', it indicates that the TS_program_map_section sent is not yet applicable and shall be the next TS_program_map_section to become valid.

section_number – the value of this 8 bit field shall be always 0x00.

last_section_number – the value of this 8 bit field shall be always 0x00.

PCR_PID – this is a 13 bit field indicating the PID of the transport stream packets which shall contain the PCR fields valid for the program specified by program_number. If no PCR is associated with a program definition for private streams, then this field shall take the value of 0x1FFF. Refer to the semantic definition of PCR in 2.4.3.5 of ISO/IEC 13818-1 for restrictions on the choice of PCR_PID value.

program_info_length – this is a 12 bit field, the first two bits of which shall be '00'. It specifies the number of bytes of the descriptors immediately following the program_info_length field.

stream_type – this is an 8 bit field specifying the type of program element carried within the packets with the PID whose value is specified by the elementary_PID. The values of stream_type are specified in Table 2-36 of ISO/IEC 13818-1.

elementary_PID – this is a 13 bit field specifying the PID of the transport stream packets which carry the associated program element.

ES_info_length – this is a 12 bit field, the first two bits of which shall be '00'. It specifies the number of bytes of the descriptors of the associated program element immediately following the ES_info_length field.

CRC_32 – this is a 32 bit field that contains the CRC value that gives a zero output of the registers in the decoder defined in Annex B of ISO/IEC 13818-1 after processing the entire transport stream program map section.

A.3.2.2.1 Descriptors in program_info_loop

A.3.2.2.1.1 Registration descriptor

See Table A.14.

Table A.14 – Registration_descriptor (descriptor_tag = 0x05)

Syntax	number of bits	Identifier
Registration_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
format_identifier	32	uimsbf
reserved	8	bslbf
version	8	bslbf
reserved	32	bslbf
}		

descriptor_tag – this 8-bit unsigned integer field shall have the value 0x05, identifying this descriptor as Registration_descriptor.

descriptor_length – this 8-bit unsigned integer field specifies the number of bytes of the Registration_descriptor immediately following descriptor_length field.

format_identifier – this field contains a 32 bit value obtained from the registration authority. The value of the format_identifier shall be 0x4d54524d (MTRM (MPEG Transport stream for Recording Media) in ASCII).

version – this 8-bit field defines version number of this specification. The first 4 bits indicate the major version number, and the last 4 bits indicate the minor version number.

EXAMPLE: Version1.0 is coded as: '0001 0000'.

EXAMPLE: Version0.7 is coded as: '0000 0111'.

reserved – These bits are reserved for future definition and are currently defined to have a value of one.

A.3.2.2.1.2 Copy_control_descriptor

The Copy_control_descriptor of D-VHS shall be carried by using the DTCP_descriptor. The DTCP_descriptor is defined in accordance with the ATSC_CA_descriptor method as defined in ATSC A/70.

The Copy_control_descriptor of D-VHS shall support the DTCP_descriptor as shown in Table A.15. The D-VHS equipment shall not change this information of the DTCP_descriptor.

Table A.15 – DTCP_descriptor (descriptor_tag = 0x88)

Syntax	Numberof bits	Identifier
DTCP_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
CA_System_ID	16	uimsbf
For(i=0;i<descriptor_length-2;i++){		
private_data_byte	8	bslbf
}		
}		

Syntax	Numberof bits	Identifier
private_data_byte {		
reserved	5	bslbf
EPN	1	bslbf
DTCP_CCI	2	bslbf
reserved	5	bslbf
Image_Constraint_Token	1	bslbf
APS	2	bslbf
}		

CA_System_ID – this field shall be set to 0x0fff (DTLA).

EPN – this field indicates the encryption plus non-assertion status that is used to stop Internet retransmission of contents that are not otherwise copy-controlled. (This field shall be interpreted in accordance with the DTCP specification. A detailed definition of the bit will be described as Version 1.0 as soon as that is defined by the specification. If the EPN bit assignment is different from that defined by the specification, the bit assignment shall comply with the specification.)

DTCP_CCI – this copy control information field indicates the digital copy generation management information as shown in Table A.16.

Table A.16 – DTCP_CCI

DTCP_CCI	Description
00	Copy-free
01	No-more-copies
10	Copy-one-generation
11	Copy-Never

Image_Constraint_Token – this field is the information indicating whether high definition resolution content shall be image-constrained when being transmitted over unprotected analog component outputs. The two states of the Image_Constraint_Token bit has the meaning shown in Table A.17.

Table A.17 – Image_Constraint_Token

Image_Constraint_Token	Description
0	The content shall be image constrained to no greater than 520 000 pixels (horizontal pixel size multiplied by vertical line size).
1	The content need not be image constrained.

APS – this field defines analog video output control for the associated program.

When DTCP_CCI is equal to 00, APS shall be set to 00.

When DTCP_CCI is equal to 10, APS defines the analog video output control for the copied program. According to the APS value, the analog output shall work as described in Table A.18.

Table A.18 – APS

APS	Description
00	Copy-free
01	APS is on: Type 1 (AGC)
10	APS is on: Type 2 (AGC + 2L Colorstripe)
11	APS is on: Type 3 (AGC + 4L Colorstripe)

reserved – these bits are reserved for future definition and are currently defined to have a value of one.

A.3.2.2.2 Descriptors in ES_info_loop

A.3.2.2.2.1 Stream_identifier_descriptor

Table A.19 – Stream_identifier_descriptor (descriptor_tag = 0x52)

Syntax	Number of bits	Identifier
Stream_identifier_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
component_tag	8	uimsbf
}		

component_tag – this field identifies the component stream to associate it with a description given in a component descriptor. Within a program map section, each stream identifier descriptor shall have a different value for this field. The value of component_tag shall be constrained as shown in Table A.20.

Table A.20 – component_tag constraints

Component	Value of component_tag
video	0x00 – 0x0F (0x00 shall indicate default ES)
audio	0x10 – 0x2F (0x10 shall indicate default ES)
subtitling data	0x30 – 0x4F (0x30 shall indicate default ES)
reserved	0x50 – 0xFF

A.3.2.2.2.2 ISO_639_language_descriptor

See Table A.21.

Table A.21 – ISO_639_language_descriptor (descriptor_tag = 0x0A)

Syntax	Number of bits	Identifier
ISO_639_language_descriptor () {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
for(i=0; i<N; i++){		
ISO_639_language_code	24	bslbf
audio_type	8	uimsbf
}		
}		

ISO_639_language_code – identifies the language or languages used by the associated program element. The ISO_639_language_code contains a 3-character code as specified by ISO 639-2. Each character is coded into 8 bits according to ISO 8859-1 and inserted in order into this 24-bit field. In the case of multilingual audio streams, the sequence of ISO_639_language_code fields shall reflect the content of the audio stream.

audio_type – the audio_type is an 8-bit field which specifies the type of stream defined in Table A.22.

Table A.22 – Audio type values

Value	Description
0x00	Undefined
0x01	Clean effects
0x02	Hearing impaired
0x03	Visual impaired commentary
reserved	Reserved

clean effects – this field indicates that the referenced program element has no language.

hearing impaired – this field indicates that the referenced program element is prepared for the hearing impaired.

visual_impaired_commentary – this field indicates that the referenced program element is prepared for the visually impaired viewer.

A.3.2.3 SIT

All D-VHS streams shall have SIT. The interval of the SIT should be no greater than 3,0 s.

Table A.23 – SIT table (PID = 0x001F table_id = 0x7F)

Syntax	Number of bits	Identifier
selection_information_section(){		
table_id	8	uimsbf
section_syntax_indicator	1	bslbf
reserved_future_use	1	bslbf
ISO_reserved	2	bslbf
section_length	12	uimsbf
reserved_future_use	16	bslbf
ISO_reserved	2	bslbf
version_number	5	uimsbf
current_next_indicator	1	bslbf
section_number	8	uimsbf
last_section_number	8	uimsbf
reserved_future_use	4	bslbf
transmission_info_loop_length	12	uimsbf
for(i=0; i<N; i++) {		
descriptor()		
}		
for(i=0;i<N;i++){		
service_id	16	uimsbf
reserved_future_use	1	bslbf
running_status	3	uimsbf
service_loop_length	12	uimsbf
for(j=0;j<N;j++){		
descriptor()		
}		
}		
CRC_32	32	rpchof
}		

table_id – this is an 8 bit field, which shall be set to 0x7F.

section_syntax_indicator – the section_syntax_indicator is a 1 bit field which shall be set to "1".

section_length – this is a 12 bit field, the first two bits of which shall be ”00”. It specifies the number of bytes of the section, starting immediately after the section_length field and including the CRC. The section_length shall not exceed 1 021 so that the entire section has a maximum length of 1 024 bytes.

version_number – this 5 bit field is the version number of the table. The version_number shall be incremented by 1 when a change in the information carried within the table occurs. When it reaches a value of 31, it wraps around to 0. When the current_next_indicator is set to "1", then the version_number shall be that of the currently applicable table. When the current_next_indicator is set to "0", then the version_number shall be that of the next applicable table.

current_next_indicator – this 1 bit indicator, when set to "1" indicates that the table is the currently applicable table. When the bit is set to "0", it indicates that the table sent is not yet applicable and shall be the next table to be valid.

section_number – this 8 bit field gives the number of the section. The section_number shall be 0x00.

last_section_number – this 8 bit field specifies the number of the last section. The last_section_number shall be 0x00.

transmission_info_loop_length – this 12 bit field gives the total length in bytes of the following descriptor loop describing the transmission parameters of the partial transport stream.

service_id – this is a 16 bit field which serves as a label to identify this service from any other service within a transport stream. The service_id is the same as the program_number in the corresponding program_map_section.

running_status – this 3 bit field indicates the running status of the event in the original stream. This is the running status of the original present event. If no present event exists in the original stream the status is considered as not running. The meaning of the running_status value is as defined in ETR 211.

service_loop_length – this 12 bit field gives the total length in bytes of the following descriptor loop containing SI related information on the service and event contained in the partial transport stream.

CRC_32 - this is a 32 bit field that contains the CRC value that gives a zero output of the registers in the decoder defined in Annex B of ISO/IEC 13818-1 after processing the entire section.

The transmission information descriptor loop of the SIT contains all the information required for controlling and managing the play-out and copying of partial transport streams. The following descriptor is proposed to describe this information.

In the service descriptor loop of the SIT, only descriptors which also may occur in the SDT or the EIT present/following are allowed.

A.3.2.3.1 Descriptors in transmission_info_loop

A.3.2.3.1.1 Partial_transport_stream_descriptor

Table A.24 – Partial_transport_stream_descriptor (descriptor_tag = 0x63)

Syntax	Number of bits	Identifier
Partial_transport_stream_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
Reserved_future_use	2	bslbf
peak_rate	22	uimsbf
Reserved_future_use	2	bslbf
minimum_overall_smoothing_rate	22	uimsbf
reserved_future_use	2	bslbf
maximum_overall_smoothing_buffer	14	uimsbf
}		

peak_rate – the maximum momentary transport packet rate (i.e. 188 bytes divided by the time interval between start times of two succeeding transport stream packets). At least an upper bound for this peak_rate should be given. A typical value is the net transponder rate (without FEC). This 22 bit field is coded as a positive integer in units of 400 bits/s.

minimum_overall_smoothing_rate – minimum smoothing buffer leak rate for the overall transport stream (all packets are covered). This 22 bit field is coded as a positive integer in units of 400 bits/s. The value 0x3FFFFF is used to indicate that the minimum smoothing rate is undefined.

maximum_overall_smoothing_buffer – maximum smoothing buffer size for the overall transport stream (all packets are covered). This 14 bit field is coded as a positive integer in units of 1 byte. The value 0x3FFF is used to indicate that the maximum smoothing buffer size is undefined.

A.3.2.3.1.2 Parental_rating_descriptor

Table A.25 – Parental_rating_descriptor (descriptor_tag = 0x55)

Syntax	Number of bits	Identifier
Parental_rating_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
for(i=0; i<N; I++){		
country_code	24	bslbf
rating	8	bslbf
}		
}		

country_code – this 24-bit field identifies a country using the 3-character code as specified in ISO 3166-1. Each character is coded into 8-bits according to ISO 8859-1 and inserted in order into the 24-bit field.

EXAMPLE: The United States of America has the 3-character code "USA", which is coded as: '0101 0101 0101 0011 0100 0001'.

rating – this 8-bit field is coded according to Table A.26, giving the recommended minimum age in years of the end user.

Table A.26 – Rating

Rating	Description
0x00	undefined
0x01 to 0x0F	minimum age = rating + 3 years
0x10 to 0xFF	future use

EXAMPLE: 0x04 implies that end users should be at least 7 years old.

A.3.2.3.2 Descriptors in ES_info_loop

A.3.2.3.2.1 Component_descriptor

See Table A.14.

Table A.27 – Component_descriptor (descriptor_tag = 0x50)

Syntax	Number of bits	Identifier
Component_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
reserved_future_use	4	slbfb
stream_content	4	uimsbf
component_type	8	uimsbf
component_tag	8	uimsbf
ISO_639_language_code	24	bslbf
for(i=0; i<N; i++){		
text_char	8	uimsbf
}		
}		

stream_content – this 4-bit field specifies the type (video, audio, or subtitling data) of stream. The coding of this field is specified in Table A.28.

component_type – this 8-bit field specifies the type of the video, audio or subtitling data component. The coding of this field is specified in Table A.28.

component_tag – this 8-bit field has the same value as the component_tag field in the stream identifier descriptor (if present in the PSI program map section) for the component stream.

ISO_639_language_code – this 24-bit field identifies the language of the component (in the case of audio or subtitling data) and of the text description which may be contained in this descriptor. The ISO_639_language_code contains a 3-character code as specified by ISO 639-2. Both code sets, i.e. ISO 639-2/B and ISO 639-2/T, may be used.

Each character is coded into 8 bits according to ISO 8859-1 and inserted in order into the 24-bit field.

text_char – this is an 8-bit field. A string of "text_char" fields specifies a text description of the component stream. Text information is coded using the character sets and methods described in A.3.5. In the case of a 2-byte code, the number of characters shall be equal to or less than 8. The total number of bytes shall be equal to or less than 16.

Table A.28 – Stream_content and component_type

stream_content	component_type	Description
0x00	0x00 – 0xFF	reserved for future use
0x01	0x00	reserved for future use
	0x01	video MP@ML (I) 4:3
	0x02	video MP@ML (I) 16:9 with pan vector
	0x03	video MP@ML (I) 16:9 without pan vector
	0x04	unused field
	0x05 – 0xA0	reserved for future use
	0xA1	video MP@ML (P) 4:3
	0xA2	video MP@ML (P) 16:9 with pan vector
	0xA3	video MP@ML (P) 16:9 without pan vector
	0xA4	unused field
	0xA5-0xB0	reserved for future use
	0xB1	video MP@HL(H14L) (I) 4:3
	0xB2	video MP@HL(H14L) (I)16:9 with pan vector
	0xB3	video MP@HL(H14L)(I)16:9 without pan vector
	0xB4	unused field
	0xB5-0xC0	reserved for future use
	0xC1	video MP@HL(H14L) (P) 4:3
	0xC2	video MP@HL(H14L) (P)16:9 with pan vector
	0xC3	video MP@HL(H14L)(P)16:9 without pan vector
	0xC4	unused field
	0xC5-0xFF	reserved for future use
0x02	0x00	reserved for future use
	0x01	audio single mono channel
	0x02	audio dual mono channel
	0x03	audio stereo (2 channel)
	0x04	audio multi-lingual, multi-channel
	0x05	audio surround sound
	0x06 – 0x3F	reserved for future use
	0x40	audio description for the visually impaired
	0x41	audio for the hard of hearing

stream_content	component_type	Description
	0x42 – 0xAF	reserved for future use
	0xB0 – 0xFE	user defined
	0xFF	reserved for future use
0x03	0x00	reserved for future use
	0x01 – 0x0F	unused field
	0x10	subtitle ETS (normal) with no monitor aspect ratio
	0x11	subtitle ETS (normal) 4:3
	0x12	subtitle ETS (normal) 16:9
	0x13	unused field
	0x14 – 0x1F	reserved for future use
	0x20	subtitle ETS (for the hard of hearing) with no monitor aspect ratio
	0x21	subtitle ETS (for the hard of hearing) 4:3
	0x22	subtitle ETS (for the hard of hearing) 16:9
	0x23	unused field
	0x24 – 0xFF	reserved for future use
0x04 – 0x0B	0x00 – 0xFF	reserved for future use
0x0C	0x00	subtitle PNG (normal) with no monitor aspect ratio
	0x01	subtitle PNG (normal) 4:3
	0x02	subtitle PNG (normal) 16:9
	0x03 – 0x 1F	reserved for future use
	0x20	subtitle PNG (for the hard of hearing) with no monitor aspect ratio
	0x21	subtitle PNG (for the hard of hearing) 4:3
	0x22	subtitle PNG (for the hard of hearing) 16:9
	0x23 – 0x3F	reserved for future use
	0x40	subtitle PNG (narrative subtitle for dubbed) with no monitor aspect ratio
	0x41	subtitle PNG (narrative subtitle for dubbed) 4:3
	0x42	subtitle PNG (narrative subtitle for dubbed) 16:9
	0x43 – 0xFF	reserved for future use
0x0D – 0x0F	0x00 – 0xFF	user defined

A.3.2.3.2.2 Short_event_descriptor

The short event descriptor provides the name of the event and a short description of the event in text form. See Table A.29.

Table A.29 – Short_event_descriptor (descriptor_tag = 0x4D)

Syntax	Number of bits	Identifier
Short_event_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
ISO_639_language_code	24	bslbf
event_name_length	8	uimsbf
for (i=0; i<event_name_length; i++){		
event_name_char	8	uimsbf
}		
text_length	8	uimsbf
for (i=0; i<text_length; i++){		
text_char	8	uimsbf
}		
}		

ISO_639_language_code – this 24-bit field contains the ISO 639-2 three character language code of the language of the following text fields. Both ISO 639-2/B and ISO 639-2/T may be used. Each character is coded into 8 bits according to ISO 8859-1 and inserted in order into the 24-bit field.

EXAMPLE: Japan has the 3-character code "jpn", which is coded as:

'0110 1010 0111 0000 0110 1110'.

event_name_length – an 8-bit field specifying the length in bytes of the event name. The total number of bytes shall be equal to or less than 80.

event_name_char – this is an 8-bit field. A string of "char" fields specifies the event name. Text information is coded using the character sets and methods described in A.3.5. In the case of a 2-byte code, the number of characters shall be equal to or less than 40.

text_length – this 8-bit field specifies the length in bytes of the following text describing the event. The total number of bytes shall be equal to or less than 160.

text_char – this is an 8-bit field. A string of "char" fields specify the text description for the event. Text information is coded using the character sets and methods described in A.3.5. In the case of a 2-byte code, the number of characters shall be equal to or less than 80.

A.3.2.3.2.3 Extended_event_descriptor

The extended event descriptor provides a detailed text description of an event, which may be used in addition to the short event descriptor. More than one extended event descriptor can be associated to allow information about one event longer than 256 bytes to be conveyed. See Table 30.

Table A.30 – Extended_event_descriptor (descriptor_tag = 0x4E)

Syntax	Number of bits	Identifier
Extended_event_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
descriptor_number	4	uimsbf
last_descriptor_number	4	uimsbf
ISO_639_language_code	24	bslbf
length_of_items	8	uimsbf
for (i=0; i<N; i++){		
Item_description_length	8	uimsbf
for (j=0; j<N; j++){		
item_description_char	8	uimsbf
}		
item_length	8	uimsbf
for (j=0; j<N; j++){		
item_char	8	uimsbf
}		
}		
text_length	8	uimsbf
for (i=0; i<N; i++){		
text_char	8	uimsbf
}		
}		

descriptor_number – this 4-bit field gives the number of the descriptor. It is used to associate information which cannot be fitted into a single descriptor. The descriptor_number of the first Extended_event_descriptor of an associated set of Extended_event_descriptors shall be "0x00". The descriptor_number shall be incremented by 1 with each additional Extended_event_descriptor in this section.

last_descriptor_number – this 4-bit field specifies the number of the last Extended_event_descriptor (that is, the descriptor with the highest value of descriptor_number) of the associated set of descriptors of which this descriptor is part.

ISO_639_language_code – this 24-bit field identifies the language of the following text fields. The ISO_639_language_code contains a 3-character code as specified by ISO 639-2. Both ISO 639-2/B and ISO 639-2/T may be used. Each character is coded into 8 bits according to ISO 8859-1 and inserted in order into the 24-bit field.

EXAMPLE: Japan has the 3-character code "jpn", which is coded as:

'0110 1010 0111 0000 0110 1110'.

length_of_items – this is an 8-bit field specifying the length in bytes of the following items.

item_description_length – this 8-bit field specifies the length in bytes of the item description. The total number of bytes shall be equal to or less than 16.

item_description_char – this is an 8-bit field. A string of "item_description_char" fields specify the item description. Text information is coded using the character sets and methods described in A.3.5. In the case of a 2-byte code, the number of characters shall be equal to or less than 8.

item_length – this 8-bit field specifies the length in bytes of the item text. The total number of bytes shall be equal to or less than 200 in this one descriptor.

item_char – this is an 8-bit field. A string of "item_char" fields specify the item text. Text information is coded using the character sets and methods described in A.3.5. In the case of a 2-byte code, the number of characters shall be equal to or less than 100.

text_length – this 8-bit field specifies the length in bytes of the non itemized extended text. This field shall be set to "0x00".

text_char – this is an 8-bit field. A string of "text_char" fields specify the non itemized extended text.

A.3.2.3.2.4 Multilingual_service_name_descriptor

The multilingual service name descriptor provides the names of the service provider and service in text form in one or more languages. See Table A.31.

Table A.31 – Multilingual_service_name_descriptor (descriptor_tag = 0x5D)

Syntax	Number of bits	Identifier
Multilingual_service_name_descriptor(){		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
for (i=0; i<N; i++) {		
ISO_639_language_code	24	bslbf
service_provider_name_length	8	uimsbf
for (j=0; j<N; j++){		
char	8	uimsbf
}		
service_name_length	8	uimsbf
for (j=0; j<N; j++){		
char	8	uimsbf
}		
}		
}		

ISO_639_language_code – this 24-bit field contains the ISO 639-2 three character language code of the language of the following text fields. Both ISO 639-2/B and ISO 639-2/T may be used. Each character is coded into 8 bits according to ISO 8859-1 and inserted in order into the 24-bit field.

EXAMPLE: Japan has the 3-character code "jpn", which is coded as:

'0110 1010 0111 0000 0110 1110'.

service_provider_name_length – this 8-bit field specifies the length in bytes of the following service provider name. The total number of bytes shall be equal to or less than 20.

service_name_length – this 8-bit field specifies the length in bytes of the following service name. The total number of bytes shall be equal to or less than 20.

char – this is an 8-bit field. A string of char fields specify the name of the service provider or service. Text information is coded using the character sets and methods described in A.3.5. In the case of a 2-byte code, the number of characters shall be equal to or less than 10.

A.3.2.3.2.5 Content_descriptor

See Table A.32.

Table A.32 – Content_descriptor (descriptor_tag = 0x54)

Syntax	Number of bits	Identifier
Content_descriptor_descriptor() {		
descriptor_tag	8	bslbf
descriptor_length	8	uimsbf
for(i=0; i<N; i++){		
content_nibble_level_1	4	uimsbf
content_nibble_level_2	4	uimsbf
user_nibble	4	uimsbf
user_nibble	4	uimsbf
}		
}		

content_nibble_level_1: this 4-bit field represents the first level of a content identifier. This field shall be coded according to Table A.33.

content_nibble_level_2: this 4-bit field represents the second level of a content identifier. This field shall be coded according to Table A.33.

user_nibble: reserved.

Table A.33 – Content_nibble level 1 and 2 assignments

Content_nibble_level_1	Content_nibble_level_2	Description
0x0	0x0 to 0xF	undefined content
	Movie/Drama	
0x1	0x0	movie/drama (general)
0x1	0x1	detective/thriller
0x1	0x2	adventure/western/war
0x1	0x3	science fiction/fantasy/horror
0x1	0x4	comedy
0x1	0x5	soap/melodrama/folkloric
0x1	0x6	romance
0x1	0x7	serious/classical/religious/historical movie/drama
0x1	0x8	adult movie/drama
0x1	0x9 to 0xE	reserved for future use
0x1	0xF	user defined
	News/Current affairs	
0x2	0x0	news/current affairs (general)
0x2	0x1	news/weather report
0x2	0x2	news magazine
0x2	0x3	documentary
0x2	0x4	discussion/interview/debate
0x2	0x5 to 0xE	reserved for future use
0x2	0xF	user defined
	Show/Game show	
0x3	0x0	show/game show (general)
0x3	0x1	game show/quiz/contest
0x3	0x2	variety show
0x3	0x3	talk show
0x3	0x4 to 0xE	reserved for future use
0x3	0xF	user defined
	Sports	
0x4	0x0	sports (general)
0x4	0x1	special events (Olympic Games, World Cup etc.)
0x4	0x2	sports magazines
0x4	0x3	football/soccer
0x4	0x4	tennis/squash
0x4	0x5	team sports (excluding football)
0x4	0x6	athletics
0x4	0x7	motor sport
0x4	0x8	water sport
0x4	0x9	winter sports
0x4	0xA	equestrian

Content_nibble_level_1	**Content_nibble_level_2**	**Description**
0x4	0xB	martial sports
0x4	0xC to 0xE	reserved for future use
0x4	0xF	user defined
	Children's/Youth programs	
0x5	0x0	children's/youth programs (general)
0x5	0x1	pre-school children's programs
0x5	0x2	entertainment programs for 6 to 14
0x5	0x3	entertainment programs for 10 to 16
0x5	0x4	informational/educational/school programs
0x5	0x5	cartoons/puppets
0x5	0x6 to 0xE	reserved for future use
0x5	0xF	user defined
	Music/Ballet/Dance	
0x6	0x0	music/ballet/dance (general)
0x6	0x1	rock/pop
0x6	0x2	serious music/classical music
0x6	0x3	folk/traditional music
0x6	0x4	jazz
0x6	0x5	musical/opera
0x6	0x6	ballet
0x6	0x7 to 0xE	reserved for future use
0x6	0xF	user defined
	Arts/Culture (without music)	
0x7	0x0	arts/culture (without music, general)
0x7	0x1	performing arts
0x7	0x2	fine arts
0x7	0x3	religion
0x7	0x4	popular culture/traditional arts
0x7	0x5	literature
0x7	0x6	film/cinema
0x7	0x7	experimental film/video
0x7	0x8	broadcasting/press
0x7	0x9	new media
0x7	0xA	arts/culture magazines
0x7	0xB	fashion
0x7	0xC to 0xE	reserved for future use
0x7	0xF	user defined
	Social/Political issues/Economics	
0x8	0x0	social/political issues/economics (general)
0x8	0x1	magazines/reports/documentary
0x8	0x2	economics/social advisory
0x8	0x3	remarkable people

Content_nibble_level_1	Content_nibble_level_2	Description
0x8	0x4 to 0xE	reserved for future use
0x8	0xF	user defined
	Children's/Youth programmes Education/ Science/Factual topics	
0x9	0x0	education/science/factual topics (general)
0x9	0x1	nature/animals/environment
0x9	0x2	technology/natural sciences
0x9	0x3	medicine/physiology/psychology
0x9	0x4	foreign countries/expeditions
0x9	0x5	social/spiritual sciences
0x9	0x6	further education
0x9	0x7	languages
0x9	0x8 to 0xE	reserved for future use
0x9	0xF	user defined
	Leisure hobbies	
0xA	0x0	leisure hobbies (general)
0xA	0x1	tourism/travel
0xA	0x2	handicraft
0xA	0x3	motoring
0xA	0x4	fitness and health
0xA	0x5	cooking
0xA	0x6	advertisement/shopping
0xA	0x7	gardening
0xA	0x8 to 0xE	reserved for future use
0xA	0xF	user defined
	Special characteristics	
0xB	0x0	original language
0xB	0x1	black and white
0xB	0x2	unpublished
0xB	0x3	live broadcast
0xB	0x4 to 0xE	reserved for future use
0xB	0xF	user defined
0xC to 0xE	0x0 to 0xF	reserved for future use
0xF	0x0 to 0xF	user defined

A.3.2.3.2.6 Partial_TS_time_descriptor

See Table A.34.

Table A.34 – PartialTS_time_descriptor (descriptor_tag = 0xC3)

Syntax	Number of bits	Identifier
PartialTS_time_descriptor() {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
event_version_number	8	uimsbf
event_start_time	40	bslbf
Duration	24	uimsbf
Offset	24	bslbf
Reserved	5	bslbf
offset_flag	1	bslbf
other_descriptor_status	1	bslbf
UTC_time_flag	1	bslbf
If(UTC_time_flag == 1){		
UTC_time	40	bslbf
}		
}		

event_version_number – this 8-bit field is the version number of the event. The event_version_number shall be incremented by 1 when a change of the event occurs within a service. When it reaches the value 255, it wraps around to 0.

event_start_time – this 40-bit field contains the start time of the event in Universal Time, Co-ordinated (UTC) and Modified Julian Date (MJD) (see A.3.6). This field is coded as 16 bits giving the 16 LSBs of MJD followed by 24 bits coded as 6 digits in 4-bit Binary Coded Decimal (BCD). If the start time is undefined, all bits of the field are set to "1".

EXAMPLE: 93/10/13 12:45:00 is coded as '0xC079124500'.

duration – A 24-bit field containing the duration of the event in hours, minutes and seconds. This field is coded as 6 digits in 4-bit Binary Coded Decimal (BCD). If the duration is undefined, all bits of the field are set to "1".

EXAMPLE: 01:45:30 is coded as '0x014530'.

offset – this 24-bit field containing the offset time in hours, minutes and seconds. This offset time is applied to event_start_time and UTC_time. This field is coded as 6 digits in 4-bit Binary Coded Decimal (BCD). If no offset time is applied, all bits of the field are set to "0".

EXAMPLE: 05:00:00 is coded as '0x050000'.

offset_flag – this 1-bit information indicates the polarity of the offset time. If this bit is set to "0", the value of offset is added to event_start_time and UTC_time. If this bit is set to "1", the value of offset is subtracted from event_start_time and UTC_time.

other_descriptor_status – this 1-bit information indicates the status of descriptors within the SIT except for this PartialTS_time_descriptor. If this bit is set to "0", the other descriptors are not changed. If this bit is set to "1", there is a possibility that one or more descriptors have been changed.

UTC_time_flag – when UTC_time_flag is set to "1", UTC_time filed is present.

UTC_time – this 40-bit field contains the current time and date in Universal Time, Co-ordinated (UTC) and Modified Julian Date (MJD) (see A.3.6). This field is coded as 16 bits giving the 16 LSBs of MJD followed by 24 bits coded as 6 digits in 4-bit Binary Coded Decimal (BCD).

EXAMPLE: 93/10/13 12:45:00 is coded as '0xC079124500'.

A.3.2.3.2.7 Caption_service_descriptor

Table A.35 – Caption_service_descriptor (descriptor_tag = 0x86)

Syntax	Number of bits	Identifier
Caption_service_descriptor () {		
descriptor_tag	8	uimsbf
descriptor_length	8	uimsbf
Reserved	3	bslbf
number_of_services	5	uimsbf
for (i=0;i<number_of_services;i++) {		
language	24	uimsbf
cc_type	1	bslbf
reserved	1	bslbf
if (cc_type==line21) {		
reserved	5	bslbf
line21_field	1	bslbf
}		
else		
caption_service_number	6	uimsbf
easy_reader	1	bslbf
wide_aspect_ratio	1	bslbf
reserved	14	bslbf
}		
}		

number_of_services – an unsigned 5-bit integer in the range 1 to 16 that indicates the number of closed caption services present in the associated video service. Note that if the video service does not carry television closed captioning, the Caption_service_descriptor() shall not be present in the SIT.

Each iteration of the "for” loop defines one closed caption service present as a sub-stream within the 9 600 bit/s closed captioning stream. Each iteration provides the sub-stream's language, attributes, and (for advanced captions) the associated service number reference. Refer to EIA 708 B for a description of the use of the service number field within the syntax of the closed caption stream.

language – a 3-byte language code per ISO 639-2 defining the language associated with one closed caption service. The ISO_639_language_code field contains a three-character code as specified by ISO 639-2. Each character is coded into 8 bits according to ISO 8859-1 (ISO Latin-1) and inserted in order into the 24-bit field.

cc_type – a flag that indicates, when set, that an advanced television closed caption service is present in accordance with EIA 708 B. When the flag is clear, a line-21 closed caption service is present. For line-21 closed captions, the line21_field field indicates whether the service is carried in the even or odd field.

line21_field – a flag that indicates, when set, that the line-21 closed caption service is associated with the field 2 of the NTSC waveform. When the flag is clear, the line-21 closed caption service is associated with field 1 of the NTSC waveform. The line21_field flag is defined only if the cc_type flag indicates line-21 closed caption service.

caption_service_number – a 6-bit unsigned integer value in the range 0 to 63 that identifies the Service Number within the closed captioning stream that is associated with the language and attributes defined in this iteration of the "for" loop. See EIA 708 B for a description of the use of the Service Number. The caption_service_number field is defined only if the cc_type flag indicates closed captioning in accordance with EIA 708 B.

easy_reader – a Boolean flag which indicates, when set, that the closed caption service contains text tailored to the needs of inexperienced readers. Refer to EIA 708 B for a description of "easy reader" television closed captioning services. When the flag is clear, the closed caption service is not so tailored.

wide_aspect_ratio – a Boolean flag which indicates, when set, that the closed caption service is formatted for displays with 16:9 aspect ratio. When the flag is clear, the closed caption service is formatted for 4:3 display, but may be optionally displayed centered within a 16:9 display.

A.3.2.4 DIT

DIT should be in the discontinuous points in the transport stream. See Table A.36.

Table A.36 – DIT table (PID = 0x001E table_id = 0x7E)

Syntax	Number of bits	Identifier
discontinuity_information_section(){		
table_id	8	uimsbf
section_syntax_indicator	1	bslbf
reserved_future_use	1	bslbf
reserved	2	bslbf
section_length	12	uimsbf
transition_flag	1	bslbf
reserved_future_use	7	bslbf
}		

table_id – this is an 8 bit field, which shall be set to 0x7E.

section_syntax_indicator – the section_syntax_indicator is a 1 bit field which shall be set to "0".

section_length – this is a 12 bit field, which is set to 0x001.

transition_flag – this 1 bit flag indicates the kind of transition in the transport stream. When the bit is set to “1”, it indicates that the transition is due to a change of the originating source. The change of the originating source can be a change of originating transport stream and/or a change of the position in the transport stream (for example in the case of a time-shift). When the bit is set to “0”, it indicates that the transition is due to a change of the selection only, i.e. while staying within the same originating transport stream at the same position.

A.3.3 Guidelines to use D-VHS SI's

A.3.3.1 PAT

The PAT shall not violate the MPEG-2 systems requirements.

The PAT only lists selected services. In addition, the network_PID reference shall take the value of the SIT_PID instead of the NIT_PID. The references to non-selected programs/services shall be removed.

A.3.3.2 PMT

The PMT shall not violate the MPEG-2 systems requirements.

The corresponding PMT section shall reflect contents of the stream.

SI tables (NIT, SDT, EIT, BAT, RST, TDT, TOT) shall not be in the stream.

A.3.3.3 SIT

The SIT shall be packetized in transport stream packets starting from the beginning of the payload, i.e. in a packet with payload_unit_start_indicator in the transport stream packet header set to “1” and with the pointer_field set to “0x00”. Furthermore, it is recommended that the SIT be packetized in a single transport stream packet (if possible). The transmission_info_loop in SIT shall contain the Partial_transport_stream_descriptor. The following loop shall contain all the service_ids of the selected services. The service_loop may contain descriptors that can be in the EIT and SDT.

A.3.3.4 DIT

At a transition, the bitstream may be discontinuous with respect to any of the SI information (including PAT and PMT). The DIT table shall be inserted at this transition point.

Whenever a partial bitstream discontinuity occurs, two transport packets belonging to PID 0x001E shall be inserted directly at the transition point, with no other packets in between. The first one shall have 184 bytes of adaptation field stuffing with the discontinuity_flag set to “1” (in order to ensure compliance to MPEG-2 continuity counting constraints for successions of transitions introduced at independent transmission/storage stages). The second of these transport packets shall contain the DIT and not have the discontinuity_flag set to “1”.

A.3.4 Service Information for ES

A.3.4.1 Audio

a) When AC-3 audio is recorded, the stream_type value used for AC3 shall be 0x81.
b) When AAC audio is recorded, the stream_type value used for AAC shall be 0x0F.
c) When DTS audio is recorded, the stream_type value used for DTS is 0x82.
d) The ISO_639_language descriptor may be used to specify the language of the associated audio element.
e) If MPEG2 audio is recorded as more than one hierarchical layer, Hierarchy_descriptor is mandatory.
f) When linear PCM audio is recorded, the stream_type value used for Linear PCM is 0x83.

A.3.4.2 Closed caption

The closed caption data may be recorded in the picture user data of MPEG2 elementary video specified EIA 708 B.

When closed caption data is recorded, Caption_service_descriptor shall be recorded. This descriptor shall not appear on events with no closed captioning service.

A.3.4.3 Subtitling

a) When subtitling data specified in PNG of ARIB STD-B24 is recorded, component_tag shall be set with Stream_identifier_descriptor. Stream_content shall be set to 0x0C and component_type shall be set to 0x00, 0x01, 0x02, 0x20, 0x21, 0x22, 0x40, 0x41, 0x42 with Component_descriptor. ISO_639_language_descriptor may be used to specify the language of the associated subtitling data element. The stream type value used for PNG of ARIB STD-B24 is 0x06.

b) When subtitling data specified in ETS 300 743 is recorded, component_tag shall be set with Stream_identifier_descriptor. Stream_content shall be set to 0x03 and component_type shall be set to 0x10, 0x11, 0x12, 0x20, 0x21 and 0x22 with Component_descriptor. ISO_639_language_descriptor may be used to specify the language of the associated subtitling data element. The stream type value used for ETS 300 743 is 0x06.

A.3.5 Coding of text characters

When ISO_639_language_code in the descriptor indicates "jpn", text information is coded using the character sets and methods described in Annex A of ARIB STD-B10. In other cases, text information is coded using the character sets and methods described in Annex A of EN 300 468.

Further consstraints and clear indications to the Annex A of ARIB STD-B10 are described in this Subclause.

A.3.5.1 Overview

Text character sets are described in Table A.37. G0, G1, G2 and G3 sets are invoked as 8-bit codes (GL, GR) by invocation. One of the coded character set is designated to the G0, G1, G2 and G3 by designation of the coded character set. A DRCS character set, a MACRO character set, and a user defined character set shall not be used.

Table A.37 – Text characters set

Text characters set	
Kanji (2-byte code)	described in A.3.5.4
Alphabet/Numeral (1-byte code)	described in ARIB STD-B5 page 57
Hiragana (1-byte code)	described in ARIB STD-B5 page 58
Katakana (1-byte code)	described in ARIB STD-B5 page 59
Additional character (2-byte code)	described in A.3.5.5

A.3.5.2 Control character

The control character shall be used as shown in Table A.38.

Table A.38 – Control character

APR	Active position return
LS1	Locking-shift one
LS0	Locking-shift zero
SS2	Single-shift two
ESC	Escape
SS3	Single-shift three
SP	Space
MSZ	Designation middle size character in case of 1-byte code or 2-byte code of alphabet/numeral character defined in ARIB STD-B5 Table 8-2, and in case of SPACE.
NSZ	Designation normal size character
XCS(CSI)	Definition of character to take its place

Control of invocation shall comply with Table A.39.

Table A.39 – Invocation

		Control		
Invocation	**Expression**	**Set**	**Object for designation**	**Method**
LS0	00/15	G0	GL	Locking-shift
LS1	00/14	G1	GL	Locking-shift
LS2	ESC 06/14	G2	GL	Locking-shift
LS3	ESC 06/15	G3	GL	Locking-shift
LS1R	ESC 07/14	G1	GR	Locking-shift
LS2R	ESC 07/13	G2	GR	Locking-shift
LS3R	ESC 07/12	G3	GR	Locking-shift
SS2	01/9	G2	GL	Single-shift
SS3	01/13	G3	GL	Single-shift

Control of designation shall comply with Table A.40.

Table A.40 – Designation

Representation	Control	
	Category	**Object for designation**
ESC 02/8 F	1 byte G set	G0
ESC 02/9 F		G1
ESC 02/10 F		G2
ESC 02/11 F		G3
ESC 02/4 F	2 byte G set	G0
ESC 02/4 02/9 F		G1
ESC 02/4 02/10 F		G2
ESC 02/4 02/11 F		G3

Final character shall be used as shown in Table A.41.

Table A.41 – Final character

Category	Characters set	Final character (F)
G set	Kanji	03/9
	Alphabet/numeral	04/10
	Hiragana	03/0
	Katakana	03/1
	additional character	03/11

A.3.5.3 Initialization

Initialization shall work at the beginning of designation and invocation as shown in Table A.42. In the case where item_description_length in Extended_event_descriptor is equall to "0", initialization shall not work.

Table A.42 – Initial designation and invocation

Designation	G0	Kanji
	G1	English/Number
	G2	Hiragana
	G3	Katakana
Invocation	GL	LS0(G0)
	GR	LS2R(G2)
Character size	–	NSZ

A.3.5.4 Kanji

The Kanji character defined by row address from 1 to 84 in ARIB STD-B5 is set without changing cell and row address as shown in Figure A.1. The undefined part of the area in the row address from 1 to 84, and part of the area in the row address from 85 to 94 are reserved.

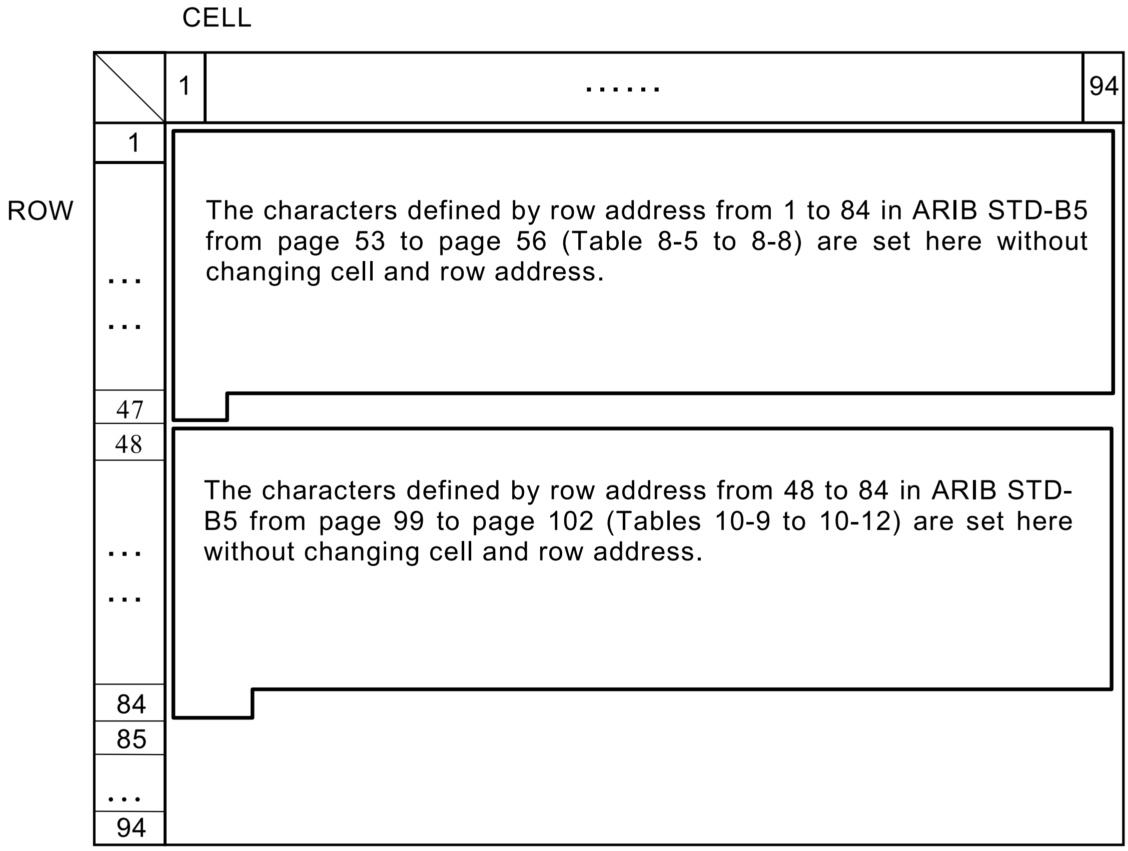


Figure A.1 – Kanji characters set structure

If JIS X 208 is revised, the characters may be set here without changing cell and row address.

A.3.5.5 Additional character

Additional characters defined by row address from 90 to 94 in ARIB STD-B5 page 101, 102 (Table 10-11,12) are set without changing cell and row address as shown in Figure A.2 and Table A.43.

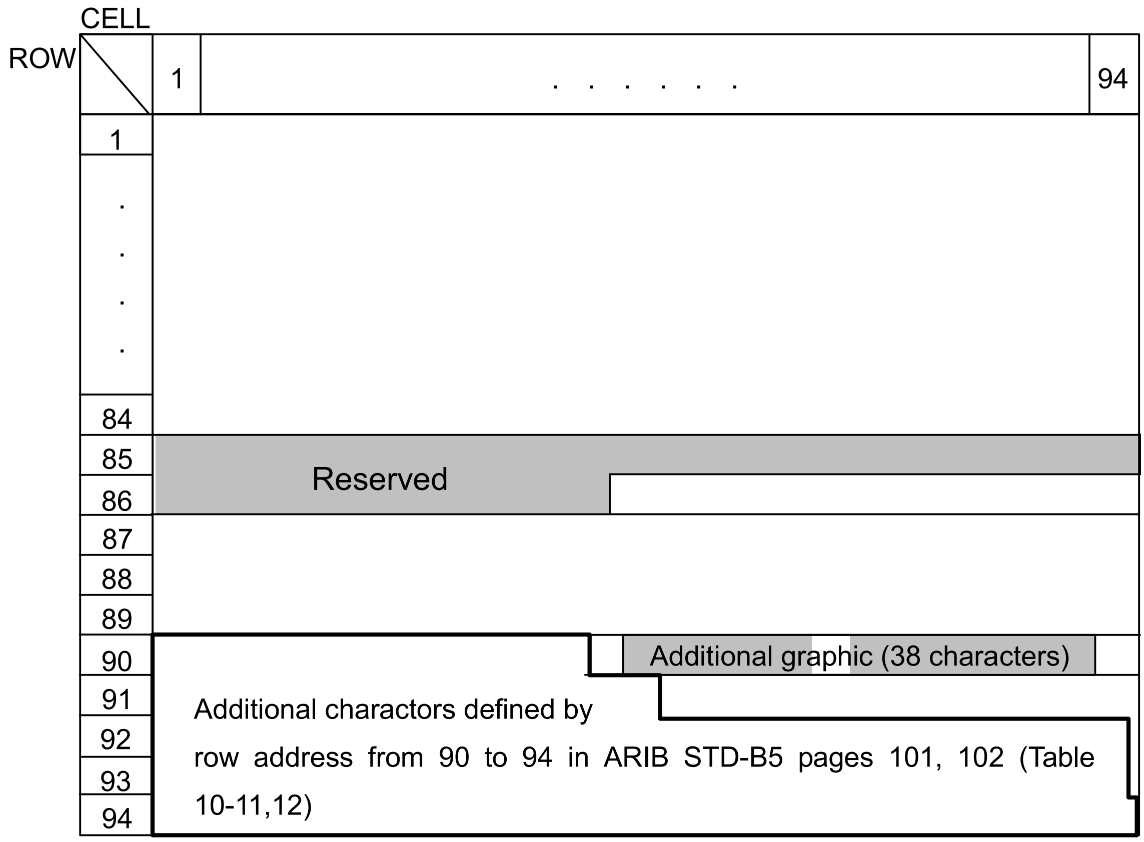
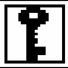


Figure A.2 – Additional character set structure

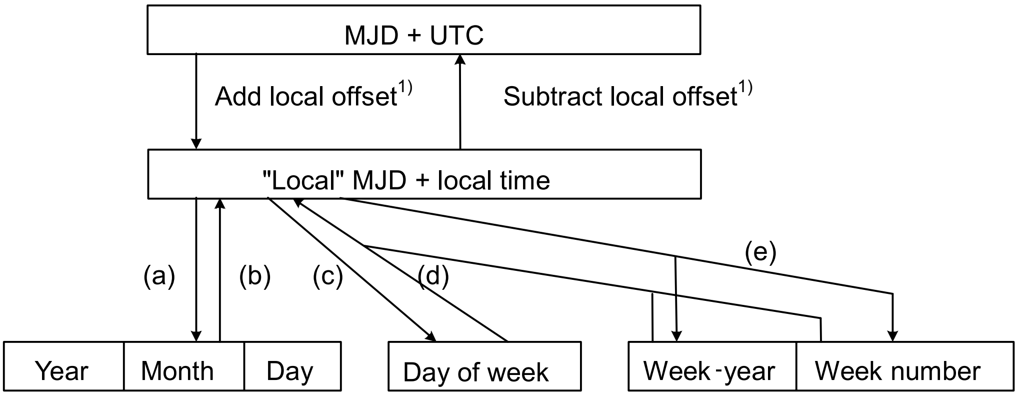
Table A.43 – Additional graphic characters set

Row	Cell	Code	Character
90	45	7A4D	10.
	46	7A4E	11.
	47	7A4F	12.
	48	7A50	HV
	49	7A51	SD
	50	7A52	P
	51	7A53	W
	52	7A54	MV
	53	7A55	手
	54	7A56	字
	55	7A57	双
	56	7A58	デ
	57	7A59	**S**
	58	7A5A	二
	59	7A5B	多
	60	7A5C	解
	61	7A5D	SS
	62	7A5E	B
	63	7A5F	N
	64		
	65		

Row	Cell	Code	Character
90	66	7A62	天
	67	7A63	交
	68	7A64	映
	69	7A65	無
	70	7A66	料
	71	7A67	
	72	7A68	前
	73	7A69	後
	74	7A6A	再
	75	7A6B	新
	76	7A6C	初
	77	7A6D	終
	78	7A6E	生
	79	7A6F	販
	80	7A70	声
	81	7A71	吹
	82	7A72	PPV
	83	7A73	㊙
	84	7A74	ほか
	85		
	86		
	87		
	88		

A.3.6 Conversion between time and date conventions (informative)

The types of conversion which may be required are summarized in Figure A.3.



Key

[1] Offsets are positive for longitudes East of Greenwich and negative for longitudes West of Greenwich.

Figure A.3 – Conversion routes between MJD and UTC

The conversion between MJD + UTC and the "local" MJD + local time is simply a matter of adding or subtracting the local offset. This process may, of course, involve a "carry" or "borrow" from the UTC affecting the MJD. The other five conversion routes shown on the diagram are detailed in the formulas a) to e) below.

For the symbols used, see Table A.44.

Table A.44 – Symbols

MJD	Modified Julian Date
UTC	Universal Time, Co-ordinated
Y	Year from 1900 (for example for 2003, Y = 103)
M	Month from January (= 1) to December (= 12)
D	Day of month from 1 to 31
WY	"Week number" Year from 1900
MN	Week number according to ISO 2015
WD	Day of week from Monday (= 1) to Sunday (= 7)
K, L , M' , W, Y'	Intermediate variables
int	Integer part, ignoring remainder
mod 7	Remainder (0 - 6) after dividing integer by 7

a) To find Y, M, D from MJD

$Y' = \text{int}\ [\ (MJD - 15\ 078{,}2) / 365{,}25\ ]$

$M' = \text{int}\ \{\ [\ MJD - 14\ 956{,}1 - \text{int}\ (Y' \times 365{,}25)\ ] / 30{,}600\ 1\ \}$

$D = MJD - 14\ 956 - \text{int}\ (Y' \times 365{,}25) - \text{int}\ (M' \times 30{,}600\ 1\ )$

If $M' = 14$ or $M' = 15$, then $K = 1$; otherwise $K = 0$

$Y = Y' + K$

$M = M' - 1 - K \times 12$

b) To find MJD from Y, M, D

If $M = 1$ or $M = 2$, then $L = 1$; otherwise $L = 0$

$MJD = 14\ 956 + D + \text{int}\ [\ (Y - L) \times 365{,}25] + \text{int}\ [\ (M + 1 + L \times 12) \times 30{,}600\ 1\ ]$

c) To find WD from MJD

$WD = [\ (MJD + 2) \bmod 7\ ] + 1$

d) To find MJD from WY, WN, WD

$MJD = 15\ 012 + WD + 7 \times \{\ WN + \text{int}\ [\ (WY \times 1\ 461 / 28) + 0{,}41]\ \}$

e) To find WY, WN from MJD

$W = \text{int}\ [\ (MJD / 7) - 2\ 144{,}64\ ]$

$WY = \text{int}\ [\ (W \times 28 / 1\ 461) - 0{,}007\ 9]$

$WN = W - \text{int}\ [\ (WY \times 1\ 461 / 28) + 0{,}41]$

See Table A.45 for an example.

Table A.45 – Example

MJD = 45218	W = 4315
Y = (19)82	WY = (19)82
M = 9 (September)	WN = 36
D = 6	WD = 1 (Monday)

NOTE These formulas are applicable between the inclusive dates 1900 March 1 to 2100 February 28.

BIBLIOGRAPHY

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3, rue de Varembé
1211 Genève 20
Switzerland

or

Fax to: **IEC**/CSC at +41 22 919 03 00

Thank you for your contribution to the standards-making process.

Nicht frankieren
Ne pas affranchir

Non affrancare
No stamp required

A Prioritaire

RÉPONSE PAYÉE

SUISSE

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International Electrotechnical Commission
3, rue de Varembé
1211 GENEVA 20
Switzerland

Q1 Please report on **ONE STANDARD** and **ONE STANDARD ONLY**. Enter the exact number of the standard: *(e.g. 60601-1-1)*

....

Q2 Please tell us in what capacity(ies) you bought the standard *(tick all that apply).* I am the/a:

- purchasing agent ❑
- librarian ❑
- researcher ❑
- design engineer ❑
- safety engineer ❑
- testing engineer ❑
- marketing specialist ❑
- other...

Q3 I work for/in/as a: *(tick all that apply)*

- manufacturing ❑
- consultant ❑
- government ❑
- test/certification facility ❑
- public utility ❑
- education ❑
- military ❑
- other...

Q4 This standard will be used for: *(tick all that apply)*

- general reference ❑
- product research ❑
- product design/development ❑
- specifications ❑
- tenders ❑
- quality assessment ❑
- certification ❑
- technical documentation ❑
- thesis ❑
- manufacturing ❑
- other...

Q5 This standard meets my needs: *(tick one)*

- not at all ❑
- nearly ❑
- fairly well ❑
- exactly ❑

Q6 If you ticked NOT AT ALL in Question 5 the reason is: *(tick all that apply)*

- standard is out of date ❑
- standard is incomplete ❑
- standard is too academic ❑
- standard is too superficial ❑
- title is misleading ❑
- I made the wrong choice ❑
- other ....

Q7 Please assess the standard in the following categories, using the numbers:
(1) unacceptable,
(2) below average,
(3) average,
(4) above average,
(5) exceptional,
(6) not applicable

- timeliness ....
- quality of writing....................................
- technical contents...................................
- logic of arrangement of contents ....
- tables, charts, graphs, figures...............
- other ....

Q8 I read/use the: *(tick one)*

- French text only ❑
- English text only ❑
- both English and French texts ❑

Q9 Please share any comment on any aspect of the IEC that you would like us to know:

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